70 IMPORTANT QUESTIONS & ANSWERS

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S. NO.	Title	No. of Questions	Page No.
1	PART-A **(Short Answer up to 25 words)**	40	1-11
2	PART-B **(Analytical/ Problem Solving questions)**	15	12-67
3	PART-C **(Descriptive/ Analytical/ Problem Solving/ Design Questions)**	15	68-118

PART-A

Que1. Define Disaster?

Ans. A serious disruption in the functioning of community or a society causing widespread material, economic, social or environmental losses which exceed the ability of the affected society to cope using its own resources. Disasters can be caused by many different kinds of **hazards**—scroll down for examples—and can have devastating impacts on people and communities.

The frequency, complexity and severity of their impacts are likely to increase in the future due to factors such as climate change, displacement, conflict, rapid and unplanned urbanization, technological hazards and public health emergencies.

Que2. Define Disaster Management.

Ans. Disaster Management is a collective term encompassing all aspects of planning for and responding to disasters, including both pre and post disaster activities. It refers to the management of both risk and consequence of disasters.
Disaster management is a process of effectively preparing for and responding to disasters. It involves strategically organizing resources to lessen the harm that disasters cause. It also involves a systematic approach to managing the responsibilities of disaster prevention, preparedness, response, and recovery.

Que3. What are three activities of disaster management?

Ans.

- Hazard analysis
- Vulnerability analysis
- Risk analysis

1) **Hazard Analysis: -** It includes the histogenesis (previous history of occurrence) of different hazard, nature and mode of occurrence of a particular hazard in a specific region or locality. These are helpful in the production and management of natural disaster.
2) **Vulnerability Analysis:-** In simple words, it is a potential for loss to an individual, community or place due to a disaster, which is affected by geographical as well as social conditions.
3) **Risk Analysis: -** It is a measure of the expected losses due to a hazardous event of a particular magnitude occurring in a given area over a specific time period.

Que4. Define Hazard?

Ans. Hazard is a potential disaster. It is an event that may lead to disaster. For example, a flood is a hazard, when it occurs. A process, phenomenon or human activity that may cause loss of life, injury or other health impacts, property damage, social and economic disruption or environmental degradation.

Hazards may be natural, anthropogenic or socio-natural in origin. A process, phenomenon or human activity that may cause loss of life, injury or other health impacts, property damage, social and economic disruption or environmental degradation.

Hazards may be single, sequential or combined in their origin and effects. Each hazard is characterized by its location, intensity or magnitude, frequency and probability. Biological hazards are also defined by their infectiousness or toxicity, or other characteristics of the pathogen such as dose-response, incubation period, case fatality rate and estimation of the pathogen for transmission.

Ques5. Define characteristics features of disaster?

Ans.
1. The disaster occurs rapidly, suddenly severe unexpectedly and it is widespread.
2. Disaster is different from emergencies and hazards and has a greater concern than them.
3. It is generally not confirmed to a local area or particular region.
4. It can occur at anywhere and anytime.
5. The intensity of environmental disaster always viewed in terms of the quantum of damages done to the human society (such as normal needs and process like food, shelter, health etc.).

Disasters can come in different forms and shapes, including sudden localized events like earthquakes, tsunamis, and tornadoes, as well as less sudden events that cover wider areas and cause damage for longer periods, such as famines, pandemics, and climate changes . Natural disasters like earthquakes, tornados, hurricanes, tsunamis, wildfires, and mudslides occur frequently and can have psychosocial outcomes including posttraumatic stress disorder, depression, anxiety, and physical health problems . Effective disaster/mass casualty first responders and leaders possess key attributes such as incident command/disaster knowledge, teamwork/interpersonal skills, performing one's role, cognitive abilities, communication skills, adaptability/flexibility, problem solving/decision-making, staying calm under stress, personal character, and overall knowledge . Disaster medicine is a discipline that combines emergency medicine and disaster management, involving the study of subject matter from multiple medical disciplines. It presents unique ethical situations and can be classified into natural and manmade disasters, with a cycle of activities including preparedness, response, recovery, and prevention/mitigation.

Ques6.Define vulnerability?

Ans. Vulnerability is the inability to resist a hazard or to respond when a disaster has occurred. For instance, people who live on plains are more vulnerable to floods than people who live in higher areas.
In actual fact, vulnerability depends on several factors, such as people's age, state of health, local environmental and sanitary conditions, as well as on the quality and state of local buildings and their location with respect to any hazard.

The conditions determined by physical, social, economic and environmental factors or processes which increase the susceptibility of an individual, a community, assets or systems to the impacts of hazards.

Ques7.Define types of vulnerability?

Ans.
1) **Economic vulnerability: -** Families with low income often live in high-risk areas around cities, because they can't afford to live in safer (and more expensive) places. This is what we call economic vulnerability.

2) **Physical vulnerability: -** A wooden house is sometimes less likely to collapse in an earthquake, but it may be more vulnerable in the event of a fire or a hurricane. This is what we call physical vulnerability.

Ques8.Define disaster risk?

Ans. Disaster risk is expressed as the likelihood of loss of life, injury or destruction and damage from a disaster in a given period of time. Disaster risk is widely recognised as the consequence of the interaction between a hazard and the characteristic that make people and places vulnerable and exposed.
Disasters are sometimes considered external shocks, but disaster risk is results from the complex interaction between development processes that generate conditions of exposure, vulnerability and hazard.
Risk = Hazard × Exposure × Vulnerability

Ques9.Define disaster risk characteristics?

Ans. 1) **Forward looking: -** The likelihood of loss of life, destruction and damage in a given period of time.
2) **Dynamic: -** It can increase or decrease according to our ability to reduce vulnerability.
3) **Invisible: -** It is comprised of not only the threat of high impact events, but also the frequent and low impact events that are often hidden.
4) **Unevenness: -** Hazards affect different areas, but the pattern of disaster risk reflects the social construction of exposure and vulnerability in different countries.
5) **Emergent and complex: -** Many processes, including climate change and globalised economic development, are creating new interconnected risks.

Ques10.Define measure to disaster risk?

Ans. We can measure disaster risk by analysing trends of, for instance previous disaster losses. These trends can help us to gauge whether disaster risk reduction is being effective or not. We can also estimate future losses by conducting a risk assessment.

Ques11.Define reduce to disaster risk?

Ans. However, if a country invests in disaster risk reduction, over time it can reduce the potential losses it faces, thus freeing up critical resources for development. Disaster risk reduction contribute to strengthening resilience and therefore to the achievement of sustainable development.

Ques12.Define Exposure?

Ans. Exposure represents the stock of property and infrastructure exposed to a hazard, and it can include socioeconomic factors. The situation of people, infrastructure, housing, production capacities and other tangible human assets located in hazard-prone areas.
Measures of exposure can include the number of people or types of assets in an area. These can be combined with the specific vulnerability and capacity of the exposed elements to any particular hazard to estimate the quantitative risks associated with that hazard in the area of interest.

Ques13.What are natural Disasters?

Ans. A natural disaster is a major adverse event resulting from natural processes of the Earth; examples are floods, hurricanes, tornadoes, volcanic eruptions, earthquakes, tsunamis, and other geological processes. A natural disaster can cause loss of life or damage property, and typically leaves some economic damage in its wake, the severity of which depends on the affected population's resilience (ability to recover) and also on the infrastructure available.

Ques14.What are Anthropogenic Disaster?

Ans. Anthropogenic hazards are hazards caused by human action or inaction. They are contrasted with natural hazards. Anthropogenic hazards may adversely affect humans, other organisms, biomes and ecosystems. The anthropogenic hazard results in the form of human intent, negligence, human error and involving a failure of man-made system.

Ques15.Define disaster and development linked?

Ans. Disaster and development are closely linked. Disasters can both destroy development initiatives and create development opportunities. Development schemes can both increase and decrease vulnerability.
The attitude was that the disasters, especially natural ones, were an act of god and as such were beyond human control; accepting death and damage to property was part of the costs. With such an attitude, most development plans were designed without consideration for the effect of disasters would have on community plans and vice versa. When a disaster did occur, the response was directed at meeting emergency needs and cleaning up.

Ques16.What is Climate Change?

Ans. Climate change refers to a change in the climate that persists for decades or longer, arising from either natural causes or human activity. Climate change is already modifying the frequency and intensity of many weather-related hazards as well as steadily increasing the vulnerability and eroding the resilience of exposed populations that depends on arable land, access to water, and stable mean temperatures and rainfall.

Ques17.What is Hazard Event?

Ans. It is the physical parameter of the **hazard event** that causes the harm. Environmental events become hazards once they threaten to affect society and/or the environment adversely. A **physical event,** such as a volcanic eruption, that does not affect human beings is a **natural phenomenon** but not a natural hazard. A natural phenomenon that occurs in a populated area is a **hazardous event.**
The manifestation of a hazard in a particular place during a particular period of time. Severe hazardous events can lead to a disaster as a result of the combination of hazard occurrence and other risk factors.

Ques18.What are Event parameter and Site parameter?

Ans. Local hazards - **Location & Probability**
• The extent to which they threaten local populations - **Severity**

• Ease with which their effects can be averted **– Manageability**
The severity of a natural hazard is quantified in terms of the magnitude of occurrence, which is an **event parameter**. It can also be done in terms of the effect of the occurrence at a particular location. This is called a **site parameter**. Both parameters may be combined in certain situations.

Ques19.What are Hazard Assessments?

Ans. A hazard assessment is a procedure designed to identify, assess, and manage health risks and safety hazards in the workplace. It also recommends appropriate safety measures to use to mitigate the identified hazards.
Hazard assessments can be a detailed and thorough process or it can be as simple as following running through a checklist or visually inspecting a working area. Hazard assessments should be conducted daily, even if there are no noticeable changes to the workspace or jobsite. This will ensure that all hazards are accounted for and all relevant hazard controls are put in place.

Ques20.Define Information Management in Disaster Risk Reduction?

Ans. In recent years, researchers and experts have been developing methods to conduct the assessment of hazards, vulnerability, and coping capacities; as well as techniques to combine such assessments in order to present them in risk map format. Such maps are essential in developing strategies to reduce the level of existing risks, and as a way to avoid a generation of new risks due to underlying social and economic risk drivers.

Ques21. What is disaster risk reduction?

Ans. Historically, dealing with disasters focused on emergency response, but towards the end of the 20th century it was increasingly recognised that disasters are not natural (even if the associated hazard is) and that it is only by reducing and managing conditions of hazard, exposure and vulnerability that we can prevent losses and alleviate the impacts of disasters. Since we cannot reduce the severity of natural hazards, the main opportunity for reducing risk lies in reducing vulnerability and exposure. Reducing these two components of risk requires identifying and reducing the underlying drivers of risk, which are particularly related to poor economic and urban development choices and practice, degradation of the environment, poverty and inequality and climate change, which create and exacerbate conditions of hazard, exposure and vulnerability.

Ques22. What is earthquake?

Ans. An earthquake is a term used to describe both a sudden slip on a fault and the resulting ground shaking and radiated seismic energy caused by the slip, or by volcanic or magmatic activity, or other sudden stress changes in the earth.

Worldwide, more than one million earthquakes occur each year, or an average of two a minute. A major earthquake in an urban area is one of the worst natural disasters that can occur. During the last four decades (1970-2017), earthquakes have been responsible for over a million deaths around the world in Armenia, China, Ecuador, Guatemala, Haiti, Iran, India, Indonesia, Japan, Mexico, Pakistan, Peru, and Turkey.

Ques23. What is the concept of Disaster Management Cycle?

Ans. The concept of Disaster Management Cycle integrates isolated attempts on the part of different actors, government and nongovernment, towards vulnerability reduction or disaster mitigation, within the enveloping domain of disaster management, as phases occurring in different time periods in disaster management continuum. This has facilitated a planned approach to disaster management in that post-disaster recovery and pre -disaster mitigation planning are perceived as integrated/related activities.

Ques24.What are the three stages of activities in disaster management?

Ans.
1. **Before a disaster**: to reduce the potential for human, material, or environmental losses caused by hazards and to ensure that these losses are minimised when disaster strikes

2. **During a disaster**: to ensure that the needs and provisions of victims are met to alleviate and minimise suffering

3. **After a disaster**: to achieve rapid and durable recovery.

Ques25. What is Rehabilitation in disaster management?

Ans. Rehabilitation implies activities that are undertaken to support the victims' return to of temporary housing and public utilities as interim measures to assist longer-term recovery through permanent housing and infrastructure. Besides physical elements, rehabilitation programmes also include economic rehabilitation through livelihood recovery and support actions and finding alternate employment options for those who cannot get back to their original occupations due to irreparable damage. Rehabilitation also includes psycho-social rehabilitation for those who are badly traumatised and need support in terms of psychosocial counseling or even medication in some cases.

Ques26. What is Reconstruction in disaster management?

Ans. Reconstruction attempts to return communities to improved pre-disaster functioning. It includes the replacement of buildings, infrastructure and lifeline facilities such as roads, bridges and communication links, so that longterm development prospects are enhanced rather than reproducing the same conditions which made an area or a population vulnerable in the first place.

Ques27.What is risk reduction in disaster management?

Ans. The Risk Reduction is chronologically the latest paradigm for mitigating the impact of disasters. The disaster risk management process is a process for good decision-making and for ensuring the best use of limited resources. It applies standard principles, process and techniques of risk management to disaster management Risk reduction can take place in 2 ways i.e. Long-term Mitigation and Short-term Preparedness. This protective process embraces measures, which enable governments, communities and individuals to respond rapidly to disaster situations to cope with them effectively.

Ques28. Why do earthquake happen?

Ans. Although the Earth looks like a just below the surface. The Earth is made of mantle, a liquid outer core and originates from inside the Earth is on a seismograph. most often along geologic faults, narrow that make up Earth's crust. pretty solid place from the surface, it's actually **four basic layers**: a solid crust, a hot, nearly solid a solid inner core.
The solid crust and top, stiff layer lithosphere isn't a continuous piece that wraps around the whole Earth like an eggshell. It's actually made up of giant puzzle pieces called shifting as they drift around on the Earth's crust is fractured into tectonic plates that have been moving very slowly over the Earth's surface for millions of years. This non-stop movement causes leads to cracks called **faults**. An **earthquake** is the sudden of the mantle make up a region called the **tectonic plates**. Tectonic plates are constantly **viscous**, or slowly flowing, mantle layer stress on Earth's crust. When the stresses plates move, it also causes movements movement of Earth's crust at a fault line of Earth—*a* **lithosphere**.

Ques29.Write Short notes on India's key Vulnerabilities?

Ans. Indian sub-continent has unique geo-climatic and socio-political conditions that make it vulnerable to both the natural as well as manmade disasters. Around 6% of the population of India is impacted annually by the exposures to disasters. They Key natural disasters in India include floods, droughts, cyclones, earthquakes, landslides and avalanches that have resulted in loss of lives and livelihoods.

According to a Planning Commission report, the key vulnerabilities of India include the following:

- Coastal States, particularly in the East Coast and Gujarat on west coast, are vulnerable to cyclones.
- 4 crore hectare land mass is vulnerable to floods and river erosion.
- 68 per cent of net sown area is vulnerable to drought.
- 55 per cent of total area is in Seismic Zones III-V and vulnerable to earthquakes of moderate to high density.

Ques30.what are different stages of cyclone?

Ans.

- **Tropical Depression:** When the maximum sustained speed of the wind measures up to 61 kilometres per hour or 33 knots, it is termed a tropical depression.

- **Tropical Storm:** When the maximum sustained speed of the wind is between 62 kilometres per hour and 88 kilometres per hour, or between 34 and 47 knots, it is termed a tropical storm.

- **Severe Tropical Storm:** When the maximum sustained wind speed measures between 89 kilometres per hour and 117 kilometres per hour, or between 48 and 63 knots, it is classified as a severe tropical storm.

- **Typhoon:** When the maximum wind sustained speed measures between 118 kilometres per hour and 220 kilometres per hour or between 64 and 120 knots, it is classified as a typhoon.

- **Super Typhoon:** When the maximum wind sustained speed measures beyond 220 kilometres per hour or beyond 120 knots, it is classified as a super typhoon.

Ques31.what is drought?

Ans. Drought is a prolonged dry period in the natural climate cycle that can occur anywhere in the world. It is a slow-onset disaster characterized by the lack of precipitation, resulting in a water shortage. Drought can have a serious impact on health, agriculture, economies, energy and the environment.

An estimated 55 million people globally are affected by droughts every year, and they are the most serious hazard to livestock and crops in nearly every part of the world. Drought threatens people's livelihoods, increases the risk of disease and death, and fuels mass migration. Water scarcity impacts 40% of the world's population, and as many as 700 million people are at-risk of being displaced as a result of drought by 2030.

Ques32.what are different keywords used to explain development?

Ans. It's a sad fact that the vast majority of contemporary architecture writing is produced by property developers. What is somewhat less sad is the reality that much of this cringeworthy drivel goes largely unnoticed by the public, even as they walk past it every day. But as it hides in plain sight, covering the hoardings that surround construction sites, this development-speak is normalising a warped representation of urban life which obscures developers' tendency to appropriate a neighbourhood's cultural value for the purpose of profit.

Ques33. What is earthquake magnitude?

Ans. Earthquake magnitude is a measure of the "size," or by an earthquake source and recorded by seismographs. (The types and nature of these waves are described in the section Seismic waves it is necessary for purposes of comparison to compress the range of wave amplitudes measured on seismograms by means of F. Richter set up a magnitude maximum seismic wave amplitude
standard seismograph (the Wood 100 km (60 miles) from the earthquake epicentre. Reduction of amplitudes observed at various distances to the amplitudes expected of empirical tables. Richter magnitudes the maximum wave amplitudes at two given distances is the same for all earthquakes and is independent of azimuth.

Ques34.what is ground shaking?

Ans. Ground shaking is the most familiar effect of earthquakes.It is a result of the passage of seismic waves through the ground , and ranges from quite gentle in small earthquakes to incredibly violent in large earthquakes. In the 27 March 1964 Alaskan earthquake, for example, strong ground shaking lasted for as much as 7 minutes! Buildings can be damaged or destroyed, people and animals have trouble standing up or moving around, and objects can be tossed around due shaking in earthquakes.

Ques35.What are landslides?

Ans. Landslides are caused by shaking of unstable slopes. They can easily destroy buildings in their roads and railroad lines, or take hilltop homes with them as they tumble. They even can dam rivers on occasion, like in the 17 August 1959 Hebgen Lake earthquake (magnitude 7.1) in Montana.

Ques36.What are fires?

Ans. Fires are a major source liquefaction can easily rupture natural gas mains and water mains, both contributing to the ignition of fires and hindering the example of this, during the where water was pouring down streets from broken water mains spot and same time, fire contributes to starting fires, by knocking do from storage containers, and tossing hot coals from barbeques and stoves. In the 1923 Kanto earthquake in Japan nearly 100,000 people died due to fires which swept the area after the damage in San Francisco from the 1906 earthquake was caused by the subsequent fires.
of damage after earthquakes.

Ques37.what is tsunami?

Ans. A series of huge waves originating in the ocean that occurs due to sudden displacement of water is called tsunami. Tsunami is a Japanese word that consists of two words tsu and nami. The word Tsunami means Tsu-harbor,name-wave.The meaning of this whole word is that there are high waves that rise at the port. We all know about the earthquake but let me tell you that if the earthquake arises from inside the sea, due to this, the waves of the sea become very high and it reaches the shore fast speed, this is called a tsunami.

Ques38.what are landslides?

Ans. If a solid rock or rock slips on a sudden slope, it is called landslide.Landslide is a worldwide process that occurs frequently in all countries of the world on a small scale, but there is a small number of large scale landslides that occur under special circumstances. Landslides can occur in a few seconds and may take a few days and months.

Ques39.What are volcanic eruptions?

Ans. A volcano is a cavity or hole from which lava, ash, gas and water vapor emerge. Volcanic activity covers all processes from the formation of magma and gas in the interior of the earth to the appearance of lava below and above the surface and becoming cool and solid.This discharge of lava can be in the form of central explosives or in the form of cracks.

Ques40.What is major power break down?

Ans. A power outage(also called a power cut,a power out,a power blackout,a power failure,a power loss, or a blackout) is the loss of the electrical power network supply to an end user.
There are many causes of power failures in an electricity network. Examples of these causes include faults at power station, damage to electric transmission lines, substations or other parts of the distribution system, a short circuit, cascading failure, fuse or circuit breaker operation.

Part B

Que.1 Discuss the various types of natural disasters and highlight the specific efforts to mitigate disasters in India.

Ans.

Disaster		
Natural Disaster		Man Made Disaster
Hydro-meteorological Based Disasters	**Geological Based Disasters**	• Textile Processing Industrial Hazards, • Major Power Break Downs, • Traffic Accidents, • Fire Hazards
• Flood, • Flash Flood, • Cloud Burst, • Drought, • Cyclone, • Forest Fires;	• Earthquake, • Tsunami, • Landslides, • Volcanic Eruptions.	

A natural disaster is a major adverse event resulting from natural processes of the Earth; examples include floods, volcanic eruptions, earthquakes, tsunamis, and other geologic processes. A natural disaster can cause loss of life or property damage, and typically leaves some economic damage in its wake, the severity of which depends on the affected population's resilience, or ability to recover.

An adverse event will not rise to the level of a disaster if it occurs in an area without vulnerable population. In a vulnerable area, however, such as San Francisco, an earthquake can have disastrous consequences and leave lasting damage, requiring years to repair.

In 2012, there were 905 natural catastrophes worldwide, 93% of which were weather-related disasters.

Overall costs were US$170 billion and insured losses $70 billion. 2012 was a moderate year. 45% were meteorological (storms), 36% were hydrological (floods), 12% were climatologically (heat waves, cold waves, droughts, wildfires) and 7% were geophysical events (earthquakes and volcanic eruptions). Between 1980 and 2011 geophysical events accounted for 14% of allnatural During World War I, an estimated 40,000 to 80,000 soldiers died as a result of avalanches during the mountain campaign in the Alps at the Austrian-Italian front, many of which were caused by artillery fire.

EARTHQUAKES

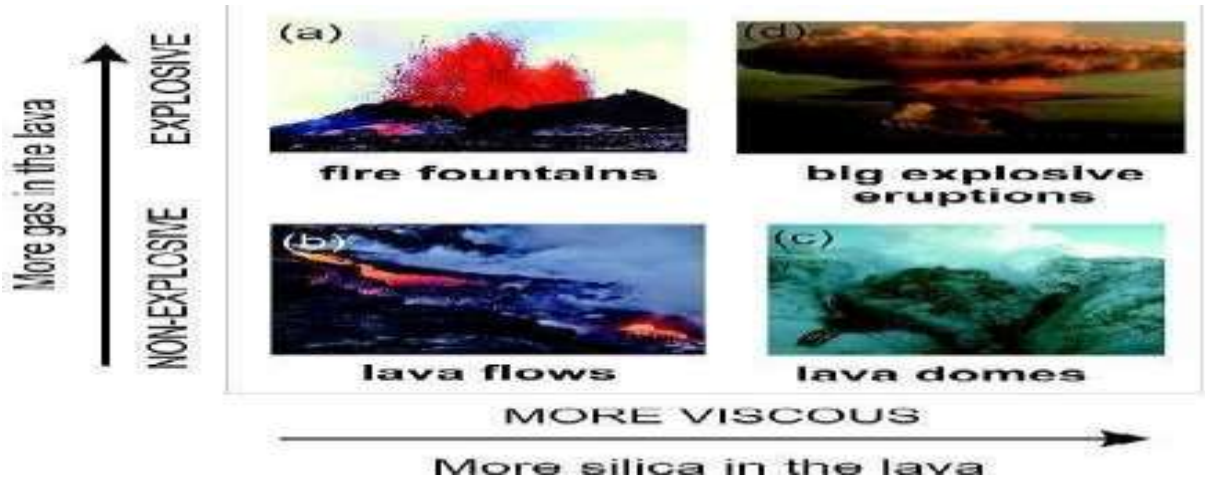
An earthquake is the result of a sudden release of energy in the Earth's crust that creates seismic waves. At the Earth's surface, earthquakes manifest themselves by vibration, shaking and sometimes displacement of the ground. The vibrations may vary in magnitude. Earthquakes are

caused mostly by slippage within geological faults, but also by other events such as volcanic activity, landslides, mine blasts, and nuclear tests. The underground point of origin of the earthquake is called the focus. The point directly above the focus on the surface is called the epicentre. Earthquakes by themselves rarely kill people or wildlife. It is usually the secondary events that they trigger, such as building collapse, fires, tsunamis.

VOLCANO

A volcano is an opening in the earth's crust through which lava, volcanic ash, and gases escape. Volcanic eruptions are partly driven by pressure from dissolved gas, much as escaping gases force the cork out of a bottle of champagne. Beneath a volcano, liquid magma containing dissolved gases rises through cracks in the Earth's crust. As the magma rises, pressure decreases, allowing the gases to form bubbles. How the magma (lava) behaves when it reaches the surface depends on both its gas content and chemical composition.

Lavas with low silica contents have low viscosities and flow freely, allowing any gas bubbles to escape readily, while lavas with high silica contents are more viscous (resistant to flow), so that any trapped gases cannot escape gradually.



1. A low viscosity (runny) lava, like basalt that contains lots of gas, forms fire-fountains

spewing spectacularly into the air and breaking into globs that solidify as they fall to the ground. Small fire-fountain eruptions produce cinder cones (like Eve Cone in northern British Columbia).

2. When runny lava contains less gas, however, it erupts in outpouring lava flows. Repeatedfire-fountain and lava flow eruptions over long time periods form gently sloping shieldvolcanoes like Anahim Peak in central British Columbia, and the volcanoes of the HawaiianIslands.
3. Andesite, dacite, and rhyolite lavas are progressively higher in silica and more viscous, so gases cannot escape gradually. If high-silica lavas contain little trapped gas, they may ooze slowly onto the surface to pile up as steep-sided lava domes.
4. When high-silica lavas contain lots of trapped gas, the pressure builds up and is released in explosive eruptions that produce volcanic ash. Some volcanoes experience both explosive and non-explosive activity, alternating explosive eruptions with periods of dome-building, forming strato volcanoes like Mount St. Helens and Mount Garibaldi.

DROUGHT

Drought is unusual dryness of soil, resulting in crop failure and shortage of water for other uses, caused by significantly lower rainfall than average over a prolonged period. Hot dry winds, high temperatures and consequent evaporation of moisture from the ground can contribute to conditions of drought.

Well-known historical droughts include:

- 1900 India killing between 250,000 to 3.25 million.
- 1921–22 Soviet Union in which over 5 million perished from starvation due to drought.
- 1928–30 Northwest China resulting in over 3 million deaths by famine.
- 1936 and 1941 Sichuan Province China resulting in 5 million and 2.5 million deaths respectively.
- The 1997–2009 Millennium Drought in Australian led to a water supply crisis across much of the country. As a result many desalination plants were built for the first time (see list).
- In 2006, Sichuan Province China experienced its worst drought in modern times with nearly 8 million people and over 7 million cattle facing water shortages.
- 12-year drought that was devastating southwest Western Australia, southeast South Australia, Victoria and northern Tasmania was "very severe and without historical precedent.
- In 2011, the State of Texas lived under a drought emergency declaration for the entire calendar year. The drought caused the Bastrop fires.

HAILSTORMS

Hailstorms are falls of rain drops that arrive as ice, rather than melting before they hit theground. A particularly damaging hailstorm hit Munich, Germany, on July 12, 1984, causing about 2 billion dollars in insurance claims.

An exceptionally clearly developed single-cell Cumulonimbus incus Big displaying the classic anvil shape; associated gusts may occur from the direct proximity to several times the height of the cloud.

TORNADOES

A tornado is a violent, dangerous, rotating column of air that is in contact with both the surface of the earth and a cumulonimbus cloud or, in rare cases, the base of cumulus. It is also referred to as a twister or a cyclone, although the word cyclone is used in meteorology in a wider sense,to refer to any closed pressure circulation. Tornadoes come in many shapes and sizes, but are typically in the form of a visible condensation funnel, whose narrow end touches the earth and is often encircled by a cloud of debris and dust. Most tornadoes have wind speeds less than 110 miles per hour (177 km/h), are approximately 250 feet (80 m) across, and travel a few miles (several kilometers) before dissipating. The most extreme tornadoes can attain wind speeds of more than 300 mph (480 km/h), stretch more than two miles (3 km) across, and stay on the

ground for dozens of miles (perhaps more than 100 km).

Well-known historical tornadoes include:

- The Tri-State Tornado of 1925, which killed over 600 people in the United States.
- The Daulatpur-Saturia Tornado of 1989, which killed roughly 1300 people in Banglades.

HEALTH DISASTER

An epidemic is an outbreak of a contractible disease that spreads through a human population.A pandemic is an epidemic whose spread is global. There have been many epidemics throughout history, such as the Black Death. In the last hundred years, significant pandemics include:

- The 1918 Spanish flu pandemic, killing an estimated 50 million people worldwide
- The 1957–58 Asian flu pandemic, which killed an estimated 1 million people
- The 1968–69 Hong Kong water flu pandemic
- The 2002-3 SARS pandemic
- The AIDS pandemic, beginning in 1959
- The H1N1 Influenza (Swine Flu) Pandemic 2009–2010

Other diseases that spread more slowly, but are still considered to be global health emergencies by the WHO, include:

IMPACT EVENTS

One of the largest impact events in modern times was the Tunguska event in June 1908.

SOLAR FLARE

A solar flare is a phenomenon where the sun suddenly releases a great amount of solar radiation, much more than normal. Some had known solar flares include:

- An X20 event on August 16, 1989
- A similar flare on April 2, 2001
- The most powerful flare ever recorded, on November 4, 2003, estimated at between X40and X45
- The most powerful flare in the past 500 years is believed to have occurred in September1859

GAMMA-RAY BURST

Gamma-ray bursts (GRBs) are flashes of gamma rays associated with extremely energetic explosions that have been observed in distant galaxies. They are the brightest electromagnetic events known to occur in the universe. Bursts can last from ten milliseconds to several minutes. The initial burst is usually followed by a longer-lived "afterglow" emitted at longer wavelengths (X-ray, ultraviolet, optical, infrared, microwave and radio).

All the bursts astronomers have recorded so far have come from distant galaxies and have been harmless to Earth, but if one occurred within our galaxy and were aimed straight at us, the effects could be devastating. Currently orbiting satellites detect an average of about one gamma-ray burst per day. The closest known GRB so far was GRB 031203.

Que2 Discuss the role of India in disaster management.

Ans.

Disaster Management Mechanism

Disaster Management can be defined as the organization and management of resources and responsibilities for dealing with all humanitarian aspects of emergencies, in particular, preparedness, response and recovery in order to lessen the impact of disasters. It means that all such measures should be taken so that hazard cannot take the form of disaster. Since we cannot prevent the coming of many natural hazards but can reduce their harmful effects through proper management, so that the loss of life and property can be minimized. As we know, the present danger is not from the natural disasters, but is the human disasters. Fiascos are the results of normal or human perils. Along these lines, through catastrophe the board, we can keep risks from being transformed into a debacle. Catastrophe the executives extensively includes the administration previously, during and after a calamity. It essentially implies arranging of different strides to decrease the result of a calamity, arranging viable reaction framework, arranging recovery and furthermore getting ready debacle tough networks in our own special layman language.

Fig: Disaster

managementThis administration can be ordered

into three stages:-

1) Pre-disaster management

It is identified with salvage even before any conceivable catastrophe emerges. Most significant motivation behind this is to relieve human misfortune. That additionally incorporates the improvement of data innovation framework; assembly of assets for vital activity, appraisal of calamity and issuance of a notice and to individuals through media, radio and so on shipping individuals in a protected spot in the event of catastrophe event and so on

2) Management during Disasters

This is one of the significant stages which rely upon the level of the pre-catastrophe the board stage. The course of speedy activity and the assistance of the casualties of the fiasco inclined regions, shipping them in safe regions are the premise of this. In this, food, dress, and wellbeing offices are additionally given to individuals.

3) Post disaster management

In this specific stage, redevelopment and remaking of influenced regions are finished. Additionally, the organization will undoubtedly assist influenced individuals by furnishing them with business or remuneration.

The idea of Disaster Management Cycle coordinates disconnected endeavours with respect to various entertainers, government and nongovernment, towards weakness decrease or calamity alleviation, inside the wrapping space of catastrophe the board, as stages happening in various time spans in misfortune the executive's continuum.

This has worked with an arranged way to deal with calamity the executives in that post-catastrophe recuperation and pre - debacle moderation arranging is seen as incorporated/related exercises.

Debacle Management endeavours are equipped towards calamity hazard the board. Debacle Risk Management "suggests the methodical course of utilizing managerial choices, association, functional abilities, and abilities to carry out approaches, techniques and adapting limits of the general public and networks to reduce the effect of normal perils and related natural and innovative fiascos. These involve all structures all exercises including underlying and non-primary measures to keep away from (anticipation) or to restrict (alleviation and readiness) unfriendly impacts to risks".

There are three key stages of activities in disaster management

1. **Before a disaster:** to reduce the potential for human, material, or environmental lossescaused by hazards and to ensure that these losses are minimised when disaster strikes.
2. **During a disaster:** to ensure that the needs and provisions of victims are met to alleviateand minimise suffering.
3. **After a disaster:** to achieve rapid and durable recovery.

Normal impression of calamity the executives are restricted to crisis alleviation and post- catastrophe restoration. This is so on the grounds that these 2 components are by a wide margin and not discrete.

Accordingly, avoidance, moderation and readiness structure pre-calamity exercises in the Disaster the board Cycle and reaction, containing help, recuperation and restoration are post- fiasco exercises. While crisis alleviation and recovery are crucial exercises, effective fiasco the board arranging should envelop the total domain of exercises and circumstances that happen previously, during and after debacles.

These stages can best be addressed as a cycle, which whenever finished public arrangement can deter future improvement of calamities by hindering the endless loop of circumstances and logical results. One of the major questions in calamity the board arranging is the distribution of assets at all phases of the debacle cycle, which upgrades the complete adequacy of hazard decrease movement and amplifies the general affect of catastrophe the executives.

This methodology has bestowed a more all encompassing discernment to catastrophe the board and has served to coordinate debacle the executives with improvement arranging in that most pre-fiasco exercises, include exercises for

weakness decrease like destitution decrease, business arrangement and so forth which are additionally standard advancement concerns.

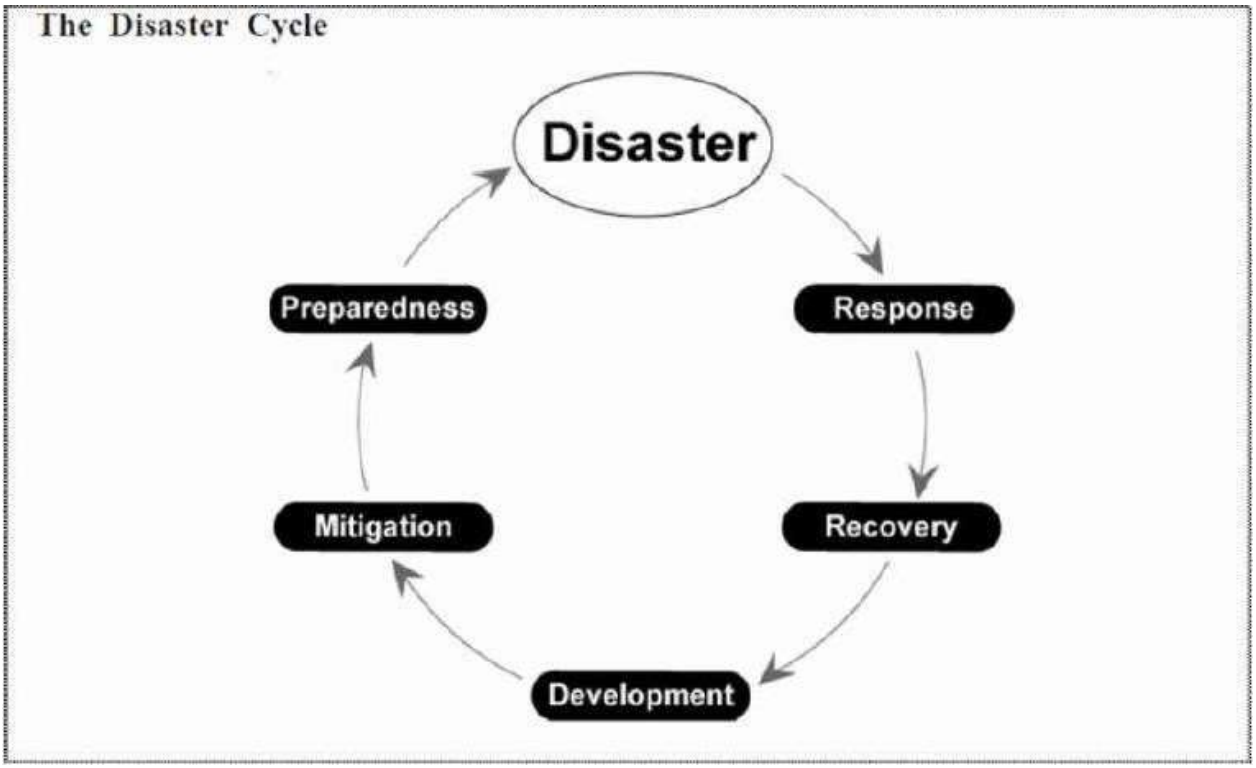


Fig: Disaster Management cycle

The different phases of disaster management are represented in the disaster cycle diagram.

1. The Disaster Event

This alludes to the constant occasion of a danger happening and influencing the 'components in danger'. The term of the occasion will rely upon the sort of danger, for instance, ground shaking may just happen for a couple of moments during a quake while flooding might occur throughouta more drawn out timeframe. Fiascos affect the actual scene. Inside a couple of moments, a whole locale is decreased to rubble in case of a quake. The effect prompts death toll and property in influenced regions; misfortunes being straightforwardly corresponded to the weakness of the district, physical and financial. Weakness is additionally financial.

More vulnerable segments of society, viz. ladies, youngsters, matured and debilitated, intellectually sick, and so forth, endure much more than their more grounded partners. Studies have additionally uncovered positive connection amongst destitution and weakness. The poor possess the most unsafe actual regions

since they are simpler to get and offer added benefits, similar to nearness to the ocean for anglers or ripe soil for ranchers close to flood inclined regions and so forth that makes them inclined to misfortunes, both of resources and life. The poor additionally come up short on the strength to recuperate from shock in the outcome of a catastrophe. This exposes the requirement for complex reaction to calamities, which assesses all friendly political and financial consequences. Issues to be addressed reach from physical to social and monetary weakness of more vulnerable areas that experience more comparative with other, better positioned.

2. Disaster Response

A Disaster is a disastrous occasion that has extreme adjusting sway. Outcomes are physical and social/human. Catastrophe Response needs to handle every single aforementioned challenge. Fiasco reaction involves re-establishing actual offices, recovery of influenced populaces, rebuilding of lost livelihoods and recreation endeavours to re-establish the framework lost or harmed. There are inborn significant exercises to be gained from catastrophe reaction. Reflectively, it exposes blemishes in endeavours relating to strategy and arranging concerning area and kind of framework and social plans to advance the social situating of the under favoured, especially as for admittance to assets of the oppressed. Calamity consequence is assessment time for the authoritative set up in that debacle reaction uncovered framework shortcomings. Calamity is a definitive trial of managerial proficiency, in the feeling of good effect on the climate, readiness, procedural straightforwardness, coordination, speed and ability.

There are inherent important lessons to be learnt with regard to administrative reforms by way of policy interventions to ensure

- Better institutional readiness
- Countering opposite pulls, for example, absence of social union inferable from nonsensical separations that successfully block reaction, in the self-appreciation help and communitarianism.
- Long-term moderation strategy to counter weaknesses, underlying and non-primary by empowering legitimate arrangements and fair execution of the equivalent.

3. Recovery

The recuperation stage includes execution of activities to advance manageable redevelopment (remaking, restoration) following a calamity. It covers long haul estimates like, remaking of houses, resources, foundation, school building, clinic

structures, and other public structures. It is an interaction attempted by a calamity influenced local area to completely re-establish itself to pre-fiasco level. Recuperation is the action that profits foundation frameworks to least working principles and guides long haul endeavours intended to return life to typical or further developed levels after a calamity. Recuperation is likewise here and there used to depict the exercises that incorporate the three covering periods of crisis help, recovery and reproduction.

Emergency Relief

Emergency relief refers to the period immediately following the disaster when steps are taken to meet the needs of survivors with regard to shelter, water, food and medical care. Activities
undertaken during and immediately following a disaster include immediate relief, rescue, and damage and needs assessment and debris clearance. Rescue and relief are critical elements of response. This would necessitate institutional/organisational improvements by way of better delegation to field agencies, improvements in decision-making and communication processes, incorporation of indigenous traditional knowledge on warning signs, a cartographic knowledge of safe and unsafe areas, survival methods, and traditional forms of insurance built around kinship and families.

The most crucial aspect in relief and rescue is communication across involved agencies. Disaster zone is often equated with a war zone, where communication is the critical factor, often, crucial,in fact, the deciding factor between success and failure.

4. Rehabilitation

Rehabilitation implies activities that are undertaken to support the victims' return to of temporary housing and public utilities as interim measures to assist longer-term recovery throughpermanent housing and infrastructure.

Besides physical elements, rehabilitation programmes also include economic rehabilitation through livelihood recovery and support actions and finding alternate employment options for those who cannot get back to their original occupations due to irreparable damage. Rehabilitation also includes psycho-social rehabilitation for those who are badly traumatised and need support in terms of psychosocial counselling or even medication in some cases.

5. Reconstruction

Reconstruction attempts to return communities to improved pre-disaster

functioning. It includes the replacement of buildings, infrastructure and lifeline facilities such as roads, bridges and communication links, so that long-term development prospects are enhanced rather than reproducing the same conditions which made an area or a population vulnerable in the first place.

6. Development

The inclusion of development as a phase in the disaster cycle is intended to ensure the naturaldisaster, societies factor hazard and vulnerability considerations into their development policiesand plans in the interest of overall progress. The rationale behind the use of the expression 'disaster management cycle' is that disaster and its management is a continuum of inter-linkedactivities. It is sometimes also referred to as the 'disaster development cycle', implying that disasters are periodic phenomena and occur regularly in such a way that there is development, followed by a disaster, then back to development till the next disaster.Sustainable development is another term that is useful in this context, implying development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It contains within it 2 key concepts of 'needs' in particular, to the essential needs of the world's poor, to which overriding priority should be given and the idea of limitations imposed by the state of technology and social organization on the environment's ability to meet the present and the future needs.

Que.3 Explain about earthquake in detail.

Ans. Earthquake Hazards

Misconceptions

Earthquakes really pose little direct danger to a person. People can't be shaken to death by an earthquake. Some movies show scenes with the ground suddenly opening up and people falling into fiery pits, but this just doesn't happen in real life.

The Effect of Ground Shaking

The first main earthquake hazard (danger) is the **effect of ground shaking**. Buildings can be damaged by the shaking itself or by the ground beneath them settling to a different level than it was before the earthquake (**subsidence**).

Buildings can even sink into the ground if soil liquefaction occurs. **Liquefaction** is the destabilization of sand or soil when **groundwater** (water underground) is forced out from between grains during the shaking of a moderate or strong earthquake. If liquefaction occurs under a building, it may start to lean, tip over, or sink several feet. The ground firms up again after the shaking has stopped. Liquefaction is a hazard in areas that have groundwater near the surface and sandy soil.

Buildings can also be damaged by strong surface waves making the ground heave and lurch. Any buildings in the path of these surface waves can lean or tip over from all the movement. The ground

shaking may also cause landslides, mudslides, and avalanches on steeper hills or mountains, all of which can damage buildings and hurt people.

Ground Displacement

The second main earthquake hazard is **ground displacement** (ground movement) along a fault. If a structure (a building, road, etc.) is built across a fault, the ground displacement during an earthquake could seriously damage or rip apart that structure.

From the figure on the left you can tell that the San Andreas Fault is a **right-lateral** transverse (strike-slip) fault because the other side of the road (on the opposite side of the fault) has moved to the right, relative to the photographer's position.

Flooding

The third main hazard is **flooding**. An earthquake can **rupture** (break) dams or levees along a river. The water from the river or the reservoir would then flood the area, damaging buildings and maybe sweeping away or drowning people.

Tsunamis and seiches can also cause a great deal of damage. A **tsunami** is what most people call a tidal wave, but it has nothing to do with the tides on the ocean. It is a huge wave caused by an earthquake under the ocean. Tsunamis can be tens of feet high when they hit the shore and can do enormous damage to the coastline. Seiches are like small tsunamis. They occur on lakes that are shaken by the earthquake and are usually only a few feet high, but they can still flood or knock down houses, and tip over trees.

Fire

The fourth main earthquake hazard is **fire**. These fires can be started by broken gas lines and power lines, or tipped over wood or coal stoves. They can be a serious problem, especially if the water lines that feed the fire hydrants are broken, too. For example, after the Great San Francisco Earthquake in 1906, the city burned for three days. Most of the city was destroyed and 250,000 people were left homeless.

Most of the hazards to people come from man-made structures themselves and the shaking they receive from the earthquake. The real dangers to people are being crushed in a collapsing building, drowning in a flood caused by a broken dam or levee, getting buried under a landslide, or being burned in a fire.

Earthquakes can cause a variety of hazards, including:

- Ground shaking: Can damage buildings and infrastructure
- Landslides and rockfalls: Can occur due to earthquakes
- Tsunamis: Huge waves caused by an earthquake under the ocean
- Flooding: Earthquakes can rupture dams or levees, causing flooding
- Fires: Can be caused by broken gas lines and power lines, or tipped over wood or coal stoves
- Liquefaction: Can occur due to earthquakes
- Avalanches: Can occur due to earthquakes
- Fissures: Can occur due to earthquakes

The extent of destruction and harm caused by an earthquake depends on several factors, including:

- The magnitude, intensity, and duration of the earthquake
- The local geology
- The time of day that it occurs
- The design and materials of buildings and industrial plants
- The risk-management measures put in place

To reduce the impacts of earthquakes, you can:

- Replace heavy roofing materials with lighter weight options
- Ensure any heavy tiles are properly secured to roof framing beneath
- Make sure your home is securely anchored to its foundations
- Avoid hazardous areas and steer clear of bookcases, heavy furniture, and appliances that may topple over

Earthquakes occur when tectonic plates move suddenly along faults in the Earth's crust, releasing stored-up energy:

Here are some details about earthquakes:

- Tectonic plates

  The Earth's outermost layer is made up of about 15 tectonic plates that move slowly relative to each other.

- Faults

  The edges of tectonic plates meet at faults, where the plates can scrape, bump, or drag against each other.

- Stress and friction

  Tectonic plates are always moving, but friction at their edges can cause them to get stuck. When the stress on the edge overcomes the friction, an earthquake occurs.

- Seismic waves

  The sudden release of energy during an earthquake causes seismic waves that travel through the Earth's crust, causing the ground to shake.

- Hypocenter and epicenter

  The location underground where the rock first breaks is called the hypocenter, and the location directly above it on the surface of the Earth is called the epicenter.

- Earthquakes and the Ring of Fire
  The Ring of Fire, a U-shaped belt around the Pacific Ocean, is a hot spot for earthquakes because it's a connection point for many tectonic plates

Earthquakes occur in distinct belts around the world, primarily along plate boundaries:

- The Circum-Pacific Belt

  Also known as the "Ring of Fire", this belt is located around the Pacific Ocean and is the most seismically active zone on Earth. It's where the Pacific plate is subducted beneath the surrounding plates, causing earthquakes in subduction zones. This belt accounts for about 81% of the world's largest earthquakes.

- The Alpide Belt

  This belt extends from Java to Sumatra, through the Himalayas, the Mediterranean, and into the Atlantic. It accounts for about 17% of the world's largest earthquakes.

- The Mid-Atlantic Ridge

  This ridge is located on the floor of the Atlantic Ocean, where two tectonic plates are spreading apart. Most of the mid-Atlantic Ridge is deep underwater.

- The Rift Valleys of East Africa
  This region is also a significant zone of seismic activity.

Earthquakes occur at faults in the Earth, which can be observed at the surface or hidden deep underground

Earthquakes can cause many hazardous effects, including:

- Ground shaking: Can damage buildings, lifelines, and the ground beneath them
- Ground displacement: Can seriously damage or rip apart structures
- Landslides: The shaking causes regions of the rock and soil to slide downhill
- Tsunamis: Huge waves caused by an earthquake under the ocean that can cause destruction both locally and at very distant locations
- Flooding: Earthquakes can rupture dams or levees along a river, causing flooding
- Soil liquefaction: A type of ground failure that can occur during an earthquake
- Fissures: Can occur during an earthquake
- Avalanches: Can occur during an earthquake
- Fires: Can occur during an earthquake

An earthquake is the shaking of the earth. It happens naturally when energy is released,

creating waves that travel in all directions. When an earthquake occurs the Earth vibrates and these vibrations are detected by seismographs. Moderate earthquakes happen every day, but big, damaging earthquakes are rare. Earthquakes are more common around the edges of tectonic plates. In India, more earthquakes occur where the Indian Plate meets the Eurasian Plate.

Some terminologies related to Earthquakes are:
Hypocenter: The location below the earth's surface where earthquake starts is called Hypocenter
Epicenter: The Location directly above it on the surface of the earth is called epicenter.
Ritcher scale: Measures magnitude of Earthquake
Mercalli scale: Measures the Earthquake intensity

Earthquakes in India
Earthquakes in India mainly happen due to the Indian tectonic plate is colliding into the Eurasian plate. The peninsular region of India is usually stable but sometimes earthquakes occur on the edges of smaller plates. For example the 1967 Koyna earthquake and the 1993 Latur earthquake happened in these areas.

India is divided into four seismic zones (II, III, IV, V) based on the level of seismicity:

Zone II: Low seismicity
Zone III: Moderate seismicity
Zone IV: High seismicity
Zone V: Very high seismicity (includes areas like the Himalayan region, northeastern states, Kutch, and Andaman & Nicobar Islands)
Zones V and IV cover the entire Himalayan region, North-East India, Western and Northern Punjab, Haryana, Uttar Pradesh, Delhi, and parts of Gujarat.

Most of the peninsular region is in a low-risk zone,

while the northern lowlands and western coastal areas are in a moderate hazard zone.

Types of Indian Earthquakes
In India, earthquakes can be categorized based on their origins and the tectonic settings. Here are the main types:

Type of Earthquake	Description	Example/Region
Tectonic Earthquakes	Caused by the movement of Earth's tectonic plates.	

– Interplate Earthquakes Occur at the boundaries between two tectonic plates. Himalayan region

– Intraplate Earthquakes Occur within a tectonic plate. Latur earthquake (1993), Maharashtra

Volcanic Earthquakes Associated with volcanic activity due to the movement of magma beneath the Earth's surface. Andaman and Nicobar Islands

Induced Earthquakes Caused by human activities such as mining, reservoir filling, geothermal energy extraction, and oil extraction.

– Reservoir-Induced Earthquakes induced by the filling of large dams. Koyna earthquake (1967), Maharashtra

Collapse Earthquakes Caused by the collapse of underground caverns or mines. Usually localized and of low magnitude. Mining areas

Explosion Earthquakes Caused by explosions, such as nuclear tests or large chemical explosions. Nuclear tests in Pokhran, Rajasthan

1. Tectonic Earthquakes

These are the most common types of earthquakes in India, caused by the movement of the Earth's tectonic plates. They can be divided into:

Interplate Earthquakes: Happen at the boundaries between two tectonic plates. For example, the Himalayan region has interplate earthquakes because the Indian and Eurasian plates collide there.

Intraplate Earthquakes: Happen within a tectonic plate not at its boundary. The 1993 Latur earthquake in Maharashtra is an example.

2. Volcanic Earthquakes

A special type of earthquake called a volcanic earthquake happens only in areas with active volcanoes. These earthquakes occur when molten rock (magma) is injected into or withdrawn from solid rock causing stress changes. This can make the ground sink or crack. These earthquakes can also happen when rock moves to fill spaces left by magma. Volcanic earthquakes do not mean the volcano will erupt they can happen at any time.

3. Induced Earthquakes

These are caused by human activities such as mining, reservoir-induced seismicity (due to the filling of large dams), geothermal energy extraction, and oil extraction. Examples include:

Reservoir-Induced Seismicity: The Koyna earthquake (1967) in Maharashtra is believed to have been induced by the filling of the Koyna dam reservoir.

4. Collapse Earthquakes

These happen when underground caves or mines collapse. They are usually small and only

affect a local area.

5. Explosion Earthquakes

These are caused by explosions, such as nuclear tests or large chemical explosions. For example, nuclear tests conducted in Pokhran, Rajasthan, generated minor seismic activity.

Earthquake Zones in India

Complete List of All Zones of Earthquakes in India:

The zones are identified using the Modified Mercalli (MM) intensity, which measures how earthquakes affect areas. After the Killari earthquake in Maharashtra in 1993 the seismic map was updated. The low danger zone called Seismic Zone I was combined with Seismic Zone II. So Zone I is no longer shown on the map.

Zone II

It is a low-intensity area, covering 40.93% of the country's land. This includes the Karnataka Plateau and the peninsula region.

Zone III

This region has moderate intensity and covers 30.79% of the country's area. It includes Kerala, Goa, and the Lakshadweep Islands, along with parts of Punjab, Rajasthan, Madhya Pradesh, Bihar, Jharkhand, Chhattisgarh, Maharashtra, Odisha, and Tamil Nadu.

Zone IV

This is called a high-intensity zone. It covers 17.49% of the country's land. It includes the rest of Jammu & Kashmir, Himachal Pradesh, Delhi, Sikkim, northern parts of Uttar Pradesh, Bihar, West Bengal, the western coast of Maharashtra, and Rajasthan.

Zone V

This is an extremely severe zone. It covers 10.79% of the country's land. It includes North Bihar, Himachal Pradesh, Uttarakhand, the Rann of Kutch in Gujarat, and the Andaman and Nicobar Islands.

Major Earthquakes in India List

Some of the devastating earthquakes have affected India. More than 58.6% of Indian Territory is vulnerable to earthquakes of moderate to very high intensity.

India's most significant earthquakes include:

Kutch Earthquake (1819) which was 8.3 magnitude

Assam Earthquake (1897)
Bihar-Nepal Earthquake (1934) of 8.4 magnitude
Koyna Earthquake (1967) of 6.5 magnitude
Uttarkashi (1991) of 6.6 magnitude
Killari (1993) of 6.4 magnitude
Bhuj (2001) of 7.7 magnitude
Jammu Kashmir (2005)

Key Regions Prone to Different Types of Earthquakes

Region	Type of Earthquake	Description
Himalayan Region	Tectonic (Interplate)	Frequent earthquakes due to the collision of the Indian and Eurasian plates.
Northeastern India	Tectonic (Interplate)	Complex tectonic interactions cause frequent earthquakes.
Western India (Kutch)	Tectonic (Intraplate)	Earthquake (2001).
Peninsular India	Tectonic (Intraplate)	Generally stable but can experience intraplate earthquakes like Latur (1993).
Andaman & Nicobar Islands	Tectonic & Volcanic	Prone to both tectonic and volcanic earthquakes.

List of Major Earthquakes in India Year-wise for UPSC

Year-wise Major Earthquakes
2015 India/Nepal Earthquake
2011 Sikkim Earthquake
2005 Kashmir Earthquake
2004 Indian Ocean Earthquake
2001 Bhuj Earthquake
1999 Chamoli Earthquake
1997 Jabalpur Earthquake
1993 Latur Earthquake
1991 Uttarkashi Earthquake
1941 Andaman Islands Earthquake
1975 Kinnaur Earthquake
1967 Koynanagar Earthquake
1956 Anjar Earthquake
1934 Bihar/Nepal Earthquake
1905 Kangra Earthquake

Causes of Earthquakes in India

Avalanches and Landslides

Tremors can make slopes unstable and collapse causing debris to fall and create landslides mainly in hilly areas. Earthquakes can also trigger avalanches, making large amounts of ice fall from snowy peaks. For example, the 2015 Nepal earthquake caused several avalanches on and near Mount Everest.

The 2011 Sikkim earthquake caused landslides and significant property damage, especially at the Singik and Upper Teesta hydroelectric projects.

Floods

Earthquakes can cause flash floods and make dams and reservoirs fail. Avalanches and landslides can block rivers, leading to flooding. For example, the 1950 Assam earthquake created a barrier in the Dihang River with huge debris, causing flash floods upstream.

Tsunami

Tsunamis are waves created when an ocean basin is disturbed displacing a large amount of water. Earthquakes can move the seafloor, causing these big waves. On December 26, 2004, an earthquake off Sumatra's coast caused the Indian Ocean Tsunami.

This happened because the Indian plate moved under the Burmese plate. Over 240000 people died in the Indian Ocean region and nearby countries. In 2011 a massive undersea earthquake in Japan caused 10-meter tsunami waves during the Tohoku earthquake. This led to a nuclear meltdown at Fukushima Daiichi, causing major global concerns due to radioactive fallout.

Impact of Earthquakes in India

Loss of Human Life and Property

Human towns and buildings suffer severe damage when the ground moves up, down, or sideways. For example, the 2015 Nepal earthquake caused massive urban destruction. This 7.8-magnitude earthquake was 8.2 kilometers deep. Many lives were lost due to uncontrolled urban growth, poorly built structures, and unscientific designs. Kathmandu's urban areas were badly hit causing 8000 deaths and $10 billion in economic loss

Alterations to the River's Course

One major effect of an earthquake is that it can change the course of a river. This happens when debris from the earthquake blocks the river's flow forcing the water to find a new path. This alteration can lead to flooding and other significant changes in the landscape.

Fountains of Mud

Earthquakes can force mud and boiling water to the surface. After the 1934 Bihar earthquake, fields were covered in knee-deep mud.

Fire Risks

Earthquakes damage gas pipelines and electric systems. This makes it much harder to put out fires caused by the earthquake.

Mitigation Measures for Earthquakes in India

The National Center for Seismology

Governmental organisations receive earthquake monitoring and hazard reports from a department of the Ministry of Earth Sciences. There are three divisions in it: Geophysical Observation System, Earthquake Hazard and Risk Assessment, and Earthquake Monitoring and services.

National Earthquake Risk Mitigation Project (NERMP)

Improving both the non-structural and structural parts of earthquake safety programs helps reduce risks in high-danger areas. In places with strong earthquakes, important safety measures are implemented. The NDMA which is in charge of the project, has made a detailed project report (DPR).

National Building Code (NBC)

It is a detailed building code and national regulation that sets rules for construction across the country. The Planning Commission first published it in 1970, and it was updated in 1983. There were three major changes: two in 1987 and one in 1997. The National Building Code of India 2005 (NBC 2005) replaced the old version. It focuses on addressing natural disaster challenges and using the best international practices.

Building Materials & Technology Promotion Council (BMTPC)

It works on projects to strengthen important buildings and raise awareness among people and government organizations. The goal is to help the public and policymakers reduce the risk to many existing public and private buildings

NDMA Guidelines for Earthquakes

In 2007, the NDMA released detailed earthquake guidelines. These rules tell State Governments, Central Ministries, and Departments what to do to make disaster management plans focused on earthquake risk. The guidelines are based on six main principles

The building of new structures that is earthquake-resistant.
Retrofitting and selective seismic strengthening of existing structures.
Enforcement and regulation.
Preparation and awareness.
Building capacity;
Emergency reaction.
Biggest Earthquakes in India
The devastating Bhuj earthquake happened on 26 January 2001 near the Pakistani border in Gujarat India. The largest earthquake in India with a magnitude of 8.6, occurred in the India-China region on 15 August 1950. It caused the death of 1530 people due to tectonic plates shifting 30 km deep.

Earthquake in the Indian Ocean
The 2004 Indian Ocean earthquake and tsunami important points

Magnitude: Between 9.1 and 9.3 on the Richter scale, making it one of the largest earthquakes ever recorded.
Duration: Faulting lasted between 8.3 and 10 minutes, unusually long for an earthquake of this magnitude.
Aftershocks: Numerous aftershocks continued for 3 to 4 months after the initial earthquake.
Energy Release: The earthquake released a massive amount of energy, causing significant geological effects.
Earth's Axis Shift: It is believed that the earthquake caused a slight shift in the Earth's axis due to the redistribution of mass.
Tsunami Generation: The seismic activity caused vertical movement of the seafloor, displacing a large volume of water and triggering a tsunami.
Impact: Indonesia was the first and hardest-hit country due to its proximity to the epicenter.
Casualties: Approximately 170000 people lost their lives, making it one of the deadliest natural disasters in recorded history

Human adjustment to and perception of earthquakes can influence how people prepare for and mitigate the effects of earthquakes:

- Risk perception

  Risk perception is influenced by factors such as fear of future earthquakes, expected damage, and expected breakdown of supplies.

- Adaptation behavior

  Adaptation behavior is how people respond to earthquake disasters. Some people may take

available approaches, while others may not.

- Policy efficacy
  Policy efficacy is a person's belief in their ability to understand and engage in earthquake risk management activities.

Here are some ways to prepare for and mitigate the effects of earthquakes:

- Create an emergency plan: Make a family emergency communications plan with an out-of-state contact and a meeting place in case you get separated.
- Prepare a supply kit: Include enough food and water for several days, a flashlight, a fire extinguisher, and a whistle.
- Secure furniture: Use brackets or seismic restraints to secure tall furniture and appliances to wall framing.
- Hang heavy items away from beds: Hang heavy items like pictures and mirrors away from beds, couches, and anywhere people sleep or sit.
- Build safer structures: Build safer structures and identify hazards.
- Educate the population: Provide education on earthquake safety.

Que.4 Explain in detail about Volcanic Disaster.

Ans. Volcano

A volcano is an opening in the earth's crust through which magma, gases and ash are released to the earth's surface. The molten rock material found in the interior of the earth is called magma. It can be noted that when magma reaches the earth’s surface, it is known as lava (Figure. 3.25). Vent is an opening or mouth of a volcano. Fumaroles are the gushing fumes through the gap in the volcano. Crater is a saucer shaped depression in the mouth of a volcano. When the crater is widened, it is called as Earthquake,Iran-Iraq border, 2017 Caldera. Volcanic ash consists of fragments of pulverized rock, minerals and volcanic glass, created during volcanic eruptions. Volcano generally erupts either through the vent (E.g. Mt. Fujiyama, Japan) or fissure (The Deccan Plateau, India).

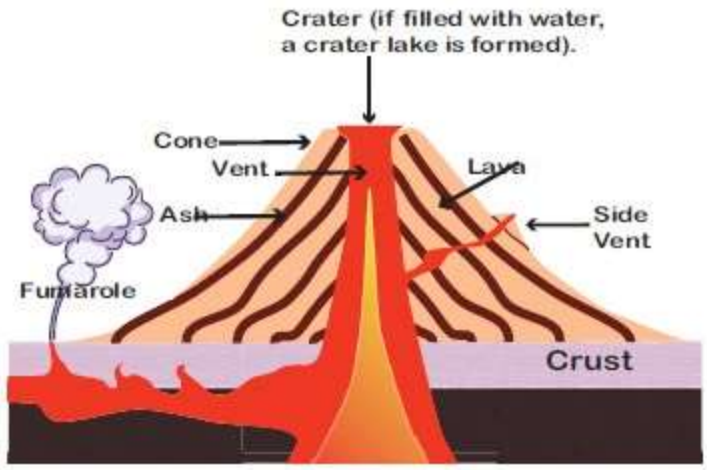


Figure 3.25 Volcano

Pumice is a volcanic rock produced when lava with a very high content of water and gases is discharged from a volcano.

Causes of Volcanic Eruptions

The following are the causes of volcanic eruptions:

Weak Zones in the Earth Crust: The parts of the earth where two tectonic plates collide against or drift apart from each other are considered very weak. Volcanoes may erupt in such zones, for example, African and Eurasian plates.

Magma Saturated with Gases: The magma, in the interior of the earth, is often found saturated with gases like carbon dioxide, and hydrogen sulfide. These gases together with water vapour make the magma highly explosive. Magma is forced out as lava on the surface of the earth due to the pressure exerted by these gases.

Types of Volcanoes

Based on the **frequency of eruption**, there are three types of volcanoes:

1. Active Volcanoes: Volcanoes which erupt frequently are called active volcanoes. Generally, their vent remains open. Mount Etna of Italy, Cotopaxi in Ecuador are some examples.

2. Dormant Volcanoes: These volcanoes may not have erupted in the recent past but there is a possibility of eruption at any time. In other words, they may lie dormant awaiting active eruption anytime. Sometimes gases and steam come out of them. They cause great destruction to life and property once they become active again. Mt. Vesuvius of Italy and Mt. Fujiyama of Japan are

examples.

3. Extinct Volcanoes: These volcanoes have exhausted their energy and have not erupted during the known geological period. The vent of these volcanoes remains closed with solidified lava. The formations such as craters may be filled with water and crater lakes may be formed. The slopes of these landforms may be covered with vegetation. Popa in Myanmar and Mt. Kenya in eastern Africa are the examples of extinct volcano.

On the **basis of nature of eruption and form** developed on the surface, they are classified into following types:

1. Shield Volcanoes: These are made up of basalt, a type of lava that is very fluid when erupted. They become explosive when water gets into the vent. They develop into a cinder cone. Hawaiian volcano is an example of this category.

2. Composite cone volcanoes: They are also called 'strato volcanoes'. They are cone-shaped volcanoes composed of layers of lava, ash and rock debris. Mount Vesuvius and Mount St. Helens are examples of composite volcanoes.

3. Cinder Cone Volcano : It forms when magma is thrown out to the surface, cooled in to ash and cinders and settled around the mouth of volcano. It is less dangerous than other volcanoes.

4. Lava Dome: Unlike composite and shield volcanoes, lava domes are of significantly smaller structure. They are formed when the lava is too viscous to flow to a great distance. As the lava dome slowly grows, the outer surface cools and hardens as the lava continues to pile within. Eventually, the internal pressure can shatter the outer surface, causing loose fragments to spill down its sides.

Fact File

The greatest volcanic explosion known to humans is perhaps Mt. Krakatau in August 1883.

Krakatau is a small volcanic island in the Sunda Straits, between Java and Sumatra.

The explosion could be heard in Australia, almost 4,000 km away.

The vibration set up enormous waves over 30 m high which drowned 36,000 people in the coastal districts of Indonesia.

Effects of Volcanic Activities

Showers of cinders and bombs can cause damage to life and properties. Sometimes ash can precipitate under the influence of rain and completely cover large areas.

The volcanic gases pose potential hazard to people, animals; agriculture, while sulfur dioxide gas can lead to acid rain and air pollution.

Positive Effects of Volcanoes

Volcanism creates new landforms. Volcanic rocks yield very fertile soil upon weathering and decomposition.

The Kimberlite rock of South Africa, the source of diamonds, is the pipe of an ancient volcano.

In the vicinity of active volcanoes, waters in the depth are heated from contact with hot magma giving rise to springs and geysers. The Puga valley in Ladakh region and Manikaran (Himachal Pradesh) are promising spots in India for the generation of geothermal electricity.

Distribution of Volcanoes across the World

Most known volcanic activity and the earthquakes occur along converging plate margins and mid-oceanic ridges. The major regions of volcanic distributions are as follows.

1. Pacific Ring of Fire

Circum-Pacific region, popularly termed the 'Pacific Ring of Fire', has the greatest concentration of active volcanoes. Volcanic belt and earthquake belt closely overlap along the 'Pacific Ring of Fire'. It is estimated to include two-thirds of the world's volcanoes.

2. Mid Atlantic Region

The Mid Atlantic Region coasts has comparatively fewer active volcanoes but many dormant or extinct volcanoes, example. St. Helena, Cape Verde Islands and the Canary Islands. But the volcanoes of Iceland and the Azores are active.

3. The Great Rift valley of Africa

In Africa some volcanoes are found along the East African Rift Valley. Kilimanjaro and Mt. Kenya are extinct volcanoes. The only active volcano in West Africa is Mt. Cameroon.

4. Mediterranean Region

Volcanoes of the Mediterranean region are mainly associated with the Alpine folds. Example, Mt. Vesuvius, Mt. Stromboli (known as the Light House of the Mediterranean Sea).

5. Other Regions

Elsewhere in the interiors of continents of Asia, North America and Europe active volcanoes are rare. There are no volcanoes in Australia.

Volcanoes in India

There are no volcanoes in the Himalayan region of India. However, Barren Island, lying 135 km north-east of Port Blair became active in 1991 and 1995.

However, the other volcanic island in Indian Territory is Narcondam (Andaman and Nicobar Islands) It is probably extinct. Its crater wall has been completely destroyed.

Volcanic eruptions can have many hazardous effects, including:

- Health hazards

  Volcanic eruptions can cause short-term health issues like eye and airway irritation, vomiting, dizziness, and headaches. Long-term exposure can lead to more serious conditions like bronchitis, lung infection, and lung disease.

- Ash fall

  Ash can travel hundreds or thousands of miles downwind from a volcano. It can be abrasive, gritty, and sometimes corrosive. Ash can be especially dangerous for infants, the elderly, and people with respiratory issues. Ash can also damage or clog drinking water and wastewater treatment facilities.

- Volcanic gases

  Volcanic gases like sulfur dioxide, carbon dioxide, and hydrogen fluoride can be harmful to health, vegetation, and infrastructure. Sulfur dioxide can cause acid rain and air pollution.

- Ballistic projectiles

  Volcanoes can hurl rocks into the air that travel like cannonballs. These rocks can land within 2 km of the vent, but can travel much further during explosive eruptions.

- Tsunamis

  Underwater volcanic eruptions can cause tsunamis, which are seismic sea waves.

- Climate change

Volcanic eruptions can impact the climate in a number of ways:

- Cooling effect: Sulfur dioxide emitted during eruptions can reflect sunlight and cool the Earth's climate.
- Ozone depletion: Sulfur dioxide can also contribute to ozone depletion.
- Greenhouse effect: Volcanic eruptions can release carbon dioxide and other halogens into the stratosphere, which can cause a greenhouse effect.

Volcanic eruptions can have many effects on the environment, including:

- Climate change: Volcanoes release large amounts of gases into the atmosphere, including sulfur dioxide and carbon dioxide. Sulfur dioxide can cause global cooling, while carbon dioxide is a greenhouse gas that can promote global warming.
- Air quality: Volcanic eruptions create smog and harmful gases that can irritate the skin, eyes, nose, and throat, and make it difficult to breathe.
- Water contamination: Volcanic eruptions can contaminate water supplies.
- Secondary events: Volcanic eruptions can cause secondary events like floods, landslides, and mudslides if there is rain, snow, or melting ice.
- Wildfires: Hot ashes from volcanic eruptions can start wildfires.
- Health threats: Volcanic eruptions can pose health threats, including suffocation, infectious diseases, respiratory diseases, burns, and traumatic injuries.
- Visibility reduction: Volcanic eruptions can reduce visibility.
- Machinery damage: Volcanic eruptions can damage machinery.
- Loss of basic amenities: Volcanic eruptions can damage power lines, gas pipelines, water and sewage pipes, leading to the loss of basic amenities like fresh water, gas, and electricity.

Que.5 Explain the phases of disaster management in detail.

Ans. The idea of Disaster Management Cycle coordinates disconnected endeavours with respect to various entertainers, government and nongovernment, towards weakness decrease or calamity alleviation, inside the wrapping space of catastrophe the board, as stages happening invarious time spans in misfortune the executive's continuum. This has worked with an arrangedway to deal with calamity the executives in that post-catastrophe recuperation and pre - debacle moderation arranging is seen as incorporated/related exercises.

Debacle Management endeavours are equipped towards calamity hazard the board. Debacle Risk Management "suggests the methodical course of utilizing managerial

choices, association, functional abilities, and abilities to carry out approaches, techniques and adapting limits of the general public and networks to reduce the effect of normal perils and related natural and innovative fiascos. These involve all structures all exercises including underlying and non-primary measures to keep away from (anticipation) or to restrict (alleviation and readiness) unfriendly impacts to risks".

There are three key stages of activities in disaster management

1. **Before a disaster:** to reduce the potential for human, material, or environmental lossescaused by hazards and to ensure that these losses are minimised when disaster strikes.
2. **During a disaster:** to ensure that the needs and provisions of victims are met to alleviateand minimise suffering.
3. **After a disaster:** to achieve rapid and durable recovery.

Normal impression of calamity the executives are restricted to crisis alleviation and post- catastrophe restoration. This is so on the grounds that these 2 components are by a wide margin and not discrete.

Accordingly, avoidance, moderation and readiness structure pre-calamity exercises in the Disaster the board Cycle and reaction, containing help, recuperation and restoration are post- fiasco exercises. While crisis alleviation and recovery are crucial exercises, effective fiasco the board arranging should envelop the total domain of exercises and circumstances that happen previously, during and after debacles.

These stages can best be addressed as a cycle, which whenever finished public arrangement can deter future improvement of calamities by hindering the endless loop of circumstances and logical results. One of the major questions in calamity the board arranging is the distribution of assets at all phases of the debacle cycle, which upgrades the complete adequacy of hazard decrease movement and amplifies the general affect of catastrophe the executives.

This methodology has bestowed a more all encompassing discernment to catastrophe the board and has served to coordinate debacle the executives with improvement arranging in that most pre-fiasco exercises, include exercises for weakness decrease like destitution decrease, business arrangement and so forth which are additionally standard advancement

concerns.

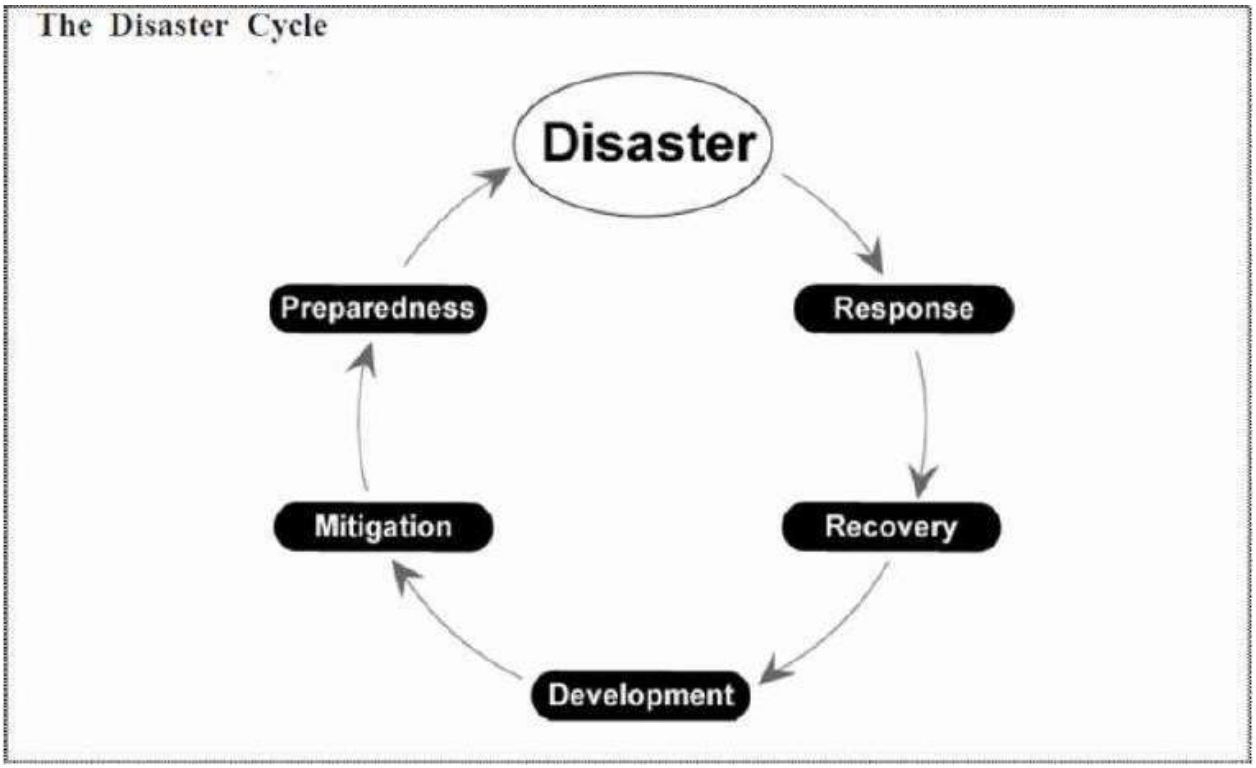


Fig: - 6.1 Disaster Management cycle

Que.6 Explain about corrective measure for sedimentation and erosion?.

Ans. here are several corrective measures for erosion, including:

- Vegetation

  Planting trees, grasses, legumes, and other vegetation can help stabilize soil and prevent erosion. Trees can help prevent landslides on steep slopes and stream bank erosion, but they may not be effective on moderately sloping hillslopes.

- Mulch

  Mulch can help protect soil from erosion, especially on footpaths with exposed soil.

- Check dams and terraces

  Check dams and terraces are effective for heavy erosion in areas with concentrated flow.

- Erosion control blankets

  These blankets can be made from natural or synthetic materials and slow down the flow of water.

- Silt fencing

Silt fencing traps sediment as water flows through it.

- Energy dissipaters

  These structures absorb the energy of flowing water.

- Articulated concrete blocks

  These blocks can be used to control erosion on artificial drainage channels and embankments of waterways.

- Mathematical modeling

  Mathematical models can help understand the mechanisms of soil erosion and resulting sediment surface runoff.

- Stormwater redirection

  Redirecting stormwater and roof runoff to areas that can settle and dissipate water can help.

- Crop rotation
  Crop rotation can help reduce soil erosion by fixing the land with alternatively planted deep-rooted crops.

Here are some corrective measures for sedimentation:

- Sediment control

  Temporary barriers like silt fences, berms, and sedimentation ponds can reduce runoff velocity and allow sediment to settle.

- Erosion control

  Maintaining healthy vegetation can reduce the amount of sedimentation transported by runoff.

- Soil stabilization

  Reducing erosion in borrow and waste areas, cut and fill slopes, and road shoulders can reduce sedimentation.

- Sediment traps

  A sediment trap is a section of the estuarine bed that is deepened to a greater depth than its surroundings.

- Hydraulic flushing

  A popular and cost-effective method for sedimentation control in sewers. It involves releasing a large volume of water quickly to create a flush wave that flushes out sediment.

- Sediment removal

  Flushing and sluicing are techniques that can remove or minimize sediment deposition.

- Adaptive strategies

  These actions mitigate the impacts of sedimentation without handling the sediment. For example, raising a dam or applying a protective coating to increase equipment resistance.

Que.7 Explain about sedimentation in detail?

Ans.

Sedimentation can cause a number of global problems, including:

- Water supply

  Sedimentation can reduce the amount of water that can be stored in reservoirs, which can threaten the sustainability of water supplies.

- Hydropower

  Sediment can damage hydroelectric turbines and other dam components, which can increase maintenance costs and decrease efficiency.

- Flooding

  Sediment can fill up storm drains and catch basins, which can lead to flooding when it rains.

- Water quality

  Sediment can make water cloudy, which can make it difficult for aquatic life to see food. It can also increase the cost of treating drinking water and cause odor and taste problems.

- Wildlife habitats

  Sediment can disrupt the natural food chain and destroy the habitats of small stream organisms, which can lead to declines in fish populations.

- Recreational access

  Sediment can make reservoirs shallower, which can reduce recreational access.

Sedimentation rates and associated storage losses can vary significantly between regions. For example, one study estimated that 120 California reservoirs had lost over 75% of their original capacity. Another study predicted that Japan's Sakuma reservoir will lose around 44% of its initial capacity by 2040.

To mitigate sedimentation, planners, developers, and operators can consider how to prevent, manage, and reduce sedimentation in water storage projects.

Although sedimentation of the world's reservoirs represents a serious threat to the sustainability of hydropower, there is limited guidance on how best to address the problem. Sedimentation

affects the safety of dams and reduces energy production, storage, discharge capacity and flood attenuation capabilities. It increases loads on the dam and gates, damages mechanical equipment and creates a wide range of environmental impacts. This article explores sedimentation issues as they pertain to hydropower facilities, dam safety and the environment; discusses sedimentation management techniques; and describes how they can be implemented to limit the impacts on hydropower.

Background

Reservoir sedimentation is a process of erosion, entrainment, transportation, deposition and compaction of sediment carried into reservoirs formed and contained by dams. In unregulated, mature rivers with stable catchments, sediment processes are relatively balanced. Construction of a dam decreases flow velocities, initiating or accelerating sedimentation,1 resulting in progressively finer materials being deposited (see Figure 1).

There are three stages in a reservoir's life:[2]

- Continuous and rapidly occurring sediment accumulation;
- Partial sediment balance, where often fine sediments reach a balance but coarse sediments continue to accumulate; and
- Full sediment balance, with sediment inflow and outflow equal for all particle sizes.

Most of the world's reservoirs are in the continuous accumulation stage.[2] Many were designed by estimating sedimentation rates in order to provide a pool with sufficient volume to achieve a specified design life. However, this design life is typically far less than what is actually achievable. Therefore, managing reservoirs to achieve a full sediment balance is essential in order to maximize their lives.

Developing regions of the world that stand to benefit most from hydroelectricity are often those with the highest sediment yields (see Figure 2 on page 8).[3] In these regions, sustainable hydropower development must involve consideration of sediment management techniques during design, construction and operation.

Sediment impacts on generation

About 0.5% to 1% of the total volume of 6,800 km^3 of water stored in reservoirs around the world is lost annually as a result of sedimentation.[2] As a result, global per capita reservoir storage has rapidly decreased since its peak at about 1980. Current storage is equivalent to levels that existed nearly 60 years ago.2

Loss of reservoir storage reduces flexibility in generation and affects the reliability of water supply. Without storage, hydropower facilities are entirely dependent on seasonal flows. These flows might not occur when energy is needed, eliminating one of the key benefits that hydropower provides over other renewables.

Sediments discharged from an upstream dam in a cascade system can increase tailwater levels, reducing power generation.[1] This would impact the generation potential of all plants in the cascade and increase the possibility of powerhouse flooding.

Sediment impacts on stability

Sediment loads are commonly idealized as a static at-rest soil pressure. The U.S. Bureau of Reclamation's design manual for small dams suggests that sediments be considered equivalent to a fluid with an implied pressure coefficient of about 0.39 and an internal friction coefficient of about 37 degrees.

However, actual reservoir sediment properties can vary considerably. Unconsolidated fine-grained sediments likely have lower shear resistance and a higher at-rest pressure coefficient, while a reservoir filled with coarser sediments may have a higher shear strength.1

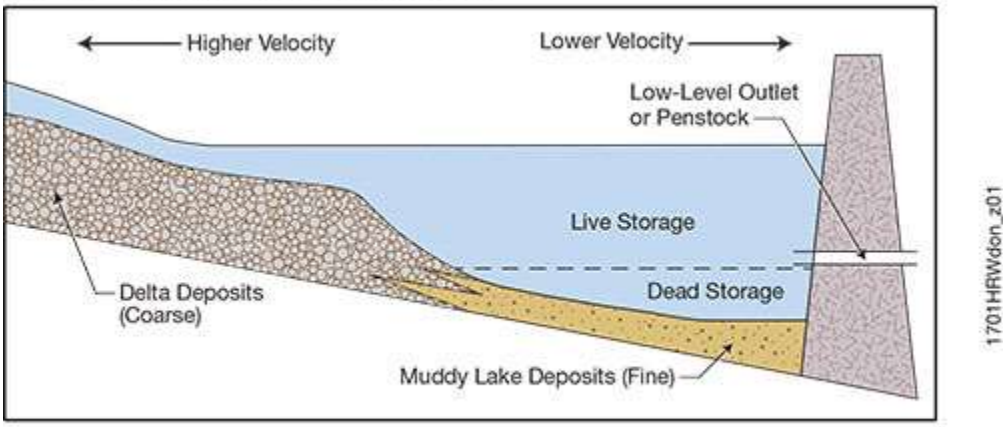
Published criteria with respect to potential changes in uplift pressures due to sedimentation often neglect the fact that fine-grained sediments may reduce uplift in the same manner as does an engineered upstream blanket. Conversely, in the case where there is a large turbid inflow, higher uplift pressures would be expected until enough particles had settled to form a blanket.

During a seismic event, it is likely that liquefied sediments would quickly return close to their original state, resulting in a rapid dissipation of pore pressures. Therefore, it may be questionable to automatically assign higher uplift pressures in this case.

Commonly used design considerations can omit some plausible load cases. For example, an underwater sediment slope failure could cause surface waves, adding additional loading, hydro-

dynamic pressure waves and an inertial loading from the dense fluidized soil-water mass. Another phenomena commonly ignored relates to turbidity currents in reservoirs. Such turbid fluid with a sediment load of 100 mg/l could be about 6% heavier than clear water.1

Figure 1 — Typical Reservoir Sediment Profile*



Typically, sedimentation in the reservoir behind a dam takes the form of progressively finer materials being deposited as the flows approach the dam.

*Adapted from Morris, G.L. and J. Fan, *Reservoir Sedimentation Manual*, McGraw-Hill, New York, 1998.

Submarine landslides are widely studied because of their potential to create tsunami waves. However, designers also need to consider the potential that failure of the steeply sloped deltaic front could increase loading and produce compression waves that may fluidize finer sediments near the toe of the landslide. As the deposition advances toward the dam, the potential for issues progressively increases.

It is often assumed that, during an earthquake, sediments fully liquefy, lose all strength and exert a dense fluid hydrostatic load on the dam. However, this degree of fluidization likely is not possible in a reservoir filled with coarse materials. Designers also often assume that the fully fluidized dense fluid contributes to hydro-dynamic pressure loading based on Westergaard's formula, ignoring the physical basis for its derivation. In fact, there is some question about the applicability of Westergaard's formula for hydro-dynamic pressures.

Designs also need to consider the degree of saturation of the sediments. There is minimal system damping under dynamic loading when reservoir sediments are fully saturated. However, significant reductions in acceleration occur when sediments are partially saturated.4,5 For rigid foundations, hydrodynamic pressures decrease slightly at the dam base when sediments are fully saturated but increase when partially saturated.[4] Partial saturation will increase he system's response to horizontal ground movement.[5] Sediment thickness is an important consideration, especially when the sediments are partially saturated.[5] Thin layers result in minimal absorption of horizontal motions, largely due to a relatively high modulus of elasticity and low attenuation coefficient.[6] Over the reservoir life cycle, this changes as sediments continue to accumulate.[7] Other important factors are sediment density, compressibility and pore water pressure.[5,7]

This dependence on sediment properties makes a strong case for their measurement and inclusion as part of the design.[4] However, designs are performed before sedimentation occurs and the same

sediments that are stable under normal conditions and absorb energy at the bottom of the reservoir could liquefy. For this reason, the use of a reservoir bottom reflection coefficient must be logically linked to assessment of the reservoir sediment behavior and ongoing monitoring.

Sediment impacts on discharge capability

Sediments will often block low-level outlets designed to allow for reservoir drawdown.[1] As sedimentation continues, clogging of spillway tunnels or other conduits may occur.[1]

Reduction of spillway capacity can occur as a result of the loss of approach depth when the sediment front reaches the dam. The reservoir becomes a delta-filled valley that takes a meandering course such that a flood wave does not spread out to allow flood routing.

Sediment impacts on equipment

Sediment can damage turbines and other mechanical equipment through erosion of the oxide coating on the blades, leading to surface irregularities and more serious material damage.[8] Sustained erosion can lead to extended shutdown time for maintenance or replacement.[8]

Que.8 Explain in detail about chemical and nuclear disaster?.

Ans. Chemical disasters may be traumatic in their impacts on human beings and have resulted in huge casualties and also large-scale damage to nature and property. The elements which are at highest risks due to chemical disaster primarily include the industrial plant, its employees & workers, hazardous chemicals vehicles, the residents of nearby settlements, adjacent buildings, occupants and surrounding community.

Brief Overview

In the year 1984, India had witnessed the world's worst chemical (industrial) disaster – the "Bhopal Gas Tragedy" (which happened on December 2nd, 1984). The Bhopal Gas tragedy was the most traumatic chemical accident in history. The tragedy led to the death of over 2500 people due to the accidental release of toxic gas Methyl IsoCyanate (MIC). The current generation continues to bear the ill-effects of the tragedy.

Such accidents are significant in terms of loss of lives, suffering, pain, injuries, damage to environment and property and loss of lives. India continued to witness a series of chemical accidents even after Bhopal had demonstrated the vulnerability of the country.

There are about 1861 Major Accident Hazard (MAH) units, spread across 298 districts and 25 states & 3 Union Territories, in all zones of the country. Besides, there are thousands of registered

and hazardous factories and unorganized sectors dealing with numerous ranges of hazardous material posing serious and complex levels of disaster risks.

Factors Causing Chemical Disasters

1. Ageing of process plants, defects in design and inadequate steps to pace with modern technologies in the Indian chemical industry has increased vulnerability to chemical disasters.
2. Organic solvents are the most common source of fires and explosions in the chemical industry.
3. A majority of the industrial accidents occur due to human error as a result of non-compliance of Standard Operating Procedures (SOPs). Piper alpha accident is a classic example of how human error can lead to chemical disasters, wherein a worker accidentally activates a pump under maintenance without a safety valve in place that leads to a gas leak and subsequent explosion.
4. There is an increased threat due to terrorist activities and sabotages. Natural disasters such as floods and earthquakes have also caused a major disaster in the chemical industry.
5. Another common cause that results in chemical disasters is improper maintenance of equipment. Regular maintenance at scheduled intervals following the manufacturer's recommendations is important for ensuring that the equipment runs smoothly and safely. The Flixborough incident is a prime example of improper maintenance which led to the death of 28 people and injured many.
6. Non-availability of an emergency response team to mitigate accidents during the transportation of hazardous chemicals have also resulted in major disasters in several locations in India.
7. Hazardous waste processing and its proper disposal need special attention as these activities can also contribute to fire, explosions and toxic releases to the environment.
8. Terrorist attack/ unrest leading to sabotage.

Chemical hazards are systems that may lead to chemical accidents, which are the unintentional or intentional release of hazardous substances that can harm people, the environment, or both:

- Causes

  Chemical accidents can be caused by natural events, human error, or deliberate acts. Examples include industrial accidents, terrorist attacks, or volcanic eruptions.

- Types

  Chemical accidents can range from small releases to major emergencies, and can be sudden or have a slow onset. They can include fires, explosions, leaks, or contamination of the food or water supply.

- Impact

Chemical accidents can cause illness, injury, disability, or death. They can also damage property and nature.

- Prevention

  Efforts to prevent chemical accidents include improved safety systems, changes in chemical use and manufacture, and monitoring of toxic waste disposal.

- Response

  If you are in a vehicle during a chemical accident, you should stop and seek shelter in a permanent building. If you must remain in your car, you should close the windows and vents, and shut off the air conditioner and heater.

The most significant chemical accident in history was the 1984 Bhopal disaster in India, where over 3,000 people died after a toxic gas was released at a pesticides factory

Toxic chemicals released by industries this year, tons

7,593,146

Sources and methods:

Facts

- Each and **every second 310 Kg of toxic chemicals are released** into our air, land, and water by industrial facilities around the world.

- This amounts to approximately **10 million tons** (**over 21 billion pounds**) of toxic chemicals released into our environment by industries each year.

- Of these, over **2 million tons** (**over 4.5 billion pounds**) **per year** are **recognized carcinogens.** This amounts to about **65 Kg** each second.

Definitions

- **toxic chemicals:** substances that can cause severe illness, poisoning, birth defects, disease, or death when ingested, inhaled, or absorbed by living organisms.

- **recognized carcinogens:** as defined by the Occupational Safety & Health Administration (view OSHA Carcinogens list)

- **release:** any spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping or disposing into the environment *(as defined by EPCRA, U.S. Emergency Planning and Community Right-to-Know Act, section, 329-8)*

- **environment**: includes water, air and land and the interrelationship which exists among and between water, air, and land and all living things.

A "release" refers to different ways that toxic chemicals from industrial facilities enter the air, water and land. Releases include spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping, or disposing into the environment.

A toxic chemical release is when toxic chemicals enter the environment through various means, such as leaking, spilling, or discharging. These releases can occur in the air, water, or land.

Type of release	**Description**
Air	Releases from point sources, like smokestacks, or non-point sources, like leaks
Water	Discharges to bodies of water, like rivers and streams, or runoff
Land	Disposal in landfills, surface impoundments, or underground injection wells
Off-site	Transfers to a facility separate from the reporting facility for disposal or other release

Toxic chemical releases can be caused by natural events, accidents, or intentional acts. They can range from small releases to major emergencies, and can have a slow onset or be sudden and acute.

Some of the chemicals released in the largest quantities by industry include:

- Chemical manufacturing: Ammonia and ethylene
- Paper manufacturing: Methanol
- Electric utilities: Sulfuric acid

If you are exposed to toxic chemicals, you should:

- Rinse your eyes with clean, cool water for several minutes
- Seek medical help
- Avoid spreading chemicals from your clothes onto your skin
- Put contaminated clothes into a closed container
- Put a warning on the bag and put it in a secure place outside until it can be disposed of

A nuclear explosion is a result of a nuclear device detonating and releasing energy rapidly. The effects of a nuclear explosion include:

- Blast: A large fireball is created that vaporizes everything inside it, including soil and water.
- Heat: The explosion produces a wave of intense heat.
- Radiation: The explosion produces a wave of radiation.
- Fallout: Radioactive material mixes with the vaporized material and falls back to Earth as particles, which can contaminate people, food, and water. Fallout is most dangerous in the first few hours after the explosion.
- Mushroom cloud: The vaporized material rises and forms a mushroom cloud.

Some notable nuclear explosions and accidents include:

- Hiroshima and Nagasaki

  The United States dropped atomic bombs on these Japanese cities at the end of World War II, killing around 109,000 people.

- Chernobyl

  This accident in Ukraine in 1986 killed around 30 people and damaged billions of dollars of property. The World Health Organization estimates that up to 4,000 additional cancer deaths may be related to the accident.

- Castle Bravo
  This hydrogen bomb test in the Marshall Islands in 1954 was the worst radiological disaster in the United States' testing history.

If you are in the area of a nuclear explosion, you should:

- Find a shelter in the middle of a large building or underground, such as a basement or subway.
- Stay inside for at least 24 hours, unless told otherwise by authorities.
- Turn off fans, air conditioners, and forced-air heating units.
- Close windows, doors, and fireplace dampers.
- Follow instructions from emergency response officials

- **Nuclear Disaster**
- A nuclear and radiation accident is defined by the International Atomic Energy Agency as "an event that has led to significant consequences to people, the environment or the facility.

Examples include lethal effects to individuals, large radioactivity release to the environment, or reactor core melt."The prime example of a "major nuclear accident" is one in which a reactor core is damaged and significant amounts of radiation are released, such as in the Chernobyl Disaster in 1986. The impact of nuclear accidents has been a topic of debate practically since the first nuclear reactors were constructed. It has also been a key factor in public concern about nuclear facilities.Some technical measures to reduce the risk of accidents or to minimize the amount of radioactivity released to the environment have been adopted. Despite the use of such measures, "there have been many accidents with varying impacts as well near misses and incidents".

- **Impact**
- Benjamin K. Sovacool has reported that worldwide there have been 99 accidents at nuclear power plants.[5] Fifty-seven accidents have occurred since the Chernobyl disaster, and 57% (56 out of 99) of all nuclear-related accidents have occurred in the USA.Serious nuclear power plant accidents include the Fukushima Daiichi nuclear disaster (2011), Chernobyl disaster (1986), Three Mile Island accident (1979), and the SL-1 accident (1961).Stuart Arm states, "apart from Chernobyl, no nuclear workers or members of the public have ever died as a result of exposure to radiation due to a commercial nuclear reactor incident."Nuclear-powered submarine mishaps include the K-19 reactor accident (1961),the K-27 reactor accident (1968),and the K-431 reactor accident (1985).Serious radiation accidents include the Kyshtym disaster,Windscale fire, radiotherapy accident in Costa Rica,radiotherapy accident in Zaragoza,radiation accident in Morocco,Goiania accident,radiation accident in Mexico City, radiotherapy unit accident in Thailand,and the Mayapuri radiological accident in India.
- **Nuclear power plant accidents**
- One of the worst nuclear accidents to date was the Chernobyl disaster which occurred in 1986 in Ukraine. That accident killed 30 people directly, as well as damaging approximately $7 billion of property. A study published in 2005 estimates that there will eventually be up to 4,000 additional cancer deaths related to the accident among those exposed to significant radiation levels.Radioactive fallout from the accident was concentrated in areas of Belarus, Ukraine and Russia. Approximately 350,000 people were forcibly resettled away from these areas soon after the accident. Benjamin K. Sovacool has reported that worldwide there have been 99 accidents at nuclear power plants from 1952 to 2009 (defined as incidents that either resulted in the loss of human life or more than US$50,000 of property damage, the amount the US federal government uses to define major energy accidents that must be reported), totaling US$20.5 billion in property damages.Fifty-seven accidents have occurred since the Chernobyl disaster, and almost

two-thirds (56 out of 99) of all nuclear-related accidents have occurred in the USA. There have been comparatively few fatalities associated with nuclear power plant accidents. Nuclear reactors become preferred targets during military conflict and, over the past three decades, have been repeatedly attacked during military air strikes, occupations, invasions and campaigns.

- **Nuclear Safety**
- Nuclear safety covers the actions taken to prevent nuclear and radiation accidents or to limit their consequences. This covers nuclear power plants as well as all other nuclear facilities, the transportation of nuclear materials, and the use and storage of nuclear materials for medical, power, industry, and military uses. The nuclear power industry has improved the safety and performance of reactors, and has proposed new safer (but generally untested) reactor designs but there is no guarantee that the reactors will be designed, built and operated correctly.Mistakes do occur and the designers of reactors at Fukushima in Japan did not anticipate that a tsunami generated by an earthquake would disable the backup systems that were supposed to stabilize the reactor after the earthquake.According to UBS AG, the Fukushima nuclear accidents have cast doubt on whether even an advanced economy like Japan can master nuclear safety. Catastrophic scenarios involving terrorist attacks are also conceivable. An interdisciplinary team from MIT have estimated that given the expected growth of nuclear power from 2005 – 2055, at least four serious nuclear accidents would be expected in that period.To date, there have been five serious accidents (core damage) in the world since 1970 (one at Three Mile Island in 1979; one at Chernobyl in 1986; and three at Fukushima-Daiichi in 2011), corresponding to the beginning of the operation of generation II reactors. This leads to on average one serious accident happening every eight years worldwide. Nuclear weapon safety, as well as the safety of military research involving nuclear materials, is generally handled by agencies different from those that oversee civilian safety, for various reasons, including secrecy. There are ongoing concerns about terrorist groups acquiring nuclear bomb-making material.

Que.9 Explain in detail about Soil Erosion**.**

Ans. What is Soil Erosion?

In this process, the soil particles are loosened or washed away in the valleys, oceans, rivers, streams or far away lands. This has been worsening due to human activities such as agriculture and deforestation.

Soil erosion is a continuous process that occurs either slowly or at an alarming rate. It results in a continuous loss of topsoil, ecological degradation, soil collapse, etc.

Let us have a detailed look at the causes, effects and prevention of soil erosion.

Cause of Soil Erosion

Following are the important causes of soil erosion:

Rainfall and Flooding

Higher intensity of rainstorms is the main cause of soil erosion. Four types of soil erosion are caused by rainfall:

- Rill erosion
- Gully erosion
- Sheet erosion
- Splash erosion

The raindrops disperse the soil, which is then washed away into the nearby streams and rivers. Regions with very heavy and frequent rainfall face a large amount of soil loss. The flowing water during floods also erodes a lot of soil by creating potholes, rock-cut basins, etc.

Agriculture

The farming practices are the major cause of soil erosion. The agricultural activities disturb the ground. The trees are cleared and the land is ploughed to sow new seeds. Since most of the crops are grown during the spring season, the land lies fallow during winters. Most of the soil is eroded during winters.

Also, the tyres of tractors make grooves on the land, making a natural pathway for water. Fine soil particles are eroded by wind.

Grazing

The grazing animals feed on the grasses and remove the vegetation from the land. Their hooves churn up the soil. They also pull out plants by their roots. This loosens the soil and makes it more prone to erosion.

Logging and Mining

A large number of trees are cut down to carry out the logging process. Trees hold the soil firmly. The canopy of the trees protects the soil from heavy rainfall. The leaf litter that protects the soil from erosion, is also lost during logging.

Mining activities also disturb the land and leave the soil more prone to erosion.

Construction

The construction of roads and buildings exposes the soil to erosion. The forests and grasslands are cleared for construction purposes, which exposes the soil making it vulnerable to erosion.

Rivers and Streams

The flowing rivers and streams carry away the soil particles leading to a V-shaped erosion activity.

Heavy Winds

During dry weather or in the semi-arid regions, the minute soil particles are carried away by the wind to faraway lands. This degrades the soil and results in desertification.

Also Read: Types of Pollution

Effects of Soil Erosion

The major effects of soil erosion include:

Loss of Arable Land

Soil erosion removes the top fertile layer of the soil. This layer is rich in the essential nutrients required by the plants and the soil. The degraded soil does not support crop production and leads to low crop productivity.

Clogging of Waterways

The agricultural soil contains pesticides, insecticides, fertilizers, and several other chemicals. This pollutes the water bodies where the soil flows.

The sediments accumulate in the water and raise the water levels resulting in flooding.

Air Pollution

The dust particles merge in the air, resulting in air pollution. Some of the toxic substances such as pesticides and petroleum can be extremely hazardous when inhaled. The dust plumes from the arid and semi-arid regions cause widespread pollution when the winds move.

Desertification

Soil erosion is a major factor for desertification. It transforms the habitable regions into deserts. Deforestation and destructive use of land worsens the situation. This also leads to loss of biodiversity, degradation of the soil, and alteration in the ecosystem.

Destruction of Infrastructure

The accumulation of soil sediments in dams and along the banks can reduce their efficiency. Thus, it affects infrastructural projects such as dams, embankments, and drainage.

Also Read: Difference between erosion and weathering

Soil Erosion Prevention

Soil erosion is a serious environmental issue. Steps should be taken to curb this problem. Following are some of the methods of soil erosion prevention:

1. Plant trees on barren lands to limit erosion of soil.
2. Add mulch and rocks to prevent the plants and grass underneath to prevent soil erosion.
3. Mulch matting can be used to reduce erosion on slopes.
4. Put a series of fibre logs to prevent any water or soil from washing away.
5. A wall at the base of the slope can help in preventing the soil from eroding.
6. Every household should have a proper drainage system so that water flows down into proper water collecting systems.

Key Points of Soil Erosion

- It is the natural process of wearing away topsoil, but human activities have accelerated the process.
- It is usually caused due to the removal of vegetation, or any activity that renders the ground dry.
- Farming, grazing, mining, construction and recreational activities are some of the causes of soil erosion.

- The effects of soil erosion are not just land degradation. It has led to a drastic increase in pollution and sedimentation in rivers that clogs the water bodies resulting in a decline in the population of aquatic organisms.
- Degraded lands lose the water holding capacity resulting in floods.

The health of the soil is of utmost importance to the farmers and the population that depends upon agriculture for food and employment. There are several challenges to resist soil erosion, but there are solutions to prevent it as well.

Que.10 Explain Man-made Hazards**?**

Ans. Man-made Hazards:

Disasters have been a part of human life since the beginning of time. Man-made hazards and catastrophes are events that reduces quality of life, health issues, and even fatalities. Some natural catastrophes have become more frequent and intense as a result of human influences.

- Man-made disasters can be intentional, by accident or due to improper management (negligence) of man made systems.
- Avoiding man-made hazards requires caution from authorities and proper disaster management plans at a local level. When members of human communities intend to inflict damage on fellow members, strict surveillance mechanisms help to avert danger.
- A hazard exists only when it affects a person exposed to it. But not all disasters come with an alarm bell.

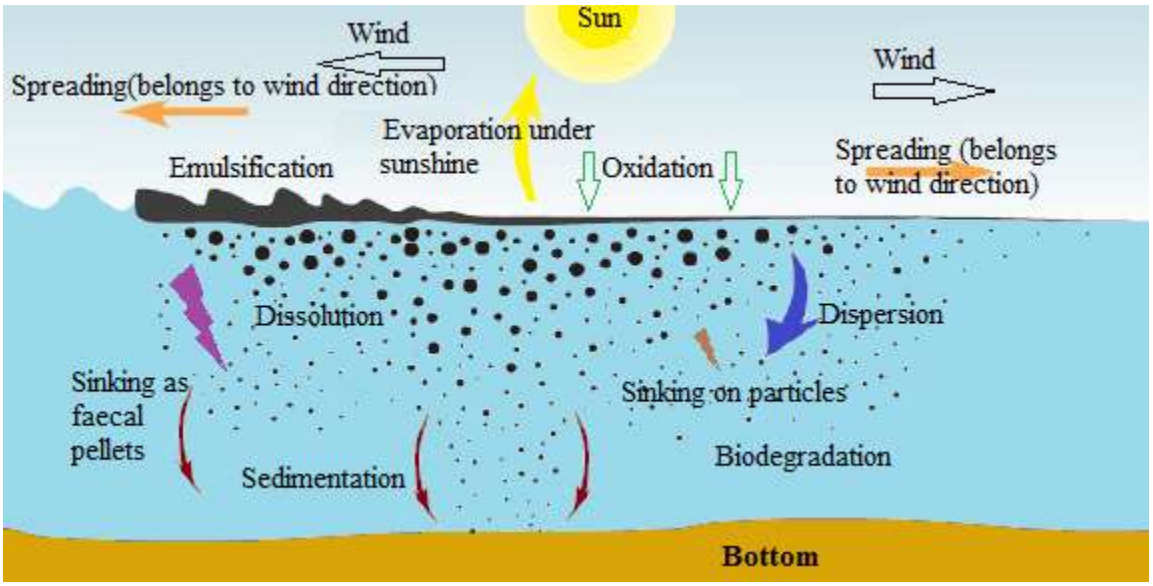
Definition of Man-made Hazards

- Man-made hazards are **disruptions of a massive scale** over a short or long period of time, caused by **human action or inaction**.
- These disasters cause economic damage, loss of life, health emergencies, etc.
- Man-made disasters can be classified into Sudden disasters, Continuing disasters, and Armed conflicts.
- Human forces are more responsible for **sudden disasters** than natural ones. Sudden hazards include the discharge of methyl isocyanate from the Union Carbide factory in Bhopal in 1984 and the 1986 emission of radioactive chemicals from the Chernobyl nuclear plant in the Soviet Union. Terrorist attacks unleashed on citizens are also sudden disasters.
- Oil spills and pollution over a long period of time are examples of continuing man-made hazards and disasters. The causal factor cannot be blocked immediately leading to a lasting impact.
- **Wars and armed conflicts** have constantly mired human history. There are domestic upheavals, territorial conflicts between countries, and intervention of extra state entities like terrorist organizations

Types of Man-made Hazards-Pro Live Tests

- Unlimited Test Re-Attempts

Oil and Chemical Spill



- People, animals, land and the marine ecosystem are all affected by an oil and chemical spill, making it one of the most destructive environmental man-made hazards that may occur.
- Such accidental spills harm and pollute vital resources in the food supply system, among other things.
- Explosives, flammable and combustible materials, poisons, and radioactive materials are examples of hazardous materials. Accidents during transportation and chemical leakages at industrial units or refineries are the most common sources of these releases.
- Most industrial accidents may be prevented by implementing and following adequate safety standards and processes. Accidents generate harmful gases and poisonous chemicals that damage living forms in neighbouring regions.
- According to estimates, Iraqi forces fled from Kuwait in **1991** and broke valves of oil wells and pipelines, causing the **Gulf War oil leak** releasing 8 million barrels of oil into the Persian gulf. A five-inch-thick oil slick covered 101 miles by 42 miles at its peak.
- The Bhopal Gas Tragedy of 1984 was the world's worst chemical (industrial) catastrophe in 1984. Methyl isocyanate, a poisonous chemical, was accidentally released, resulting in the death of nearly 2500 individuals.

Terrorism

- To intimidate, coerce, or demand ransom, terrorists employ force or violence against people or property in violation of criminal laws.
- In order to convince individuals that their government is helpless to combat terrorism, terrorists often use threats to generate fear among the population.
- From terrorist threats to assassinations to kidnappings and hijackings, bomb scares and bombings, and cyber warfare to the use of chemical, biological and nuclear weapons, terrorism may take many forms.
- The **9/11 terrorist attacks** that took place on the Twin Towers (New York) in USA was

the turning point in counter terrorism policy. The United States took a lead in suppressing the growth of terrorist networks in Iraq and Afghanistan.

- In November 2008, Mumbai, Maharashtra, India, was the scene of numerous terrorist operations. Lashkar-e-Taiba, a Pakistani terrorist group, carried out the assaults in Mumbai. A number of landmarks in Mumbai's south were attacked by terrorists, including the Chhatrapati Shivaji train station and Leopold Café, two hospitals, and a theatre.

Wars

- Wars are fully fledged armed attacks between states that wreak immense havoc to life. It leads to huge economic losses, deaths and psychological distress.
- The two World Wars of the 20th century saw millions lose their life and loved ones.
- Wars are usually the culminations of long standing rifts between countries that spiral out due to trigger events.
- **Nuclear bombing of Hiroshima and Nagasaki** on **6th and 9th August of 1945** wreaked unforeseen damage on the cities.
- The form and platform for wars have shifted from being geographical-physical conflicts to cyberspace. Governments and extra state actors have the ability to disrupt normal functioning of lives by attacking important facilities like power plants, refineries etc.
- The Kargil War of 1999 was fought between India and Pakistan along the Line of Control. Hundreds of soldiers from both sides succumbed to death.

Nuclear Accidents

- The International Atomic Energy Agency (IAEA) defines a nuclear accident as an incident that has resulted in severe implications for the people, the environment, or the nuclear plant. Radioactive fallout and reactor core melt are potentially high damage events.
- The **Chernobyl disaster in 1986 and the Fukushima Daiichi accident in 2011** highlighted the consequences of any mishap in nuclear power plants.

Human Accelerated Hazards

- Environmental issues like pollution, climate change, ozone layer depletion are intensified due to human intervention.
- Landslides and meteorological phenomena like tornadoes, storms are classified under natural disasters. But several factors like construction of roads, destruction of green cover etc can accelerate these natural disasters. More than 5000 lost their lives in the **Uttarakhand floods of 2013**.

Ques.11 Explain Immediate relief in disaster management and Assessment **?**

Ans. Immediate relief in disaster management is the emergency act that takes place during and immediately after a disaster. It includes:

- **Warning and evacuation**

- **Search and rescue**
- **Providing immediate assistance**
- **Assessing damage**
- **Debris clearance**
- **Emergency accommodation**

Other aspects of disaster management include:

- **Preparedness**

  Initiatives to save lives, reduce property damage, and mitigate the impact of a crisis

- **Recovery**

  The process of returning the situation to how it was before the crisis, including clean up, identifying support, and long-term monitoring

- **Crisis counseling**

  Programs to help disaster-impacted individuals and communities recover through outreach and psycho-educational services

- **Financial assistance**

  Emergency relief organizations provide immediate financial support to people in financial crisis

- **Disaster risk reduction**
  Includes prevention, mitigation, and preparedness

Assessment

The purpose of assessments in the initial phase of a disaster is to provide information that can guide our emergency services in activities such as search and rescue missions, pinpoint location and nature of secondary threats, provide information about the status offacilities needed to treat or support the survivors

The first step after stabilizing the situation by providing sufficient relief is to assess the damage. A meticulously executed assessment exercise would provide an ideal base for the rehabilitation efforts. This exercise is best carried out through multi-disciplinary teams which go into all aspects of damage (social, economical, psychological) in participation with the local community. Based on the assessment of the damage and the needs, a recovery strategy has to be formulated. The strategy should include all interventions - economic, social, political and psychological. The resources should be identified and the roles and responsibilities of all concerned should be defined.

Assessment surveys are used in disaster management to gather information and feedback to help guide emergency services and tailor relief efforts:

- **Needs assessment**

  Surveys can help identify the immediate needs of affected populations, such as food, shelter, and medical care.

- **Risk assessment**

  Surveys can help gauge the level of risk perception, preparedness, and understanding of disaster protocols within communities.

- **Post-disaster evaluation**

  Surveys can help assess the effectiveness of response efforts by gathering feedback from affected individuals.

- **Community engagement**
  Surveys can help involve communities in the planning process by seeking their input.

Ques.12 Explain Rescue training for search and operation at the national and regional level**?**

Ans. 1) Rescue training for search and operation at the national and regional level can include:

- **Understanding hazards**: Learning about the risks and hazards in a structural collapse environment
- **Survey of affected areas**: Learning how to survey areas that have been affected
- **Basic life support**: Learning how to perform basic life support measures and provide pre-hospital care
- **Understanding rescue support systems**: Learning about rescue support systems at the regional, national, and international levels
- **Linking community response**: Learning how to connect community-based response to local emergency services

Search and rescue operations involve three separate steps:

- **Sizeup**: Assessing the situation and developing a safe action plan
- **Search**: Locating victims and documenting their location
- **Rescue**: Using procedures and methods to extricate the victims

Emergency response training can also be important for people working in offices, businesses, industries, and factories. This training can help people learn how to respond to emergency situations and can be the difference between life and death.

INTRODUCTION

Search, rescue and evacuation processes need to be carried out immediately after a disaster strikes a certain area or building. These are the most immediate critical operations that are usually performed by the local volunteers, voluntary organisations and the emergency agencies. Light Search and Rescue is a procedure carried out at primary stages, initially to find out persons with

injuries in lightly damaged buildings, or even without any injuries and needing assistance, and to help them exit. If the condition worsens and the local groups are not able to control the situation, then the specialist groups within emergency agencies have to be called in for professional help, and at times even the Defence Forces including the Army, the Navy, the Air Force and the Coast Guard are called on for help. Primarily, Search and Rescue Operations are undertaken to save the maximum possible number of victims who are trapped in an area affected by a disaster. The basic aim of all such operations is to ensure the survival of the maximum possible number of affected people. A plan is worked out with the help of local people through surveys and then appropriate steps are taken by the various teams involved to carry out the operations. Besides physical rescue, the aim is also a systematic and organised approach in a post-disaster situation riddled with chaos and confusion.

SIGNIFICANCE OF SEARCH AND RESCUE (SAR)

Search and rescue, often known by the acronym SAR, is the process of identifying the location of disaster victims who may be trapped or isolated, and bringing them to safety and providing them with medical attention after a disaster strikes. Search and rescue generally involves the local people who are well versed with the local terrain and can be instrumental in searching and accessing the trapped victims. SAR teams rely on sniffer dogs that are specially trained to smell out human beings trapped under the rubble. Heavy machines such as cranes and earthmovers are used to remove heavy rubble; and special equipment to delicately remove fallen structural elements and reach out inside heaps of rubble with visual or sound equipment for locating survivors. In case of floods and cyclones, boats and helicopters are used to carry out the search and rescue operations by forming teams and carrying out SAR operations in the entire area systematically, each team covering its assigned sectors. After the search, rescue and evacuation, some important steps are required to be taken in order to provide relief to the evacuees.

Prime amongst these are:

Medical Aid Water and food Shelter

Temporary subsistence supplies

Health and sanitation

Information,

and Security.

PHASES OF SAR

SAR is carried out for different categories of affected people depending on the complexities of the processes involved. This is illustrated as follows:

1) The immediate stage is searching and rescuing people with injuries among people trapped in the affected area that we see at first sight.

2) The second stage is the rescue of persons whom we are able to contact but are unable to reach. These are people who may be trapped under heaps of rubble from a collapsed building, or people stuck on trees or high-rise buildings, or trapped inside basements.

3) The third stage includes searching those areas where survivors are likely to be trapped generally. These are called likely survival points.

4) The fourth stage is the last stage in which debris leading to likely survival points is cleared.

Ques.13 Explain Land use zoning**?**

Ans. Land use zoning is a tool that can help prevent and mitigate disasters by integrating natural hazard assessment into land use planning. Land use zoning can help reduce the risk of disasters

by:

- **Identifying risk areas**

  Using risk assessment and mapping to identify areas with a high potential for natural hazards

- **Enforcing land use criteria**

  Using local regulations, building codes, and material specifications to enforce land use decisions

- **Monitoring and evaluating**

  Using disaster risk management indicators to monitor and evaluate the effectiveness of land use zoning

- **Involving stakeholders**

  Including relevant actors and organizational mechanisms in the decision-making process

Here are some other ways that land use planning can help with disaster management:

- **Planning for emergency measures**: Organizing emergency measures and coordinating recovery operations
- **Displacing populations**: Displacing populations if necessary
- **Reconstructing**: Reconstructing after a disaster

Land use planning can help reduce disaster losses in new development areas. It can also help strengthen local resilience by empowering vulnerable populations and designing socially acceptable solutions.

Zoning plays a key role in disaster management by dividing areas into zones based on the risk of disasters and the allowed land uses:

- **Zoning for flood risk**

  Zoning can help minimize the impact of flooding on people and infrastructure by dividing urban areas into zones with different development restrictions. For example, construction may be prohibited in high-risk areas like floodplains.

- **Zoning for seismic hazard**

  Zoning can help reduce the impact of earthquakes by designating areas with different densities based on geologic considerations. For example, areas with a high risk of ground failure from liquefaction may be designated as very low density residential.

- **Zoning for disaster prevention**

  Zoning can help prevent disasters by relating natural hazard assessments to land-use planning. For example, micro-zoning can help determine land-use and building criteria by analyzing the probability of natural hazards in a given site.

Zoning can also help with: Organizing emergency measures, Coordinating recovery operations, Improving the capacity of the poor and most disadvantaged, and Promoting gender equality

Ques.14 Explain about Hazard zonation maps**?**

Ans. Hazard zonation maps are created by dividing an area into zones based on the likelihood of a hazard occurring. The process involves analyzing the physical characteristics of an area and considering environmental and triggering factors.

Here are some steps for preparing hazard zonation maps:

- **Identify factors**: Consider environmental factors like slope angle, soil type, and geological structures, as well as triggering factors like rainfall intensity and earthquakes.
- **Analyze data**: Use satellite and other data to map the factors.
- **Assign weights**: Use software to translate the factors into weights.
- **Create zones**: Use the weights to create zones with varying degrees of hazard.
- **Overlay maps**: Use GIS to overlay the derived maps to create the hazard zonation map.
- **Clarify in the field**: Make adjustments based on field clarifications.

Hazard zonation maps can help with disaster mitigation by: Identifying high-risk areas, Developing early warning systems, Implementing mitigation measures, and Developing land-use policies and regulations

Introduction:

Disaster preparedness refers to measures taken in advance to minimize the impact of natural or man-made disasters on human life and property. Disaster preparedness is crucial for effective disaster management. Hazard zonation mapping is an important tool that can aid in disaster mitigation, especially in the case of landslides.

Hazard zonation mapping can help in disaster mitigation in the case of landslides in the following ways:

- **Identification of High-Risk Zones:** Hazard zonation mapping can help identify areas that are susceptible to landslides.
- By analyzing the physical characteristics of an area, such as slope angle, geology, soil type, rainfall, and vegetation cover, different zones can be identified with varying degrees of susceptibility to landslides. This information can be used to prioritize disaster mitigation efforts in high-risk zones.
- **Development of Early Warning Systems:** In areas identified as high-risk zones, early warning systems can be developed to alert residents to evacuate in case of a landslide.
- By using hazard zonation mapping, authorities can identify critical areas where early warning systems can be installed, such as near rivers or steep slopes.
- **Implementation of Mitigation Measures:** Hazard zonation mapping can help disaster managers to take proactive measures to mitigate the impact of landslides.
- In high-risk zones, measures such as constructing retaining walls, stabilizing slopes, and creating drainage systems can be implemented to prevent or minimize the risk of

landslides.

- **Development of Land-Use Policies and Regulations:** By identifying high-risk zones, hazard zonation mapping can help in the development of land-use policies and regulations.
- Restrictions can be put in place to limit the use of high-risk areas for certain activities, such as building construction, mining, or logging. Such policies can help to prevent or minimize the risk of landslides and other disasters in vulnerable areas.

Ques.15 Explain about Pre-disaster preparedness**?**

Ans. Pre-disaster preparedness is the practice of preparing for and planning to respond to a disaster. The goal of pre-disaster preparedness is to reduce the impact of disasters and support civil protection operations.

Some pre-disaster preparedness activities include:

- **Identifying disaster-prone areas**
- **Studying disaster-prone areas**
- **Collecting information about disaster intensity**
- **Developing early warning and evacuation systems**
- **Formulating emergency preparedness plans**
- **Conducting education and training**
- **Creating mass awareness about disaster management**
- **Updating egress plans**
- **Assembling a team to create a business continuity plan**

Pre-disaster preparedness can help communities build resilience, respond to disasters, and recover more effectively. It can also help communities access state and federal disaster resources more quickly

Managing a disaster is about mitigating, or avoiding, the risks of loss, providing prompt and adequate assistance to those affected, and ensuring a rapid and effective recovery. The concept of the Disaster Management Cycle integrates isolated attempts on the part of different government and nongovernment actors, towards vulnerability reduction or disaster mitigation, within the enveloping domain of disaster management. Governments, businesses, and civil society engage in the Disaster Management Cycle as a means of planning for and mitigating disasters, responding to disasters during and after they occur, and reclaiming the loss following a disaster. By acting appropriately at all points in the cycle, we are better able to prepare, give
better warnings, reduce vulnerability, or prevent disasters during the subsequent iteration of the cycle. The Disaster Management Cycle can be divided into three stages : the pre-disaster stage, the during-disaster stage, and the postdisaster stage.

Pre-Disaster Stage:

- Preparedness, Prevention and Mitigation are the major activities in the pre-disaster stage. Prevention is always better than cure, so this program adheres to the principle. In this stage, various preventive measures and activities are undertaken well in advance so as to respond to disasters in an effective way.
- If the early warnings can be seriously heeded, much of the disaster could have been be avoided. The focus should therefore be on preparing communities and buildings to withstand disasters. There are events of cyclones in India quite frequently, and warnings are generally given before the event occurs. If operative activities can be undertaken well in advance, then it becomes easy to prevent huge loss in terms of lives and property, the aftermath phase.

During-disaster Stage:

- Response and Relief are the important activities in the during-disaster stage. It starts in the aftermath of a disaster.
- The immediate aspects of an aid response include the search, rescue, and evacuation, identification of dead bodies, the disposal of debris, providing first aid, food, water, shelter, safety and security, health care, and sanitation, and restoring basic facilities.

Post-disaster Stage:

- The major activities in the postdisaster phase include: Rehabilitation, Reconstruction and Recovery. Disasteraffected communities will be made resilient and able to return to normalcy through these activities**.**
- Generally, this phase takes a long time as the efforts are made to restore all essential facilities to pre-disaster status.
- The major focus of this phase is on measures that could pave the way for the long-term recovery of social, economic, and physical structures, as well as processes in such a way that future disasters are unable to impact severely and irreversibly.

Disaster preparedness programmes aim to improve the technical and management capabilities of governments, organisations, and communities in order to reach a suitable degree of readiness to respond to any emergency crisis. These precautions, which are described as being logistically prepared to handle disasters, can be strengthened by having reaction mechanisms and processes, practising them, creating long- and short-term strategies, educating the public, and constructing early warning systems.

The 4 Phases of Disaster Management

Think of disasters as recurring events that take place in four main phases:

1. **Mitigation**

2. **Preparedness**

3. **Response**

4. **Recovery**

All organizations are in at least one phase at any given time. Understanding these four phases empowers your organization to prepare for and respond to crises in a smarter, more informed way. Making the right decisions gives your organization the best chance at survival and recovery following a climate-related event.

Let's take a closer look at what the disaster management phases mean and how a facility condition assessment helps you in each one.

Phase 1: Mitigation

Meaning: To prevent future emergencies and take steps to minimize their effects

The mitigation phase occurs **before** a disaster takes place. Here, an organization will take steps to protect people and property, while also decreasing risks and consequences from a given disaster situation. The organization's main goal is to reduce vulnerability to disaster impacts (such as property damage, injuries, and loss of life).

Examples of mitigation may can include measures such as:

- Clearing space around buildings to create a defensible space against fires
- Adding levees or improving property drainage to protect from flooding
- Securing furniture to floors and walls to help prevent damage/injuries during earthquakes
- Re-locating structures to less disaster-prone areas

Phase 2: Preparedness

Meaning: To take actions ahead of time to be ready for an emergency

The **preparedness** phase also occurs **before** a disaster takes place. Here, an organization attempts to understand how a disaster might affect overall productivity and the bottom line. The organization will also provide appropriate education while putting preparedness measures into place.

Examples of preparedness include hosting training, education, drills, tabletop exercises, and full-scale exercises on disaster preparedness. This ensures that stakeholders know what to do in the event of an emergency. Updating egress plans also falls under preparedness.

Organizations may also assemble a team to create a business continuity plan and list of resources needed to recover from a disaster.

Phase 3: Response

Meaning: To protect people and property in the wake of an emergency, disaster, or crisis
The **response** phase occurs in the **immediate aftermath** of a disaster. Organizations must focus their attention on addressing immediate threats to people, property, and business. Occupant safety and wellbeing largely depends on your preparedness levels before disaster strikes.
The most notable example of the response phase is ensuring that people are out of harm's way. The organization can then move on to assess damage, implement disaster response plans, triage cleanup efforts, and start resource distribution as necessary.
Businesses will also need to navigate building closures, preliminary damage assessments, and hampered communication with stakeholders (like staff, vendors, and suppliers) due to shutdowns. As the response period progresses, focus will typically shift from immediate emergency issues to conducting repairs, restoring utilities, re-establishing operations, and cleaning up. The organization will also need to begin planning the reconstruction of damaged infrastructure.

Phase 4: Recovery

Meaning: To rebuild after a disaster in an effort to return operations back to normal
The **recovery** phase takes place **after** a disaster. This phase is the restoration of an organization following any impacts from a disaster. By this time, the organization has achieved at least some degree of physical, environmental, economic and social stability.
The recovery phase of a disaster can last anywhere from **six months to a year** (or even longer depending on the severity of the incident).
An example of recovery is creating strategic protocols and action plans to address the most serious impacts of a disaster. The protocols should give clear steps to follow for various disaster events and cover multiple scenarios.
For example, a flood protocol could include:

- The organization's internal plan to mitigate the effects of water (fan deployment, etc.)
- The preferred water remediation service and their contact information
- If that vendor is unavailable, secondary and tertiary services to contact

In this phase, the organization works to obtain new resources, rebuild or create partnerships, and implement effective recovery strategies. The organization also takes steps to reduce financial burdens, rebuild damaged structures, and reduce vulnerability to future disasters.

Part-C

Ques1.Explain the **role of media** in disaster risk management and response**?**

Ans. The **role of media** in disaster risk management and response is crucial in shaping public understanding, disseminating critical information, and facilitating communication between authorities, communities, and stakeholders. Media plays a multifaceted role across the phases of disaster management, including preparedness, response, recovery, and mitigation.

Key Roles of Media in Disaster Management:

1. Disseminating Early Warnings and Alerts

One of the most important functions of media is the **dissemination of early warnings** before a disaster strikes. Quick and accurate communication of impending hazards helps in saving lives and property.

- **Real-time Alerts**: Media platforms (TV, radio, social media, websites) broadcast real-time alerts for natural disasters such as hurricanes, tsunamis, floods, earthquakes, and wildfires.
- **Coordination with Authorities**: Media channels work closely with meteorological departments, emergency management agencies, and governments to broadcast official warnings and instructions.
- **Wide Reach**: With access to various channels like radio, television, and social media, media ensures that warnings reach even remote or marginalized communities, often in local languages or dialects.

Example: In the U.S., during hurricanes, news networks provide constant updates on storm paths, evacuation orders, and safety measures, while agencies like FEMA disseminate official alerts.

2. Raising Public Awareness and Education

Media plays a significant role in educating the public about **disaster risks, preparedness measures, and mitigation strategies** before disasters happen.

- **Public Service Campaigns**: Media can run public service announcements (PSAs) or campaigns that inform people about potential hazards and how to prepare, such as creating emergency kits, identifying evacuation routes, and understanding hazard maps.
- **Disaster Drills and Preparedness**: Coverage of community drills, workshops, and training programs increases awareness and encourages public participation.
- **Highlighting Vulnerabilities**: Media can focus attention on areas, populations, or infrastructure that are particularly vulnerable to disasters, raising awareness about the need for action or investment in risk reduction.

Example: In Japan, the media often runs educational programming on earthquake safety, helping citizens understand what to do during seismic events, including "drop, cover, and hold on" drills.

3. Facilitating Response and Recovery Operations

During and after a disaster, media serves as a critical platform for facilitating **response and recovery operations** by providing information, mobilizing resources, and offering a communication link between affected populations and aid organizations.

- **Real-time Reporting**: Media outlets provide live coverage of the disaster's impact, helping governments, NGOs, and humanitarian organizations understand the extent of the damage and where help is needed.
- **Emergency Instructions**: Media broadcasts important information such as the location of emergency shelters, medical aid centers, food and water distribution points, and guidelines for reuniting families.
- **Fundraising and Aid Mobilization**: Media often plays a pivotal role in organizing fundraising campaigns, generating donations, and facilitating the mobilization of humanitarian aid from both domestic and international communities.

Example: After the 2010 Haiti earthquake, global news networks extensively covered the disaster, which spurred massive international aid efforts, with donations funneled through media campaigns.

4. Holding Authorities Accountable

The media acts as a watchdog by monitoring the actions of **governments, relief organizations, and other agencies**, ensuring that disaster response and recovery efforts are transparent, timely, and effective.

- **Investigative Journalism**: Media outlets investigate and report on mismanagement, delays, or inefficiencies in disaster response, pushing authorities to take corrective actions.
- **Ensuring Transparency**: Media ensures that disaster relief funds and resources reach the intended beneficiaries by tracking and reporting on how aid is distributed.
- **Highlighting Neglected Areas**: Media coverage can spotlight regions or communities that may be overlooked during disaster response efforts, urging authorities to provide support.

Example: In the aftermath of Hurricane Katrina in 2005, media outlets in the U.S. played a significant role in exposing the inadequacies of government response, highlighting the plight of vulnerable communities in New Orleans.

5. Countering Misinformation and Panic

In times of disaster, misinformation or rumors can spread quickly, leading to confusion or panic. The media plays a key role in countering **false information** by providing accurate and verified updates.

- **Debunking Rumors**: Media outlets work to fact-check and correct false reports, particularly during the chaos of a disaster, preventing the spread of panic.
- **Providing Expert Opinions**: Journalists often turn to disaster management experts,

scientists, and local officials to provide accurate, evidence-based explanations about the hazard, response efforts, and recovery plans.

- **Promoting Calm**: Media can help maintain public order by broadcasting messages that encourage calm, cooperation, and adherence to official guidelines.

Example: During the COVID-19 pandemic, media outlets provided critical information on public health measures, dispelling myths about the virus and sharing expert advice on how to stay safe.

6. Facilitating Two-Way Communication with Communities

Media platforms, especially social media, allow for **two-way communication**, enabling individuals and communities to interact with authorities, disaster response teams, and news outlets.

- **Crowdsourcing Information**: Citizens can report on-the-ground conditions, damage, or personal needs using social media or SMS, which can help authorities direct resources more efficiently.
- **Engaging the Public**: Through platforms like Twitter, Facebook, and WhatsApp, media outlets and emergency agencies can engage directly with the public, answer questions, and clarify instructions.
- **Citizen Journalism**: During disasters, ordinary people often use social media to share videos, photos, and real-time updates, contributing to the overall understanding of the disaster's impact.

Example: During the Nepal earthquake in 2015, social media users shared images and videos of the destruction, which helped coordinate relief efforts and assisted in locating missing people.

7. Post-Disaster Recovery and Reconstruction Coverage

After the immediate response phase, media continues to play an important role in **post-disaster recovery and reconstruction** by keeping the public informed and engaged with long-term recovery efforts.

- **Tracking Recovery Progress**: Media can keep the spotlight on recovery initiatives, ensuring that the rebuilding process remains transparent, equitable, and sustainable.
- **Highlighting Success Stories**: By reporting on communities that successfully rebuild and recover, media can inspire other regions to adopt best practices and increase resilience.
- **Pushing for Policy Change**: Media coverage of disasters often leads to discussions about policies on land use, infrastructure, disaster preparedness, and climate change, potentially leading to reforms.

Example: After the Indian Ocean tsunami in 2004, media coverage of the rebuilding process in countries like Indonesia, Sri Lanka, and Thailand highlighted both challenges and successes in long-term recovery.

8. Advocacy for Policy and Risk Reduction

Media plays a crucial role in advocating for stronger disaster risk reduction (DRR) measures and influencing public opinion to support long-term **policy changes** that improve resilience.

- **Highlighting Risk Reduction Initiatives**: By covering government and community-based efforts to reduce risks (e.g., improved building codes, early warning systems), media encourages further investment in these initiatives.
- **Raising Awareness of Climate Change**: Media often covers the link between climate change and the increasing frequency and severity of natural disasters, advocating for policies to mitigate these risks.
- **Promoting Resilience**: Media can promote a culture of resilience by showcasing how communities prepare for and recover from disasters, motivating others to take proactive measures.

Ques2.Explain in detail about Integrated Planning in the context of **Contingency Management and Preparedness?**

Ans. Integrated Planning in the context of **Contingency Management and Preparedness** is a systematic approach to disaster risk management that coordinates efforts across multiple sectors, stakeholders, and levels of government to ensure an effective response to potential emergencies. This process ensures that resources, plans, and actions are in place before a disaster strikes, reducing the impact and speeding up recovery.

Key Elements of Integrated Planning in Contingency Management and Preparedness:

1. Multi-sectoral Coordination

Integrated planning involves collaboration across various sectors (e.g., health, transportation, water resources, energy, and public safety) to ensure a comprehensive response to emergencies. Each sector contributes its specific expertise and resources, ensuring a well-rounded approach to disaster management.

- **Example**: In flood-prone areas, integrated planning involves urban planners, environmentalists, public health officials, and engineers to design flood defenses, establish early warning systems, and prepare evacuation plans.

2. Risk and Vulnerability Assessment

Assessing risks and identifying vulnerabilities are central to contingency management and preparedness. This involves the use of data and technologies, such as **GIS (Geographical Information Systems)**, to map hazard-prone areas and evaluate the resilience of communities and infrastructure.

- **Hazard Mapping**: Identifying high-risk zones for floods, earthquakes, or other disasters.
- **Vulnerability Analysis**: Understanding which populations (e.g., elderly, low-income groups) or infrastructures (e.g., hospitals, schools) are most at risk.

3. Contingency Planning

Contingency planning involves creating detailed action plans for specific disaster scenarios. These plans outline how different agencies, organizations, and communities will respond to an emergency, who will be in charge, and how resources will be mobilized.

- **Scenario Planning**: Developing multiple plans for different types of disasters (earthquakes, pandemics, industrial accidents).
- **Chain of Command**: Clearly defining roles and responsibilities for all involved agencies to avoid confusion during emergencies.

4. Resource Allocation and Stockpiling

Integrated planning also focuses on ensuring that resources—such as food, medical supplies, communication tools, and rescue equipment—are pre-positioned and readily available for emergency situations.

- **Resource Mapping**: Identifying and mapping local, regional, and national resources to be deployed during an emergency.
- **Stockpiling**: Building reserves of critical supplies, such as food, water, and medical equipment, in locations that are easily accessible during a crisis.

5. Early Warning Systems and Monitoring

Integrated planning incorporates the development of early warning systems that alert communities and responders of imminent threats. These systems use data from **remote sensing**, weather forecasting, and communication networks to provide real-time information.

- **Real-time Data Collection**: Monitoring systems such as weather stations, seismic monitors, and river level sensors.
- **Public Alert Systems**: Using SMS, radio, TV, and sirens to inform the public about impending risks.

6. Preparedness Training and Drills

For contingency plans to be effective, the people and organizations involved need to be well-prepared. This involves regular training sessions, public awareness campaigns, and simulation exercises to test and refine the contingency plans.

- **Community Drills**: Conducting earthquake, fire, or flood drills in schools, hospitals, and workplaces to familiarize people with evacuation routes and safety protocols.
- **First Responder Training**: Ensuring that emergency services like police, firefighters, and medical personnel are well-trained and ready to act swiftly in any scenario.

7. Public Engagement and Communication

A critical part of integrated planning is ensuring that communities understand the risks they face and how to respond in the event of a disaster. This involves public education programs, dissemination of preparedness information, and community participation in planning efforts.

- **Risk Communication**: Providing communities with clear and actionable information on disaster risks and how to prepare.
- **Community-based Disaster Management**: Encouraging local participation in contingency planning to ensure plans are culturally appropriate and address local needs.

8. Legal and Institutional Frameworks

Effective integrated planning requires a supportive legal and institutional framework that defines the roles and responsibilities of different actors. Laws and regulations must clearly mandate disaster preparedness and response actions.

- **Disaster Management Policies**: National and local policies that set out the responsibilities of government agencies and the private sector in disaster preparedness.
- **Inter-agency Cooperation**: Ensuring that government departments, NGOs, and the private sector work together under a unified command structure during emergencies.

9. Financial Planning and Budgeting

Securing financial resources is critical to ensure preparedness efforts are sustainable. Governments, agencies, and businesses need to allocate budgets for risk reduction measures, preparedness activities, and contingency funding for immediate response and recovery operations.

- **Pre-disaster Financing**: Establishing funds to be used for disaster risk reduction, preparedness, and early recovery.
- **Insurance and Risk Transfer**: Using insurance mechanisms to mitigate financial losses in case of a disaster.

10. Monitoring, Evaluation, and Updating Plans

Once contingency plans are in place, they must be regularly reviewed and updated to remain effective. This involves monitoring for new risks, assessing the effectiveness of drills and training, and ensuring that all stakeholders are up to date with the latest information and strategies.

- **Lessons Learned**: After every drill or real disaster, assessing what worked well and what needs improvement in future responses.
- **Plan Revisions**: Updating contingency plans based on new technologies, population changes, or emerging hazards (e.g., climate change).

Ques3.What are the various Measures of adjustment to natural hazards?

Ans. Measures of adjustment to natural hazards, particularly through mitigation, are strategies and actions aimed at reducing the impact of natural hazards on human life, property, and infrastructure. Mitigation focuses on minimizing the potential consequences of disasters, either by reducing vulnerability or preventing the hazard itself from becoming a disaster. These measures involve long-term investments in planning, infrastructure, and policies to create more resilient communities.

Here's an overview of various mitigation measures for different types of natural hazards:

1. Structural Mitigation

Structural mitigation refers to physical changes or constructions made to reduce the damage from natural hazards.

a. Flood Mitigation

- **Dams and Reservoirs**: Constructing dams to control river flow and store excess water during heavy rains or storms to prevent downstream flooding.
- **Levees and Floodwalls**: Building barriers along rivers and coastal areas to prevent floodwaters from inundating communities.
- **Stormwater Management Systems**: Implementing urban drainage systems and permeable pavements to control runoff and reduce flash floods.
- **Floodplain Zoning**: Restricting development in flood-prone areas through zoning laws and regulations to prevent loss of life and property.

b. Earthquake Mitigation

- **Seismic-Resistant Buildings**: Designing and constructing buildings with reinforced materials, flexible joints, and shock absorbers to withstand earthquake tremors.
- **Retrofitting**: Strengthening existing structures, particularly critical infrastructure such as hospitals, schools, and bridges, to meet updated earthquake-resistant standards.
- **Base Isolation Systems**: Installing base isolation techniques in buildings, which allows the structure to move independently from the ground during an earthquake, reducing the impact.
- **Zoning and Land-Use Planning**: Avoiding construction near fault lines and high-risk seismic zones through regulations and urban planning.

c. Volcanic Eruption Mitigation

- **Lava Diversion Channels**: Constructing barriers or channels to divert lava flow away from populated areas.
- **Volcanic Monitoring Systems**: Installing real-time monitoring systems to track seismic activity, gas emissions, and ground deformation near volcanoes, helping authorities predict eruptions.
- **Building Codes**: Reinforcing structures in areas prone to ash fall, making roofs and buildings strong enough to withstand the weight of ash accumulation.

d. Tsunami Mitigation

- **Tsunami Walls and Barriers**: Building coastal defense walls that can reduce the force of incoming tsunamis and minimize damage to coastal communities.
- **Elevated Buildings**: Designing buildings in tsunami-prone areas with elevated foundations to protect them from the force of waves.
- **Mangrove and Coral Reef Restoration**: Restoring natural ecosystems like mangroves and coral reefs that act as natural barriers against tsunamis by dissipating wave energy.

e. Landslide Mitigation

- **Slope Stabilization**: Reinforcing steep slopes with retaining walls, terracing, or planting deep-rooted vegetation to reduce the risk of landslides.
- **Drainage Systems**: Installing drainage systems to divert water away from slopes, preventing water accumulation that can trigger landslides.
- **Debris Flow Barriers**: Constructing barriers or catchment structures in areas prone to landslides and debris flows to protect buildings and infrastructure.

2. Non-Structural Mitigation

Non-structural mitigation involves policies, planning, education, and regulatory measures to reduce vulnerability without physical infrastructure changes.

a. Land-Use Planning and Zoning

- **Avoiding High-Risk Areas**: Prohibiting or limiting development in areas prone to hazards like floodplains, coastal zones vulnerable to storm surges, or fault lines.
- **Greenbelt Creation**: Establishing open spaces and parks in hazard-prone areas to absorb the impact of hazards without significant property or human losses.

b. Building Codes and Regulations

- **Strengthened Building Codes**: Updating building codes to ensure that structures are designed to withstand the specific hazards of an area, such as earthquakes, hurricanes, or tornadoes.
- **Enforcing Compliance**: Ensuring that developers, builders, and homeowners comply with mitigation regulations and standards through inspections and penalties for non-compliance.

c. Insurance Mechanisms

- **Disaster Insurance**: Promoting disaster insurance for individuals and businesses in high-risk areas, helping them recover financially after a disaster.
- **Government Risk Pools**: Establishing national or regional insurance schemes to spread the financial risk of disasters across a larger population.

d. Public Awareness and Education

- **Hazard Education Campaigns**: Conducting public education initiatives on disaster preparedness, evacuation procedures, and the importance of mitigation measures to raise awareness among the population.
- **Community Engagement**: Involving local communities in planning and implementing mitigation strategies, ensuring they understand the risks and how to reduce them.

e. Early Warning Systems

- **Real-Time Monitoring**: Installing advanced early warning systems for hazards such as hurricanes, floods, and earthquakes to provide timely alerts to at-risk populations.
- **Public Alert Systems**: Setting up sirens, radio alerts, and mobile messaging systems to disseminate warnings quickly in case of impending hazards.

3. Ecosystem-Based Mitigation

Ecosystem-based mitigation leverages natural processes and ecosystems to protect communities from natural hazards.

a. Wetland Restoration

- **Flood Control**: Wetlands act as natural sponges, absorbing excess water during floods and reducing downstream flooding. Restoring and conserving wetlands helps mitigate the effects of flooding.

b. Mangrove and Coastal Vegetation Restoration

- **Storm Surge Protection**: Mangroves and coastal vegetation provide natural barriers against storm surges and coastal erosion, protecting shorelines and reducing the force of incoming waves.

c. Forestation and Reforestation

- **Erosion and Landslide Control**: Planting trees in areas prone to landslides and soil erosion can help stabilize slopes and reduce the likelihood of landslides and flash floods.

d. Coral Reef Protection

- **Wave Energy Reduction**: Coral reefs act as natural breakwaters, reducing the energy of waves and protecting coastlines from erosion and storm surges.

4. Technological and Data-Driven Mitigation

Using technology and data to predict hazards and improve mitigation strategies can significantly enhance resilience.

a. Geospatial and Satellite Technologies

- **Risk Mapping**: Using Geographic Information Systems (GIS) and satellite remote sensing to map hazard-prone areas, including floodplains, fault lines, and areas vulnerable to wildfires, which helps in planning and mitigation efforts.

b. Early Warning Systems

- **Real-Time Hazard Monitoring**: Integrating early warning systems with technologies like Doppler radar (for tornadoes), seismic sensors (for earthquakes), and remote sensing to monitor hazards and provide timely alerts.

c. Big Data and Artificial Intelligence

- **Predictive Modeling**: Using AI and big data to analyze historical data and environmental conditions to predict potential hazards such as floods, wildfires, or hurricanes, improving the ability to plan and mitigate effectively.

5. Policy and Institutional Frameworks

Governments and international organizations play a key role in disaster mitigation by implementing policies and frameworks that prioritize hazard reduction.

a. International Frameworks

- **Sendai Framework for Disaster Risk Reduction (2015-2030)**: This global agreement promotes disaster risk reduction through mitigation measures, capacity-building, and community resilience.

b. National Disaster Risk Reduction Strategies

- Governments should develop comprehensive disaster risk reduction strategies that include preparedness, mitigation, and recovery plans, ensuring that mitigation is integrated into national development planning.

c. Capacity Building

- **Training and Education**: Providing training to government officials, urban planners, engineers, and community leaders on how to implement and enforce mitigation strategies.

Ques4.Explain in detail how different hazards and disasters are predicted**?**

Ans. Predicting hazards and disasters is a crucial component of disaster risk reduction, enabling early warning systems and timely actions that can save lives and reduce damage. Various scientific methods, technologies, and tools are used to predict different types of natural hazards, each with its own challenges and uncertainties. The effectiveness of hazard prediction depends on

understanding the hazard, monitoring environmental conditions, and applying predictive models.

Here's an overview of how different hazards and disasters are predicted:

1. Meteorological Hazards

These hazards include hurricanes, tornadoes, floods, and extreme weather events, which are often influenced by atmospheric conditions.

a. Weather Forecasting

- **Meteorological Models**: Advanced weather prediction models use data from satellites, weather stations, and radar systems to track atmospheric patterns. Models like the Global Forecast System (GFS) and the European Centre for Medium-Range Weather Forecasts (ECMWF) are widely used to predict weather events.
- **Hurricane/Typhoon Prediction**: Meteorologists use satellite imagery, atmospheric pressure data, and sea surface temperatures to predict the formation and path of tropical storms. Early prediction of hurricane intensity and path is critical for evacuation planning.
- **Flood Forecasting**: Hydrometeorological models predict the likelihood of floods based on rainfall patterns, river water levels, soil saturation, and snowmelt. River gauges and remote sensing are used to monitor conditions in real time.

b. Tornado Prediction

- **Doppler Radar**: Tornadoes are challenging to predict far in advance, but Doppler radar systems are used to detect severe thunderstorms and rotating storm patterns (supercells) that can lead to tornado formation.
- **Severe Weather Alerts**: Predicting the timing and location of tornadoes relies on short-term forecasting, typically providing warnings within minutes to hours of formation.

2. Geological Hazards

Geological hazards include earthquakes, tsunamis, volcanic eruptions, and landslides. Prediction in this category is often more difficult due to the complexities of earth systems.

a. Earthquake Prediction

- **Seismic Monitoring**: Earthquake prediction is one of the most difficult challenges in disaster prediction. Seismologists use seismic activity data from networks of seismometers to identify early warning signs like foreshocks or increased seismic activity. However, predicting the exact time, location, and magnitude of an earthquake remains elusive.
- **Earthquake Early Warning Systems (EEWS)**: While precise predictions are rare, early warning systems can detect the initial, weaker seismic waves (P-waves) before the more destructive waves (S-waves) arrive, providing a few seconds to minutes of warning. Countries like Japan and Mexico have sophisticated EEWS.

b. Tsunami Prediction

- **Seismic and Sea-Level Monitoring**: Tsunamis are often triggered by underwater earthquakes. Networks of seismic sensors, combined with deep-sea buoys (e.g., DART buoys) that monitor ocean floor pressure and sea level changes, help predict tsunamis and issue warnings.
- **Tsunami Simulation Models**: Once an underwater earthquake is detected, tsunami models simulate the potential wave heights, speeds, and coastal impact based on the earthquake's location, magnitude, and depth.

c. Volcanic Eruption Prediction

- **Volcanic Monitoring**: Scientists monitor volcanic activity through seismic data, gas emissions, ground deformation, and thermal imaging. Swarms of small earthquakes and increased gas emissions (e.g., sulfur dioxide) can indicate rising magma.
- **Eruption Models**: Predictive models simulate how magma might rise through a volcano's structure, estimating the likelihood and potential size of an eruption. Early warning systems are in place near active volcanoes (e.g., Mount Etna, Mount St. Helens).

d. Landslide Prediction

- **Slope Monitoring and Rainfall Data**: Landslides are often triggered by heavy rainfall, earthquakes, or volcanic activity. Scientists use ground sensors, satellite imagery, and historical data to monitor areas prone to landslides.
- **Geotechnical Models**: These models assess slope stability, soil moisture levels, and the likelihood of failure, which can help predict landslides, especially in mountainous or deforested areas.

3. Hydrological Hazards

Hydrological hazards primarily refer to hazards related to water bodies, including river floods, storm surges, and droughts.

a. Flood Prediction

- **River and Coastal Monitoring**: River flood prediction uses data from hydrological models that track rainfall, river levels, and watershed conditions to predict how much water will flow into rivers and whether they will overflow. Coastal floods are predicted by monitoring sea levels and storm surge potential.
- **Flash Flood Prediction**: These are short-term predictions made based on heavy rainfall, particularly in urban or steep areas, where water cannot absorb into the ground quickly. Weather radar and real-time precipitation data are key tools.

b. Drought Prediction

- **Climate and Soil Monitoring**: Drought prediction relies on long-term climate models,

rainfall monitoring, and soil moisture data. It focuses on assessing precipitation deficits over time, changes in temperature, and evaporation rates.

- **Early Drought Warning Systems**: These systems use indicators such as the Standardized Precipitation Index (SPI) and the Palmer Drought Severity Index (PDSI) to assess and predict drought conditions.

4. Biological Hazards

Biological hazards, such as disease outbreaks and pandemics, can also be predicted using various public health and epidemiological tools.

a. Epidemiological Models

- **Disease Surveillance**: Health organizations monitor disease incidence data to detect abnormal increases in cases (e.g., influenza, COVID-19, dengue).
- **Predictive Disease Models**: Mathematical models, such as SIR (Susceptible-Infected-Recovered), use data on disease spread, transmission rates, and population movement to predict future outbreaks.
- **Environmental Factors**: In diseases like malaria and dengue, predictions are made based on environmental conditions like temperature, humidity, and precipitation, which affect the spread of vector-borne diseases.

5. Climatological Hazards

Climatological hazards include long-term environmental changes such as heatwaves, cold waves, and wildfires.

a. Heatwave and Cold Wave Prediction

- **Climate Models**: Long-range weather and climate models predict the likelihood of extreme temperatures by analyzing atmospheric pressure, jet streams, and long-term weather patterns.
- **Extreme Temperature Alerts**: Meteorological agencies issue heatwave and cold wave alerts based on these forecasts, providing guidelines to protect vulnerable populations.

b. Wildfire Prediction

- **Fire Weather Forecasting**: Meteorologists assess temperature, humidity, wind conditions, and dry vegetation to predict the likelihood of wildfires. Predictive tools use satellite data to monitor fire-prone areas.
- **Wildfire Behavior Models**: Predictive models simulate how a wildfire might spread based on fuel availability, weather, and topography. These models help emergency services plan fire containment strategies.

6. Technologies for Hazard Prediction

- **Satellite Remote Sensing**: Satellites like Landsat, MODIS, and Sentinel provide real-time data on land use, sea surface temperatures, vegetation cover, and atmospheric conditions to monitor hazard-prone areas.
- **Geographical Information Systems (GIS)**: GIS combines various data layers (e.g., topography, hydrology, population density) to create hazard maps, showing areas at high risk of disasters.
- **Artificial Intelligence (AI) and Machine Learning**: AI models can analyze vast datasets to detect patterns and predict hazards more accurately. AI is increasingly used in areas such as flood prediction, wildfire detection, and epidemic modeling.

Ques5.Explain in detail about Disaster management cycle?

Ans. The concept of Disaster Management Cycle integrates isolated attempts on the part of different actors, government and nongovernment, towards vulnerability reduction or disaster mitigation, within the enveloping domain of disaster management, as phases occurring in different time periods in disaster management continuum. This has facilitated a planned approach to disaster management in that post- disaster recovery and pre -disaster mitigation planning are perceived as integrated/related activities.
Disaster Management efforts are geared towards disaster risk management. Disaster Risk Management "implies the systematic process of using administrative decisions, organisation, operational skills, and capacities to implement policies, strategies and coping capacities of the society and communities to lessen the impact of natural hazards and related environmental and technological disasters. These comprise all forms all activities including structural and non-structural measures to avoid (prevention) or to limit (mitigation and preparedness) adverse effects to hazards".
There are three key stages of activities in disaster management

1. Before a disaster: to reduce the potential for human, material, or environmental losses caused by hazards and to ensure that these losses are minimised when disaster strikes

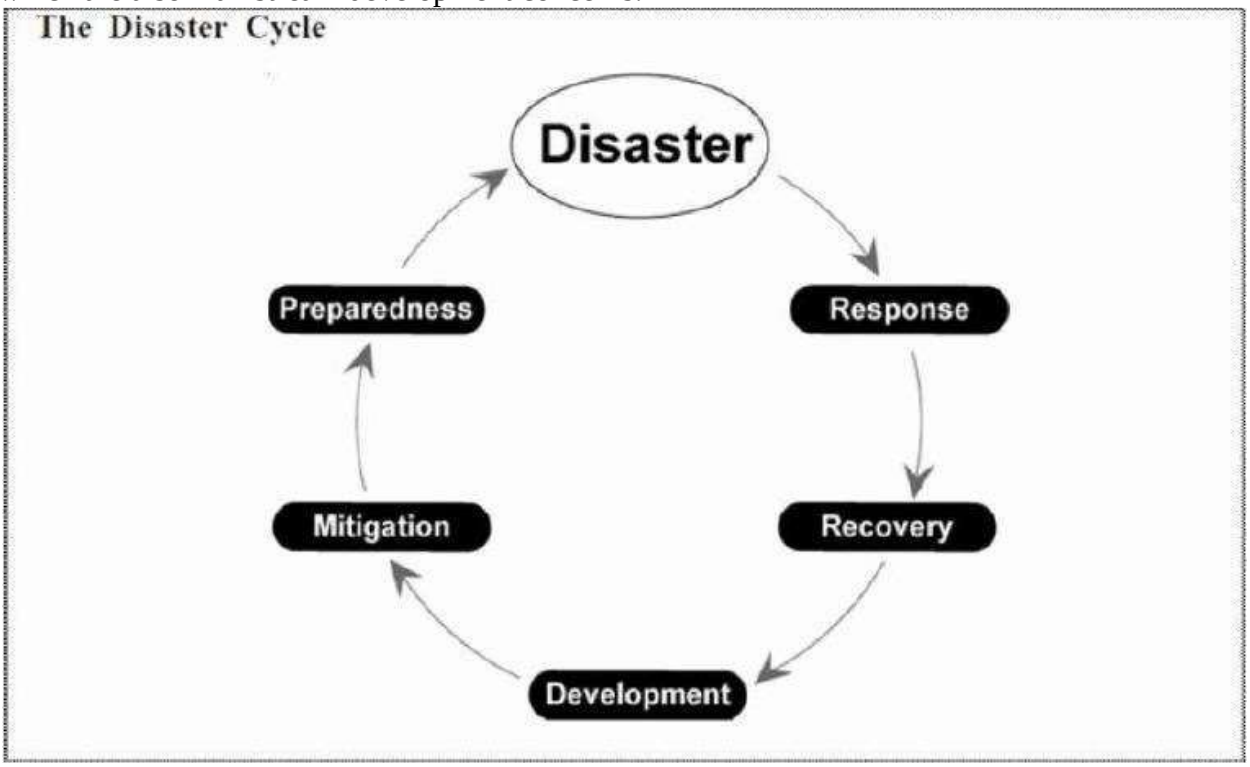
2. During a disaster: to ensure that the needs and provisions of victims are met to alleviate and minimise suffering

3. After a disaster: to achieve rapid and durable recovery.

Common perception of disaster management is limited to emergency relief and postdisaster rehabilitation. This is so because these 2 elements are by far and not separate. Thus, prevention, mitigation and preparedness form pre-disaster activities in the Disaster management Cycle and response, comprising relief, recovery and rehabilitation are post-disaster activities. Whilst emergency relief and rehabilitation are vital activities, successful disaster management planning must encompass the complete realm of
activities and situations that occur before, during and after disasters. These phases can best be represented as a cycle, which if followed through public policy can obstruct future development of disasters by impeding the vicious cycle of cause and effect. One of the key issues in disaster management planning is the allocation of resources at all stages of the disaster cycle, which

optimises the total effectiveness of risk reduction activity and maximises the overall impact of disaster management.

This approach has imparted a more holistic perception to disaster management and has served to integrate disaster management with development planning in that most predisaster activities, involve activities for vulnerability reduction like poverty reduction, employment provision etc. which are also mainstream development concerns.



The different phases of disaster management are represented in the disaster cycle diagram.

1. The Disaster Event

This refers to the real-time event of a hazard occurring and affecting the 'elements at risk'. The duration of the event will depend on the type of threat, for example, ground shaking may only occur for a few seconds during an earthquake while flooding may take place over a longer period of time. Disasters have tremendous modifying impact on the physical landscape. Within a few minutes, an entire region is reduced to rubble in the event of an earthquake. The impact leads to loss of life and property in affected areas; losses being directly correlated to the vulnerability of the region, physical and socioeconomic.

Vulnerability is also socio-economic. Weaker sections of society, viz. women, children, aged and handicapped, mentally infirm, etc., suffer a lot more than their stronger counterparts. Studies have also

unearthed positive correlation between poverty and vulnerability. The poor inhabit the most hazardous physical areas because they are easier to procure and offer added advantages, like proximity to sea for fishermen or fertile soil for farmers near flood prone areas etc., that makes them prone to losses, both of assets and life. The poor also lack the resilience to recover from

shock in the aftermath of a disaster. This brings to light the need for multi-faceted response to disasters, which takes account of all social
political and economic ramifications. Issues to be addressed range from physical to social and economic vulnerability of weaker sections that suffer more relative to other, better placed.

2. Disaster Response

A Disaster is a cataclysmic event that has severe modifying impact. Consequences are both physical and social/ human. Disaster Response has to tackle all aforesaid challenges. Disaster response entails restoring physical facilities, rehabilitation of affected populations, restoration of lost livelihoods and reconstruction efforts to restore the infrastructure lost or damaged. There are inherent important lessons to be learnt from disaster response. Retrospectively, it brings to light flaws in efforts pertaining to policy and planning with respect to location and type of infrastructure and social schemes to improve the social positioning of the under privileged, particularly with respect to access to resources of the underprivileged. Disaster aftermath is evaluation time for the administrative set up in that disaster response exposes system
weaknesses. Disaster is the ultimate test of administrative efficiency, in the sense of positive impact on the environment, preparedness, procedural simplicity, logistics, speed and expertise.

There are inherent important lessons to be learnt with regard to administrative reforms by way
of policy interventions to ensure

• Better institutional preparedness
• Countering contrary pulls such as lack of social cohesion owing to irrational differentiations that effectively impede response, in the sense of self- help and 'communitarianism'
• Long- term mitigation policy to counter vulnerabilities, structural and non- structural by enabling legal provisions and honest implementation of the same.

3. Recovery
The recovery phase involves implementation of actions to promote sustainable redevelopment (reconstruction, rehabilitation) following a disaster. It covers long-term measures like, rebuilding of houses, assets, infrastructure, school building, hospital buildings, and other public buildings. It is a process undertaken by a disaster-affected community to fully restore itself to pre-disaster level. Recovery is the activity that returns infrastructure systems to minimum operating standards and guides long-term efforts designed to return life to normal or improved levels after a disaster. Recovery is also sometimes used to describe the activities that encompass the three overlapping phases of emergency relief, rehabilitation and reconstruction

Ques6.Explain in detail about provision of immediate relief measures to disaster-affected people**?**

Ans. The provision of immediate relief measures to disaster-affected people is a critical phase in disaster management that ensures the survival and safety of individuals in the aftermath of a disaster. This involves rapid deployment of resources and services to meet basic needs, provide

medical care, and stabilize the situation. Here's an overview of key immediate relief measures:

1. Search and Rescue Operations

- **Rescue Teams**: Trained personnel, including local emergency responders, military, and international rescue teams, are deployed to locate and save survivors trapped in collapsed buildings, under debris, or in hazardous areas (e.g., floodwaters, landslides).
- **Use of Technology**: Drones, thermal imaging, and sniffer dogs are often used to enhance the effectiveness of search and rescue efforts.

2. Provision of Emergency Shelter

- **Temporary Shelters**: Setting up temporary shelters in safe zones for those displaced by the disaster. This could involve tents, community centers, schools, or large public buildings.
- **Safe Housing**: Ensuring the shelters are weather-resistant, and provide security and privacy for vulnerable groups such as women, children, and the elderly.
- **Organized Camps**: In large-scale disasters, properly managed camps may be set up with services like food distribution, water, sanitation, and medical care.

3. Food and Water Supply

- **Emergency Food Distribution**: Providing ready-to-eat meals (MREs), non-perishable food items, or community kitchens to ensure that affected people have access to basic nutrition.
- **Clean Drinking Water**: Immediate access to clean water is crucial to prevent dehydration and waterborne diseases. This could involve:
    - Bottled water distribution.
    - Installation of temporary water purification systems.
    - Setting up water tankers or mobile desalination units.

4. Medical Aid and Health Services

- **First Aid and Trauma Care**: Mobilizing medical teams to provide first aid, attend to injuries, and manage trauma cases. This is often a high priority for those injured during the disaster.
- **Mobile Clinics**: Deploying mobile medical units to provide care in areas where health facilities have been damaged or are inaccessible.
- **Vaccination and Disease Prevention**: Providing vaccinations, especially in flood-affected areas where waterborne diseases (e.g., cholera, typhoid) are a risk.
- **Mental Health Support**: Offering psychological first aid and counseling services to help people cope with trauma and emotional distress caused by the disaster.

5. Provision of Sanitation and Hygiene

- **Temporary Sanitation Facilities**: Installing portable toilets and ensuring waste disposal

to maintain hygiene in disaster-hit areas and avoid the spread of diseases.

- **Hygiene Kits**: Distributing hygiene kits containing soap, disinfectants, hand sanitizers, feminine hygiene products, and other personal hygiene items to prevent the outbreak of diseases.

6. Distribution of Essential Non-Food Items

- **Blankets and Clothing**: Providing clothing, blankets, and bedding to protect people from the elements, especially in colder climates or during nighttime.
- **Basic Tools**: Distributing tools (e.g., flashlights, batteries) to aid in navigating post-disaster conditions, particularly in areas without power.

7. Communication and Information Dissemination

- **Emergency Hotlines**: Setting up helplines and communication channels to help affected individuals contact authorities and request assistance or report missing family members.
- **Information Sharing**: Providing accurate, timely information about the disaster's status, ongoing relief efforts, and safe areas through radio, SMS, and social media platforms.
- **Family Reunification**: Facilitating systems to reunite family members separated during the disaster, often in coordination with organizations like the Red Cross.

8. Security and Protection

- **Law Enforcement**: Deploying police or military personnel to ensure law and order, prevent looting, and protect vulnerable populations from potential exploitation or violence.
- **Protection for Vulnerable Groups**: Ensuring that women, children, elderly, and people with disabilities have special protections and are given priority in accessing shelters, food, and medical aid.

9. Restoration of Critical Services

- **Restoration of Power**: Providing generators or solar-powered lights in areas without electricity to help sustain medical services, communication systems, and shelters.
- **Mobile Charging Stations**: Offering phone charging stations in shelters to help survivors contact relatives and receive updates.
- **Water and Fuel Supply**: Ensuring the delivery of fuel for emergency vehicles and generators, as well as water for drinking and sanitation.

10. Psychosocial Support and Community Engagement

- **Crisis Counseling**: Offering initial psychological support through crisis counselors or community volunteers to help people manage shock, grief, or trauma.
- **Engaging Local Communities**: Involving affected communities in relief efforts, as they often know their needs best and can provide support in the distribution and management of resources.

11. Coordination of Relief Efforts

- **Government and Local Authorities**: National and local disaster management agencies typically lead relief efforts, coordinating between various responders and managing resource distribution.
- **Non-Governmental Organizations (NGOs)**: International and local NGOs often play a vital role in providing immediate relief, as they can mobilize resources and teams quickly. Examples include the Red Cross, Médecins Sans Frontières, and the World Food Programme (WFP).
- **International Assistance**: In large-scale disasters, international organizations and foreign governments provide financial aid, supplies, and personnel to support relief operations.

12. Financial Assistance

- **Cash Transfers or Vouchers**: Providing immediate cash or voucher systems to help affected individuals purchase essential items from local markets, which can help stimulate the local economy and give people agency in their recovery.

13. Transportation and Logistics

- **Air and Sea Lifts**: Organizing helicopters, boats, or trucks to deliver aid supplies to isolated or difficult-to-reach areas.
- **Evacuation Services**: Providing transportation to move people from dangerous zones to safer shelters, often facilitated by local authorities or the military.

Ques7.Explain in detail about Natural disaster reduction and management**?**

Ans.

Natural disaster reduction and management involve the coordinated efforts to minimize the impact of disasters such as earthquakes, floods, hurricanes, tsunamis, and wildfires, while also preparing for and responding to them efficiently. The goal is to protect lives, reduce economic losses, and ensure faster recovery. Here's an overview of key aspects related to natural disaster reduction and management:

1. Disaster Risk Reduction (DRR)

Disaster Risk Reduction refers to proactive measures to prevent or minimize the damage caused by natural disasters. It focuses on long-term strategies to reduce vulnerability and exposure to hazards. Key components include:

a. Risk Assessment and Mapping

- **Hazard Identification**: Understanding the types of disasters common to a region (e.g., earthquakes, floods).
- **Vulnerability Assessment**: Assessing which populations, structures, or areas are most at risk.
- **Risk Mapping**: Creating maps showing high-risk zones to guide planning and mitigation strategies.

b. Infrastructure Development

- **Building Codes**: Enforcing building standards that can withstand hazards (e.g., earthquake-resistant buildings).
- **Flood Control Measures**: Constructing levees, dams, and drainage systems to manage flood risks.
- **Urban Planning**: Ensuring that critical infrastructure (e.g., hospitals, emergency services) are not located in high-risk areas.

c. Environmental Management

- **Sustainable Land Use**: Proper land-use planning can reduce risks such as landslides and flooding.
- **Ecosystem-based Approaches**: Protecting wetlands, forests, and mangroves, which act as natural buffers against floods and storms.

d. Early Warning Systems

- **Monitoring and Forecasting**: Using technology to monitor environmental changes (e.g., seismic activity, weather patterns).
- **Alert Systems**: Establishing effective communication channels to warn communities about impending disasters, allowing for timely evacuations.

2. Disaster Preparedness

Disaster preparedness involves planning, organizing, and equipping communities to respond effectively when a disaster strikes. Key elements include:

a. Community Education and Awareness

- **Public Awareness Campaigns**: Educating communities about potential hazards and how to prepare for them (e.g., evacuation routes, emergency kits).
- **School Programs**: Integrating disaster preparedness into school curriculums to ensure that children and youth are aware of safety measures.

b. Training and Capacity Building

- **Emergency Drills**: Regularly conducting drills for different types of disasters (e.g., earthquake drills, fire drills) to improve response times and coordination.
- **Training Emergency Personnel**: Ensuring that rescue teams, health workers, and first responders are trained in disaster response and management.

c. Emergency Response Plans

- **Local and National Plans**: Governments and local authorities develop disaster management plans that outline roles, responsibilities, and resources.
- **Evacuation Routes**: Planning and maintaining clear evacuation routes, shelters, and transportation options.

3. Disaster Response

When a disaster occurs, a quick and coordinated response is crucial to minimize casualties and damage. Response efforts include:

a. Search and Rescue Operations

- Mobilizing trained personnel to locate and rescue victims trapped in debris or dangerous areas, such as collapsed buildings or flooded regions.

b. Medical Assistance and Relief

- **Medical Response Teams**: Providing immediate healthcare, including trauma care and epidemic prevention, in affected areas.
- **Shelters and Relief Supplies**: Setting up temporary shelters, and providing food, water, sanitation, and other essential supplies.

c. Communication and Coordination

- **Crisis Communication**: Maintaining effective communication between responders, government agencies, and affected populations to ensure coordinated efforts.
- **Humanitarian Aid**: Mobilizing international and local resources for relief operations, including financial, technical, and logistical support.

4. Post-Disaster Recovery and Rehabilitation

After the immediate response phase, recovery and rehabilitation focus on rebuilding and restoring communities and systems to a state of normalcy. This includes:

a. Reconstruction

- **Housing and Infrastructure**: Rebuilding homes, roads, and public infrastructure with improved resilience to future disasters.
- **Restoring Services**: Ensuring that essential services like electricity, water, and sanitation

are restored quickly.

b. Economic and Livelihood Recovery

- **Financial Aid and Insurance**: Providing financial assistance or insurance payouts to individuals and businesses affected by the disaster.
- **Job Creation**: Implementing programs that offer employment and livelihood support to help people recover economically.

c. Mental Health and Social Support

- **Counseling Services**: Offering psychological support to survivors, particularly those dealing with trauma or the loss of loved ones.
- **Community Engagement**: Involving local communities in decision-making processes related to rebuilding efforts, ensuring that their needs are prioritized.

5. International Cooperation

- **Global Frameworks**: International agreements like the **Sendai Framework for Disaster Risk Reduction** (2015-2030) promote global collaboration in reducing disaster risks and building resilience.
- **Foreign Aid and Relief Operations**: Many countries rely on international assistance, including financial, technical, and logistical support, during disaster response and recovery phases.

6. Use of Technology and Innovation

- **Remote Sensing and GIS**: Geographic Information Systems (GIS) and satellite remote sensing are used to track disasters, map risk zones, and monitor recovery efforts.
- **Artificial Intelligence (AI) and Machine Learning**: These technologies are increasingly being used for predicting disasters, optimizing response efforts, and analyzing large datasets.
- **Drones and Robotics**: Drones help with search and rescue, damage assessments, and delivering supplies to hard-to-reach areas.

7. Climate Change Adaptation

- **Resilient Infrastructure**: Building climate-resilient infrastructure that can withstand increased flooding, storms, and extreme weather conditions.
- **Sustainable Development**: Integrating climate change adaptation into national development plans to reduce the long-term risks of climate-induced disasters.

Ques 8. Describe about Centers for Natural Disaster Reduction in India **?**

Ans. Centers for Natural Disaster Reduction in India

India is prone to a variety of natural disasters, including floods, earthquakes, droughts, and cyclones. To mitigate the impacts of these disasters, various national and state-level institutions and centers have been established to focus on disaster risk reduction (DRR), preparedness, and response. Below is an overview of key centers for natural disaster reduction in India.

1. National Disaster Management Authority (NDMA)

- **Established**: 2005 under the Disaster Management Act.
- **Objective**: NDMA is the apex body responsible for formulating policies, plans, and guidelines for disaster management in India.
- **Functions**:
  - Develop and implement the National Disaster Management Plan.
  - Coordinate the efforts of various ministries, departments, and agencies involved in disaster management.
  - Promote awareness and training programs at all levels.

2. National Institute of Disaster Management (NIDM)

- **Established**: 2003.
- **Objective**: NIDM focuses on disaster risk reduction through education, research, and training.
- **Functions**:
  - Conduct training programs and workshops for government officials, NGOs, and community leaders.
  - Develop knowledge resources and best practices for disaster management.
  - Undertake research and studies related to disaster management.

3. Central Water Commission (CWC)

- **Established**: 1974.
- **Objective**: The CWC is responsible for water resource management and flood control in India.
- **Functions**:
  - Monitor river flows and issue flood forecasts.
  - Develop flood management plans and strategies.
  - Conduct research and development on flood mitigation technologies.

4. Geological Survey of India (GSI)

- **Established**: 1851.
- **Objective**: GSI is responsible for the geological mapping and assessment of natural hazards, particularly earthquakes and landslides.
- **Functions**:
  - Conduct geological surveys and research on seismic activity.
  - Provide data and information for earthquake risk assessment and management.
  - Collaborate with other organizations to develop mitigation strategies.

5. Indian Meteorological Department (IMD)

- **Established**: 1875.
- **Objective**: IMD provides weather forecasts and early warning systems for natural disasters such as cyclones and floods.
- **Functions**:
  - Issue alerts and advisories on severe weather events.
  - Conduct research on climate change and its impact on disaster frequency and intensity.
  - Collaborate with state governments for disaster preparedness.

6. National Center for Disaster Management (NCDM)

- **Objective**: NCDM focuses on developing technologies and systems for disaster management and reduction.
- **Functions**:
  - Conduct research and development on disaster prediction and management technologies.
  - Collaborate with other national and international organizations for knowledge sharing and capacity building.
  - Develop early warning systems and risk assessment models.

7. State Disaster Management Authorities (SDMAs)

- **Objective**: Each state in India has its own SDMA to manage disasters at the state level.
- **Functions**:
  - Formulate state-specific disaster management plans and policies.
  - Coordinate disaster response and recovery efforts within the state.
  - Promote community-based disaster management initiatives and training programs.

8. National Disaster Response Force (NDRF)

- **Established**: 2006.
- **Objective**: NDRF is a specialized force responsible for disaster response and rescue operations.
- **Functions**:
  - Conduct search and rescue operations during disasters.
  - Provide assistance in relief and rehabilitation efforts.
  - Train state-level disaster response teams and communities in disaster management.

9. Research Institutions and Universities

Several research institutions and universities in India conduct research on disaster management and risk reduction, including:

- **Tata Institute of Social Sciences (TISS)**: Conducts research on social vulnerability and disaster resilience.
- **Indian Institute of Technology (IIT)**: Various IITs conduct research on engineering solutions for disaster risk reduction, such as earthquake-resistant structures.
- **National Institute of Urban Affairs (NIUA)**: Focuses on urban disaster risk management and planning for resilient cities.

Ques9.Explain in detail Role of Panchayats in Disaster Mitigation ?

Ans. Role of Panchayats in Disaster Mitigation

1. Introduction

Panchayati Raj institutions (PRIs) are the grassroots level of governance in rural India, playing a critical role in local self-government. The Panchayati Raj system empowers local communities to participate in decision-making processes, which is crucial for disaster management and mitigation. Given the increasing frequency and severity of natural disasters in India, the role of Panchayats in disaster mitigation has become essential for enhancing community resilience and preparedness.

2. Objectives

- **Empower local communities to actively participate in disaster risk reduction efforts.**
- **Enhance the capacity of Panchayats to plan and implement effective disaster management strategies.**
- **Foster collaboration between local governance structures, communities, and governmental agencies.**

3. Functions of Panchayats in Disaster Mitigation

a. Planning and Preparedness

- **Disaster Risk Assessment**: Panchayats can conduct assessments to identify vulnerabilities and risks specific to their communities, creating a baseline for planning.
- **Contingency Planning**: Developing disaster response and contingency plans tailored to local needs, including evacuation routes, shelter locations, and resource allocation.

b. Community Awareness and Education

- **Capacity Building**: Organizing training sessions and workshops for community members on disaster preparedness, first aid, and response strategies.
- **Awareness Campaigns**: Implementing awareness programs to educate residents about disaster risks, preparedness measures, and response protocols.

c. Resource Management

- **Local Resource Mobilization**: Identifying and mobilizing local resources (human, financial, and material) for disaster response and recovery efforts.
- **Infrastructure Development**: Ensuring that local infrastructure, such as roads, drainage systems, and buildings, are designed to withstand disasters through proper planning and construction practices.

d. Collaboration and Coordination

- **Engagement with Stakeholders**: Collaborating with government agencies, NGOs, and other organizations to coordinate disaster management efforts and share resources.
- **Community Participation**: Involving local communities in decision-making processes to ensure that their needs and perspectives are incorporated into disaster management strategies.

e. Monitoring and Evaluation

- **Post-Disaster Assessment**: Conducting evaluations after disaster events to assess the effectiveness of response efforts and identify areas for improvement.
- **Feedback Mechanisms**: Establishing channels for community feedback to enhance future disaster preparedness and response initiatives.

4. Case Studies

a. Kerala's Panchayat System in Flood Response (2018)
During the devastating floods in Kerala in 2018, local Panchayats played a crucial role in disaster response. They facilitated the establishment of relief camps, coordinated rescue operations, and ensured that essential services reached affected communities. Their local knowledge and networks significantly contributed to the efficiency of the response efforts.

b. Rajasthan's Community-Based Disaster Management
In Rajasthan, Panchayats have been actively involved in community-based disaster management programs that focus on building resilience to droughts and floods. Through training and capacity-building initiatives, local leaders have empowered communities to develop their disaster preparedness plans, improving their overall resilience.

5. Challenges Faced by Panchayats in Disaster Mitigation

- **Lack of Resources**: Many Panchayats face financial constraints and lack access to necessary resources for effective disaster management.
- **Limited Capacity and Training**: Insufficient training and knowledge among local leaders and members can hinder effective planning and response.
- **Policy Gaps**: Inconsistencies between national and local disaster management policies can create confusion and limit the effectiveness of local initiatives.
- **Coordination Issues**: Difficulty in coordinating with multiple agencies and stakeholders can complicate disaster response efforts.

6. Recommendations for Strengthening the Role of Panchayats

- **Capacity Building Programs**: Enhance training programs for Panchayat members and local leaders to improve their knowledge and skills in disaster risk management.
- **Resource Allocation**: Ensure adequate funding and resources are allocated to Panchayati Raj institutions for disaster preparedness and response.
- **Strengthening Policies**: Establish clear policies that outline the roles and responsibilities of Panchayats in disaster management and integrate them into national and state disaster management frameworks.
- **Encouraging Community Participation**: Foster community involvement in disaster management processes to ensure that local knowledge and needs are incorporated into planning and response efforts.

Ques10.Write Short notes on Environmental Policies and Programs in India**?**
Ans. Environmental Policies and Programs in India: Institutions and National Framework

1. Introduction

India faces significant environmental challenges, including pollution, deforestation, biodiversity loss, and climate change. To address these challenges, the Indian government has established a comprehensive framework of environmental policies and programs, supported by various institutions at the national, state, and local levels. This document provides an overview of key environmental policies, programs, and the institutions responsible for their implementation.

2. Key Environmental Policies in India

a. The Environment (Protection) Act, 1986

- This act serves as the primary legislation for environmental protection in India. It provides a framework for the protection and improvement of the environment and establishes authorities for enforcement.
- It empowers the government to regulate industrial operations, manage hazardous substances, and set standards for emissions and discharges.

b. National Policy on Environment, 2006

- This policy aims to ensure sustainable development while protecting the environment. It emphasizes the need for participatory management of natural resources and addresses issues related to pollution, biodiversity, and climate change.
- The policy promotes a holistic approach to environmental management, integrating social and economic dimensions.

c. Forest (Conservation) Act, 1980

- This act regulates the diversion of forest land for non-forest purposes. It aims to protect and conserve forest cover and biodiversity in India.
- It requires prior approval from the central government for the conversion of forest land.

d. Wildlife Protection Act, 1972

- This act provides for the protection of wildlife and their habitats. It establishes protected areas, including national parks and wildlife sanctuaries, and prohibits hunting and poaching of endangered species.

e. National Biodiversity Action Plan (NBAP), 2008

- The NBAP aims to protect India's rich biodiversity and promote sustainable use of biological resources. It includes strategies for conservation, sustainable management, and equitable sharing of benefits.

f. National Clean Air Program (NCAP), 2019

- Launched to tackle air pollution in cities, NCAP aims to reduce particulate matter (PM) levels and improve air quality through a multi-faceted approach, including monitoring, public awareness, and implementing pollution control measures.

3. Key Environmental Programs in India

a. Swachh Bharat Mission (Clean India Mission)

- Launched in 2014, this program aims to improve sanitation and waste management across the country. It focuses on eliminating open defecation and promoting solid waste management practices in urban and rural areas.

b. National Afforestation Program (NAP)

- This program aims to increase forest cover and improve ecosystem services through afforestation and reforestation initiatives. It involves community participation in forest management.

c. National River Conservation Plan (NRCP)

- The NRCP focuses on cleaning and conserving major rivers in India, including the Ganges and Yamuna. It includes measures to control pollution, restore water quality, and promote sustainable river management practices.

d. Pradhan Mantri Ujjwala Yojana

- Launched in 2016, this scheme aims to provide clean cooking fuel (LPG) to rural households, reducing reliance on traditional biomass and improving indoor air quality.

e. National Solar Mission

- Part of the National Action Plan on Climate Change, this mission aims to promote the use of solar energy and increase solar power generation capacity, contributing to India's renewable energy goals.

4. Key Institutions for Environmental Governance

a. Ministry of Environment, Forest and Climate Change (MoEFCC)

- The central ministry responsible for formulating and implementing environmental policies and programs. It oversees various environmental regulations, coordinates between different government agencies, and engages with stakeholders.

b. Central Pollution Control Board (CPCB)

- A statutory organization under the MoEFCC, the CPCB monitors and regulates air and water quality, sets standards for pollution control, and provides technical assistance to state pollution control boards.

c. State Pollution Control Boards (SPCBs)

- These boards operate at the state level, enforcing environmental laws and regulations, monitoring pollution, and implementing pollution control measures.

d. Forest Research Institute (FRI)

- An autonomous institution under the Ministry of Environment, responsible for research and training in forestry, biodiversity conservation, and sustainable forest management.

e. National Biodiversity Authority (NBA)

- Established under the Biological Diversity Act, 2002, the NBA is responsible for implementing the provisions of the act, promoting biodiversity conservation, and facilitating access to biological resources.

f. National Green Tribunal (NGT)

- Established in 2010, the NGT adjudicates environmental disputes and provides a specialized forum for the expeditious disposal of cases related to environmental protection and conservation.

Ques11.Explain in detail about Sustainable Rural Development**?**

Ans. Sustainable Rural Development: A Remedy to Disasters

1. Introduction

Sustainable rural development (SRD) refers to the process of improving the quality of life in rural areas while ensuring environmental integrity and social equity. In many countries, including India, rural areas are highly vulnerable to various disasters such as floods, droughts, landslides, and other climate-related events. Implementing sustainable practices in rural development can enhance community resilience, reduce vulnerability, and serve as an effective remedy to mitigate the impacts of disasters.

2. Objectives

- **Enhance community resilience to natural disasters through sustainable practices.**
- **Promote environmental conservation while improving rural livelihoods.**
- **Foster social equity and inclusivity in rural development initiatives.**
- **Integrate disaster risk reduction (DRR) strategies into rural development planning.**

3. Importance of Sustainable Rural Development

Sustainable rural development is crucial for:

- **Reducing Vulnerability**: By improving infrastructure, livelihoods, and resource management, rural communities can better withstand the impacts of disasters.
- **Enhancing Food Security**: Sustainable agricultural practices can ensure a stable food supply, reducing the risk of food insecurity during disasters.
- **Conserving Natural Resources**: Responsible management of land, water, and forests can help maintain ecosystem services that buffer against disasters.
- **Promoting Social Cohesion**: Strengthening community networks fosters collaboration and collective action during crises.

4. Strategies for Sustainable Rural Development as a Disaster Remedy

a. Disaster Risk Reduction (DRR) Integration

- **Community-Based DRR**: Involve local communities in identifying risks, developing contingency plans, and implementing mitigation measures.
- **Training and Capacity Building**: Educate communities on disaster preparedness and response strategies, enabling them to take proactive measures.

b. Sustainable Agriculture Practices

- **Agroecology**: Encourage farming techniques that enhance biodiversity, soil health, and water conservation. Practices like crop rotation, intercropping, and organic farming reduce dependence on chemical inputs and increase resilience to climate variability.

- **Diversification of Livelihoods**: Promote diverse income sources through agroforestry, livestock rearing, and alternative livelihoods, which can cushion communities against agricultural failures.

c. Infrastructure Development

- **Resilient Infrastructure**: Invest in building infrastructure that can withstand disasters, such as flood-resistant roads, irrigation systems, and housing.
- **Access to Services**: Improve access to health care, education, and emergency services, which are vital for community resilience during disasters.

d. Ecosystem Conservation and Restoration

- **Natural Resource Management**: Implement community-led conservation initiatives to protect forests, wetlands, and watersheds that provide essential ecosystem services and act as buffers against disasters.
- **Reforestation and Afforestation**: Promote tree planting initiatives to stabilize soil, reduce erosion, and enhance carbon sequestration.

e. Community Participation and Empowerment

- **Inclusive Decision-Making**: Engage marginalized groups, including women and indigenous communities, in planning and decision-making processes to ensure their voices are heard.
- **Strengthening Social Networks**: Foster community cohesion through social programs, cooperatives, and support groups that enhance collective resilience.

5. Case Studies

a. Sikkim's Organic Farming Initiative
Sikkim, India's first fully organic state, has demonstrated that sustainable agriculture can lead to enhanced resilience against climatic shocks. The organic farming approach has improved soil health and crop yields, reducing vulnerability to pests and climate-related stresses.

b. The Role of Mangroves in Coastal Areas
In coastal regions of India, such as Odisha, the restoration of mangroves has provided a natural barrier against storm surges and coastal erosion. These ecosystems enhance biodiversity while also protecting livelihoods dependent on fishing and agriculture.

c. Community-Based Disaster Risk Reduction in Uttarakhand
In Uttarakhand, community-led initiatives that integrate DRR into development planning have improved disaster preparedness and response capabilities. Training programs and awareness campaigns have significantly increased local capacity to handle disasters.

6. Challenges and Barriers

Despite the potential of sustainable rural development, several challenges hinder its implementation:

- **Lack of Resources**: Limited financial and technical resources for rural development initiatives can impede progress.
- **Policy Gaps**: Inconsistencies between disaster management and rural development policies can lead to ineffective strategies.
- **Climate Change**: Increasing climate variability can exacerbate vulnerabilities and complicate planning efforts.
- **Cultural Barriers**: Resistance to change and the adoption of new practices can limit the effectiveness of sustainable initiatives.

Ques12.Explain in detail about volcanic eruptions, causes and impact of volcanoes?

Ans. Volcanoes are ruptures in the crust of our planet Earth that allow hot gases, molten lava and some rock fragments to erupt by opening and exposing the magma inside. In this piece of article, we will be discussing how and why volcanoes erupt.

Causes of Volcanic Eruption

We know that the mantle of the Earth is too hot, and the temperature ranges from 1000° Celsius to 3000° Celsius. The rocks present inside melt due to high pressure and temperature. The melted substance is light in weight. This thin lava comes up to the crust since it can float easily. Since the density of the magma between the area of its creation and the crust is less than the enclosed rocks, the magma gets to the surface and bursts.
The magma is composed of andesitic and rhyolitic components along with water, sulfur dioxide, and carbon dioxide in dissolved form. By forming bubbles, excess water is broken up with magma. When the magma comes closer to the surface, the level of water decreases and the gas/magma rises in the channel. When the volume of the bubbles formed is about 75%, the magma breaks into pyroclasts and bursts out.
The three main causes of volcanic eruptions are:
The buoyancy of the magma
Pressure from the exsolved gases in the magma
Increase in pressure on the chamber lid

Volcanic Eruption Types

Volcanoes are segmented as per the style of eruption and the form which is created on the earth's surface.

Shield Volcanoes: The size of these volcanoes is greater than that of any currently active volcano on Earth's surface. These volcanoes are composed of basalt. In the event that water gets inside the vent, it will explode. Other than that, these volcanoes are distinguished by mild explosivity. The

lava that is venturing upwards does so in the shape of a fountain and emerges from the top of the cone before changing into a cinder cone.

Composite Volcanoes: The eruptions of lavas that are cooler and more viscous than basalt give rise to composite volcanoes. Explosive eruptions are how these volcanoes typically occur. With lava, a substantial amount of pyroclastic debris and ashes are easily removed. Layers are formed as a result of the material gathering close to the vent apertures. The largest composite volcanoes in the globe that have erupted in the past include the Mayon Volcano in the Philippines, Mount Fuji in Japan, etc.

Caldera: The most explosive volcanoes on the earth are regarded as calderas. Most of these volcanoes are explosive. They're more likely to collapse in on themselves during an eruption than to build something. Calderas are the name for the ruined depressions.

Flood Basalt Provinces: Volcanoes in the Flood Basalt Province spew fluid lava that can readily travel great distances. Thick basalt lava flows have blanketed many areas of the world. Volcanoes of the Mid-Ocean Ridge: These volcanoes typically erupt in oceanic regions. There is a network of mid-ocean ridges that span the ocean basins for more than 70000 kilometres. The majority of eruptions routinely occur in the centre of this ridge.

Volcanic Eruption Causes

Depending on each seismology, the Earth's mantle within the crust is divided into several portions. These range from 8 to 35 km to 410 km for the upper mantle, 400 to 660 km for the transition zone, and 660 to 2891 km for the lower mantle. From the crust to the mantle site, the conditions drastically alter. Temperatures can reach 1000oC, and pressures increase dramatically. Within the crust of the Earth, these viscous, molten rocks are gathered in large chambers.

Magma seeks to find cracks and areas of weakness in the mantle because it is lighter than the surrounding rock and floats upward towards the surface. The molten rock is known as magma when it is below the surface and explodes as ash when it rises. When the lava flows readily, it spreads far and creates wide shield volcanoes. It takes on the shape of a classic cone volcano when it is too thick. The magma rises to the surface and erupts because its density between the region of its formation and the crust is lower than that of the surrounding rocks. Along with water, dissolved carbon dioxide, sulphur dioxide, and andesitic and rhyolitic minerals make up the magma. Magma and extra water are broken up by the formation of bubbles. The water level drops and the gas/magma rise in the channel as the magma gets closer to the surface. The magma ruptures into pyroclasts and bursts out when the volume of the created bubbles reaches about 75%. These are the top three contributors to volcanic eruptions:

The magma's buoyancy

The magma's expelled gases are under pressure.

Pressure on the chamber lid rising

Lava domes are the result of lava building up inside a volcano and exploding if it is very thick. We are aware that the Earth's mantle is very warm, with temperatures ranging from 1000° to 3000° Celsius. Due to the intense pressure and heat, the inner rocks melt. The melted material weighs little. Due to its high level of flotation, this thin lava rises to the crust. When it reaches the surface, a volcano's highest point eventually erupts. With each eruption, the volcanic vent is covered with rocks, lava, and ash. The viscosity of the magma has a significant impact on the eruption's nature.

Volcanic Eruption Impact

Rock is formed on the surface of the Earth by volcanic eruptions.

Climate is impacted by gases and dust particles released into the atmosphere during volcanic eruptions.

Over millions of years, during periods of extremely high volcanism, volcanoes have also contributed to global warming by releasing greenhouse gases into the atmosphere.

Volcanoes are located in specific locations on Earth, but as gases, dust, and ash enter the atmosphere, their impacts may be more broadly dispersed.

This is because eruptions in the tropics can affect the climate in both hemispheres because of the patterns of air circulation.

Ques13.Explain Ecological Planning for Sustainability and Sustainable Development in India**?**

Ans. Ecological Planning for Sustainability and Sustainable Development in India

1. Introduction

Ecological planning is a systematic approach that integrates ecological principles with land-use planning to promote sustainability and sustainable development. In India, a country characterized by its diverse ecosystems, rapid urbanization, and economic growth, ecological planning is crucial for addressing environmental challenges, preserving biodiversity, and ensuring the well-being of communities. This document explores the significance, principles, and strategies of ecological planning for achieving sustainability in India.

2. Objectives

- **Promote sustainable land-use practices that balance ecological integrity with socio-economic needs.**
- **Enhance resilience to climate change and environmental degradation through adaptive planning.**
- **Integrate community participation and traditional knowledge into planning processes.**

- **Encourage the conservation of biodiversity and natural resources for future generations.**

3. Importance of Ecological Planning in India

- **Biodiversity Conservation**: India is one of the world's 17 megadiverse countries, housing a rich variety of flora and fauna. Ecological planning helps in conserving these vital ecosystems.
- **Climate Change Mitigation**: Strategic planning can enhance natural carbon sinks, promote renewable energy sources, and reduce vulnerability to climate impacts.
- **Sustainable Livelihoods**: Ecological planning can promote practices that sustain local economies, such as sustainable agriculture, forestry, and tourism.
- **Pollution Reduction**: Effective planning can help manage waste, reduce emissions, and protect water resources.

4. Key Principles of Ecological Planning

1. **Holistic Approach**: Consideration of social, economic, and environmental factors in planning processes to ensure balanced development.
2. **Sustainability**: Long-term viability of natural resources and ecosystems, ensuring that development meets present needs without compromising future generations.
3. **Community Involvement**: Engaging local communities in the planning process to incorporate their knowledge, needs, and aspirations.
4. **Adaptive Management**: Flexibility in planning to adjust to changing environmental conditions and socio-economic dynamics.
5. **Ecosystem Services**: Recognizing and valuing the services provided by ecosystems (such as clean water, air, and pollination) in planning and decision-making.

5. Strategies for Ecological Planning in India

a. Integrated Land-Use Planning

- Develop land-use plans that consider ecological boundaries and the carrying capacity of ecosystems.
- Promote zoning regulations that protect ecologically sensitive areas, such as forests, wetlands, and coastal zones.

b. Sustainable Agriculture Practices

- Encourage agroecological practices that enhance soil health, reduce chemical inputs, and promote biodiversity.
- Implement agroforestry and organic farming to improve livelihoods while conserving ecosystems.

c. Conservation of Natural Resources

- Establish protected areas and biodiversity hotspots to safeguard critical habitats and endangered species.
- Promote community-based resource management initiatives that empower local communities in conservation efforts.

d. Urban Planning for Sustainability

- Develop smart cities with green infrastructure, efficient public transportation, and sustainable waste management practices.
- Incorporate green spaces, parks, and urban forests to enhance urban biodiversity and improve quality of life.

e. Climate Resilience Planning

- Implement climate adaptation strategies, such as building flood defenses and restoring wetlands, to enhance resilience to climate impacts.
- Promote renewable energy sources (solar, wind, and biomass) to reduce dependency on fossil fuels and mitigate greenhouse gas emissions.

6. Community Participation and Traditional Knowledge

Engaging local communities in the planning process is essential for successful ecological planning. Incorporating traditional knowledge and practices can lead to more effective and culturally appropriate solutions. Strategies include:

- **Participatory Mapping**: Involving communities in mapping their resources and identifying ecologically sensitive areas.
- **Capacity Building**: Providing training and resources to empower local communities in sustainable practices and decision-making.
- **Collaboration with NGOs**: Partnering with non-governmental organizations that work with local communities to promote sustainable development initiatives.

7. Policy Framework and Implementation

The following policies and initiatives are critical for promoting ecological planning in India:

- **National Biodiversity Action Plan (NBAP)**: A framework to conserve biodiversity and promote sustainable use of biological resources.
- **National Environmental Policy (NEP)**: Guidelines for promoting sustainable development and environmental protection.
- **Smart Cities Mission**: A government initiative to promote sustainable urban development through innovative solutions.
- **Pradhan Mantri Fasal Bima Yojana**: Crop insurance scheme aimed at providing financial support to farmers in the event of crop loss due to natural disasters.

8. Challenges and Barriers

- **Rapid Urbanization**: The pace of urbanization often outstrips the capacity for sustainable planning, leading to environmental degradation.
- **Policy Gaps**: Lack of coherence between policies and regulations can hinder effective ecological planning.
- **Resource Constraints**: Limited financial and human resources can impede the implementation of sustainable practices and technologies.
- **Climate Change**: Increasing unpredictability of weather patterns complicates planning efforts.

Ques14.Explain in detail Regional Survey of Disasters in the Hills of India**?**

Ans. A Regional Survey of Disasters in the Hills of India

1. Introduction to Hill Disasters in India

The hilly regions of India are characterized by their unique geography, which includes steep slopes, narrow valleys, and fragile ecosystems. While these areas are rich in biodiversity and cultural heritage, they are also prone to various natural disasters, including landslides, earthquakes, flash floods, and glacial lake outburst floods (GLOFs). These disasters pose significant threats to local communities, infrastructure, and the environment, making disaster management a critical concern.

2. Objectives of the Survey

- **Assess the types and frequency of disasters affecting hilly regions in India.**
- **Identify the primary causes and contributing factors to these disasters.**
- **Evaluate the impacts of these disasters on communities, infrastructure, and ecosystems.**
- **Recommend strategies for disaster risk reduction and management.**

3. Study Areas

This survey will focus on key hilly regions in India known for experiencing significant disasters:

- **Himalayan Region**: States such as Uttarakhand, Himachal Pradesh, and Jammu & Kashmir.
- **Northeast India**: States like Sikkim, Arunachal Pradesh, and Assam.
- **Western Ghats**: Parts of Maharashtra and Karnataka.

4. Methodology

A comprehensive survey of disasters in the hills will include the following methodologies:

a. Data Collection and Analysis

- **Historical Data Review**: Collect and analyze records of past disasters, including frequency, intensity, and impacts on communities.
- **Remote Sensing and GIS**: Utilize satellite imagery and Geographic Information Systems (GIS) for mapping disaster-prone areas, landslide susceptibility, and flood risk zones.

b. Field Surveys

- **Community Surveys**: Conduct surveys in affected communities to gather data on disaster impacts, preparedness, and recovery efforts.
- **Infrastructure Assessment**: Evaluate the condition of infrastructure, including roads, bridges, and buildings, before and after disaster events.

c. Stakeholder Engagement

- **Interviews and Workshops**: Engage with local authorities, disaster management agencies, NGOs, and community members to gather insights on disaster management practices and community resilience.

5. Types of Disasters in Hill Regions

a. Landslides

- **Description**: The downward movement of rock, earth, or debris on a slope, often triggered by heavy rainfall, earthquakes, or human activities.
- **Impact Areas**: Frequent in hilly states like Uttarakhand, Himachal Pradesh, and parts of the Northeast.

b. Earthquakes

- **Description**: Sudden shaking of the ground caused by tectonic movements, leading to significant destruction in vulnerable hilly areas.
- **Impact Areas**: Regions along the Himalayan belt and Northeast India, which are seismically active.

c. Flash Floods

- **Description**: Sudden floods caused by intense rainfall, often occurring in steep, mountainous terrain.
- **Impact Areas**: Areas with poor drainage systems, including parts of Uttarakhand and Himachal Pradesh.

d. Glacial Lake Outburst Floods (GLOFs)

- **Description**: Sudden release of water from glacial lakes, often triggered by the melting of glaciers due to climate change.
- **Impact Areas**: Himalayan regions with glacial lakes, such as Sikkim and Ladakh.

6. Causes of Disasters in Hill Regions

a. Natural Factors

- **Geological Instability**: The hilly terrain is prone to landslides and earthquakes due to geological formations and fault lines.
- **Heavy Rainfall**: Intense rainfall during monsoon seasons increases the risk of landslides and flash floods.

b. Climate Change

- **Glacial Melting**: Rising temperatures contribute to accelerated glacier melting, increasing the risk of GLOFs.
- **Erratic Weather Patterns**: Climate change leads to unpredictable weather, resulting in extreme events.

c. Human Activities

- **Deforestation**: Removal of vegetation increases soil erosion and destabilizes slopes, heightening the risk of landslides.
- **Urbanization**: Inadequate planning and construction in hilly areas can exacerbate disaster risks.

7. Impacts of Disasters in Hill Regions

a. Human Displacement and Loss of Life

- Disasters often result in casualties and displacement of communities, leading to long-term socio-economic impacts.

b. Damage to Infrastructure

- Roads, bridges, and buildings are frequently damaged, leading to costly repairs and disruptions in connectivity.

c. Economic Losses

- Agriculture, tourism, and local businesses suffer significant losses, impacting livelihoods and local economies.

d. Environmental Degradation

- Ecosystems, including forests and water bodies, may be disrupted, leading to loss of biodiversity and increased vulnerability.

8. Case Studies

a. Uttarakhand Floods (2013)

- **Overview**: Catastrophic floods triggered by heavy rainfall and landslides, resulting in widespread destruction.
- **Impacts**: Approximately 5,700 people were killed, and infrastructure worth billions was damaged.

b. Sikkim Earthquake (2011)

- **Overview**: A powerful earthquake (magnitude 6.9) struck Sikkim and surrounding areas, causing significant damage.
- **Impacts**: Loss of life, destruction of infrastructure, and disruption of services.

c. Landslides in Himachal Pradesh (2021)

- **Overview**: Heavy rainfall led to multiple landslides, causing road blockages and fatalities.
- **Impacts**: Damage to homes and infrastructure, and disruption of travel and commerce.

9. Mitigation Strategies

To effectively manage and mitigate the impacts of disasters in hilly regions, the following strategies can be recommended:

a. Disaster Preparedness and Response Planning

- Develop comprehensive disaster management plans that include early warning systems and community training programs.

b. Infrastructure Resilience

- Design and construct infrastructure that can withstand landslides and earthquakes, including proper drainage systems.

c. Ecosystem Conservation

- Promote afforestation and sustainable land-use practices to stabilize slopes and reduce the risk of landslides.

d. Community Engagement and Awareness

- Foster awareness and preparedness among local communities through education and participation in disaster risk reduction initiatives.

Ques15. Explain about Regional Survey of Land Subsidence, Coastal Disasters and Cyclonic Disasters **in India**

Ans. 1. Introduction to Land Subsidence in India

Land subsidence is a significant geological phenomenon affecting various regions of India, primarily driven by natural processes and anthropogenic activities. The impacts of subsidence can lead to infrastructure damage, agricultural challenges, and environmental degradation. Understanding the causes and consequences of land subsidence in India is crucial for sustainable development and urban planning.

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2. Objectives of the Survey

- **Assess the extent and distribution of land subsidence across different regions of India.**
- **Identify primary causes of subsidence specific to these regions.**
- **Evaluate impacts on infrastructure, agriculture, and natural resources.**
- **Recommend strategies for monitoring and mitigating subsidence risks.**

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3. Study Areas

This survey can focus on several regions of India known for significant land subsidence issues, including:

- **Gujarat**: Notable for subsidence related to groundwater extraction, particularly in areas like Ahmedabad and Surat.
- **Punjab and Haryana**: Agricultural regions experiencing subsidence due to excessive groundwater pumping.
- **Tamil Nadu**: The coastal city of Chennai faces subsidence due to urbanization and groundwater extraction.
- **Delhi**: The capital territory is also affected by subsidence due to construction activities and groundwater depletion.

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4. Methodology

A comprehensive regional survey should include the following methodologies:

a. Remote Sensing Techniques

- **Satellite Imagery**: Utilize data from satellites (e.g., Sentinel-1, Landsat) for large-scale monitoring of surface deformation using InSAR (Interferometric Synthetic Aperture Radar).
- **Geographical Information Systems (GIS)**: Integrate remote sensing data with GIS for mapping subsidence-prone areas.

b. Field Measurements

- **GPS Surveys**: Conduct ground-based GPS measurements in identified subsidence zones to monitor vertical displacement.
- **Leveling Surveys**: Implement traditional leveling techniques to measure elevation changes over time.

c. Geotechnical Investigations

- **Borehole Drilling**: Collect soil samples to analyze compressibility and compaction characteristics.
- **Groundwater Monitoring**: Install piezometers to monitor groundwater levels and assess their role in subsidence.

d. Historical Data Analysis

- **Review of Past Studies**: Analyze previous research and reports on subsidence in the selected regions.
- **Land Use Change Analysis**: Study historical land use changes through remote sensing data to correlate with subsidence trends.

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5. Causes of Land Subsidence in India

a. Natural Causes

- **Soil Compaction**: Natural processes such as sediment consolidation and compaction due to geological factors.
- **Tectonic Activity**: Certain regions experience subsidence due to tectonic movements and seismic activity.

b. Anthropogenic Causes

- **Groundwater Extraction**: Over-extraction of groundwater for agricultural and urban purposes is a major cause of subsidence in states like Punjab and Gujarat.
- **Mining Activities**: Regions with coal and mineral mining, such as Jharkhand and Chhattisgarh, experience land subsidence due to underground voids.
- **Urbanization**: Rapid urban growth, especially in cities like Delhi and Chennai, leads to increased load on the ground and excessive groundwater withdrawal.

6. Impacts of Land Subsidence

a. Infrastructure Damage

- Roads, buildings, and bridges are at risk of structural damage, leading to costly repairs and safety concerns.

b. Flooding and Water Management Issues

- Lowered land elevation increases vulnerability to flooding, especially in low-lying areas.

c. Agricultural Challenges

- Soil characteristics may change, impacting crop yields and irrigation practices, particularly in agricultural states.

d. Ecosystem Disruption

- Natural habitats and wetlands may be altered or destroyed due to subsidence, leading to loss of biodiversity.

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7. Case Studies

a. Ahmedabad, Gujarat

- **Overview**: Significant subsidence due to over-extraction of groundwater.
- **Impacts**: Infrastructure damage, increased flood risk, and changes in land use.

b. Punjab and Haryana

- **Overview**: Extensive groundwater pumping for agriculture leading to land subsidence.
- **Impacts**: Declining agricultural productivity and water quality issues.

c. Chennai, Tamil Nadu

- **Overview**: Urban subsidence due to groundwater extraction and heavy construction.
- **Impacts**: Damage to buildings and increased flooding during monsoon seasons.

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8. Mitigation Strategies

To address land subsidence effectively, the following strategies can be recommended:

- **Sustainable Groundwater Management**: Implement policies to regulate groundwater extraction and promote rainwater harvesting.
- **Urban Planning and Regulation**: Integrate subsidence risk assessments into urban planning and land use policies.
- **Monitoring Programs**: Establish comprehensive monitoring networks using remote sensing and field measurements to track subsidence trends over time.
- **Public Awareness and Education**: Engage local communities in understanding the causes and consequences of subsidence, promoting sustainable practices.

A Regional Survey of Coastal Disasters in India

1. Introduction to Coastal Disasters in India

India's long coastline of approximately 7,500 kilometers is highly susceptible to various coastal disasters, including cyclones, tsunamis, coastal erosion, and rising sea levels. These disasters pose significant risks to coastal communities, infrastructure, ecosystems, and livelihoods. Understanding the types, causes, impacts, and mitigation strategies associated with coastal disasters is crucial for effective management and preparedness.

2. Objectives of the Survey

- **Assess the types and frequency of coastal disasters affecting different regions of India.**
- **Identify the primary causes and contributing factors to coastal disasters.**
- **Evaluate the impacts of coastal disasters on communities, infrastructure, and ecosystems.**
- **Recommend strategies for disaster risk reduction and management.**

3. Study Areas

This survey will focus on key coastal regions in India known for experiencing significant coastal disasters:

- **East Coast (Bay of Bengal)**: States like Odisha, Andhra Pradesh, and Tamil Nadu are frequently affected by cyclones and storm surges.
- **West Coast (Arabian Sea)**: States like Maharashtra, Gujarat, and Kerala are prone to cyclones and coastal erosion.
- **Andaman and Nicobar Islands**: Vulnerable to tsunamis and extreme weather events.

4. Methodology

A comprehensive regional survey of coastal disasters will include various methodologies:

a. Data Collection and Analysis

- **Historical Data Review**: Collect and analyze historical records of coastal disasters, including frequency, intensity, and impacts on communities.
- **Remote Sensing**: Use satellite imagery and GIS to assess changes in coastal land use, erosion patterns, and damage caused by disasters.

b. Field Surveys

- **Community Surveys**: Conduct surveys in affected coastal communities to gather data on disaster impacts, preparedness, and recovery efforts.
- **Infrastructure Assessment**: Evaluate the condition and resilience of coastal infrastructure, including roads, housing, and critical facilities.

c. Stakeholder Engagement

- **Interviews and Workshops**: Engage with local authorities, disaster management agencies, NGOs, and community members to gather insights and perspectives on coastal disasters and management strategies.

5. Types of Coastal Disasters

a. Cyclones

- **Description**: Severe tropical storms characterized by strong winds, heavy rainfall, and storm surges.
- **Impact Areas**: The eastern coast (Odisha, Andhra Pradesh) experiences frequent cyclones, leading to flooding and destruction of infrastructure.

b. Tsunamis

- **Description**: Large ocean waves generated by underwater earthquakes, volcanic eruptions, or landslides.
- **Impact Areas**: The Andaman and Nicobar Islands and parts of the eastern coast were affected by the 2004 Indian Ocean tsunami.

c. Coastal Erosion

- **Description**: The loss of coastal land due to wave action, currents, and human activities.
- **Impact Areas**: Coastal regions in Gujarat, Maharashtra, and Tamil Nadu are experiencing significant erosion, threatening habitats and human settlements.

d. Rising Sea Levels

- **Description**: Gradual increase in sea levels due to climate change, melting ice caps, and thermal expansion of seawater.
- **Impact Areas**: Low-lying areas in states like West Bengal and Kerala are at risk of inundation and saltwater intrusion.

6. Causes of Coastal Disasters

a. Natural Factors

- **Geological Factors**: Coastal regions are located in seismically active zones, increasing vulnerability to tsunamis and earthquakes.
- **Meteorological Factors**: Favorable conditions for cyclone formation, such as warm sea surface temperatures and atmospheric instability.

b. Human Activities

- **Coastal Development**: Urbanization, industrialization, and infrastructure projects can exacerbate vulnerability to coastal disasters.
- **Deforestation and Land Use Changes**: Altering natural landscapes can disrupt coastal ecosystems and increase erosion.

7. Impacts of Coastal Disasters

a. Human Displacement and Loss of Life

- Coastal disasters often lead to casualties and displacement of communities, resulting in long-term socio-economic impacts.

b. Damage to Infrastructure

- Roads, bridges, homes, and critical facilities are frequently damaged, leading to costly repairs and prolonged disruptions.

c. Economic Losses

- Fisheries, agriculture, and tourism sectors suffer significant losses, impacting local economies and livelihoods.

d. Environmental Degradation

- Coastal ecosystems, including mangroves and coral reefs, are often damaged, leading to loss of biodiversity and disruption of natural processes.

8. Case Studies

a. Cyclone Fani (2019)

- **Overview**: A powerful tropical cyclone that struck Odisha, causing extensive damage to infrastructure, agriculture, and loss of life.
- **Impacts**: Over 1.5 million people were evacuated; damages were estimated at billions of dollars.

b. 2004 Indian Ocean Tsunami

- **Overview**: A devastating tsunami that affected the Andaman and Nicobar Islands and the eastern coast of India.
- **Impacts**: Approximately 230,000 people were killed across the region, with significant destruction of coastal communities.

c. Coastal Erosion in Kerala

- **Overview**: Ongoing coastal erosion threatens homes, infrastructure, and livelihoods in coastal districts.
- **Impacts**: Loss of land, damage to fishing boats, and declining tourism due to beach erosion.

9. Mitigation Strategies

To reduce the impact of coastal disasters, the following strategies can be recommended:

a. Disaster Preparedness and Response Planning

- Develop and implement comprehensive disaster management plans, including early warning systems and community training programs.

b. Coastal Zone Management

- Establish regulations for sustainable coastal development, ensuring the protection of ecosystems and communities.

c. Ecosystem Restoration

- Promote the restoration of mangroves, wetlands, and coral reefs to enhance natural resilience against coastal disasters.

d. Community Engagement and Education

- Foster awareness and preparedness among coastal communities through education and participation in disaster risk reduction initiatives.

A Regional Survey of Cyclonic Disasters in India

1. Introduction to Cyclonic Disasters in India

Cyclonic disasters are one of the most significant natural hazards affecting India, particularly along its eastern and western coasts. India experiences around five to six tropical cyclones each year, primarily during the monsoon season, resulting in widespread destruction, loss of life, and disruption of livelihoods. Understanding the patterns, impacts, and mitigation strategies associated with cyclonic disasters is crucial for enhancing disaster resilience in affected regions.

2. Objectives of the Survey

- **Assess the frequency and distribution of cyclonic disasters across different coastal regions of India.**
- **Identify the primary causes and contributing factors to cyclonic disasters.**
- **Evaluate the impacts of cyclonic disasters on communities, infrastructure, and ecosystems.**
- **Recommend strategies for disaster preparedness, response, and recovery.**

3. Study Areas

This survey will focus on key regions in India known for experiencing significant cyclonic activity:

- **East Coast**: States such as Odisha, Andhra Pradesh, Tamil Nadu, and West Bengal.
- **West Coast**: States like Gujarat and Maharashtra.
- **Island Regions**: The Andaman and Nicobar Islands.

4. Methodology

A comprehensive survey of cyclonic disasters will include the following methodologies:

a. Data Collection and Analysis

- **Historical Data Review**: Collect and analyze records of cyclones, including frequency, intensity, landfall locations, and impacts over the years.
- **Remote Sensing and GIS**: Use satellite imagery and Geographic Information Systems (GIS) to analyze cyclone tracks, rainfall distribution, and damage assessments.

b. Field Surveys

- **Community Surveys**: Conduct surveys in affected communities to gather data on disaster impacts, preparedness, and recovery efforts.
- **Infrastructure Assessment**: Evaluate the condition of infrastructure, including roads, buildings, and public facilities, before and after cyclonic events.

c. Stakeholder Engagement

- **Interviews and Workshops**: Engage with local authorities, disaster management agencies, NGOs, and community members to gather insights on cyclonic disaster management and community resilience.

5. Types of Cyclonic Disasters

Cyclonic disasters can be classified based on their intensity and the damage they cause:

a. Tropical Cyclones

- **Description**: These storms form over warm ocean waters and can bring strong winds, heavy rainfall, and storm surges.
- **Example**: Cyclone Fani (2019) in Odisha.

b. Severe Cyclones

- **Description**: Cyclones that reach wind speeds exceeding 120 km/h, causing extensive damage.
- **Example**: Cyclone Gaja (2018) in Tamil Nadu.

c. Super Cyclones

- **Description**: Extremely intense cyclones with wind speeds greater than 150 km/h, leading to catastrophic impacts.
- **Example**: Super Cyclone Yasi (1991) affecting Odisha.

6. Causes of Cyclonic Disasters

a. Natural Factors

- **Ocean Temperature**: Warm sea surface temperatures are a key factor in cyclone formation.
- **Atmospheric Conditions**: Low-pressure systems and favorable wind patterns contribute to cyclone development.

b. Climate Change

- **Rising Sea Levels**: Increased sea levels exacerbate storm surges and flooding during cyclones.
- **Extreme Weather Patterns**: Climate change influences the frequency and intensity of cyclonic events.

7. Impacts of Cyclonic Disasters

a. Human Displacement and Loss of Life

- Cyclones often lead to casualties and the displacement of communities, resulting in long-term socio-economic impacts.

b. Infrastructure Damage

- Roads, bridges, buildings, and critical infrastructure are frequently damaged, leading to costly repairs and prolonged disruptions.

c. Economic Losses

- Sectors such as agriculture, fisheries, and tourism suffer significant losses, impacting local economies and livelihoods.

d. Environmental Degradation

- Cyclones can lead to loss of coastal ecosystems, including mangroves and wetlands, disrupting natural habitats and biodiversity.

8. Case Studies

a. Cyclone Fani (2019)

- **Overview**: A powerful cyclone that struck Odisha, causing extensive damage to infrastructure, agriculture, and loss of life.
- **Impacts**: Over 1.5 million people were evacuated; damages were estimated at around $9.5 billion.

b. Cyclone Amphan (2020)

- **Overview**: A super cyclone that affected West Bengal and Odisha, causing significant destruction.
- **Impacts**: Estimated damages of $13 billion, impacting millions of people and causing extensive flooding.

c. Cyclone Gaja (2018)

- **Overview**: A severe cyclone that struck Tamil Nadu, leading to widespread destruction.
- **Impacts**: Significant agricultural losses and damage to infrastructure, particularly in coastal districts.

9. Mitigation Strategies

To effectively manage and mitigate the impacts of cyclonic disasters, the following strategies can be recommended:

a. Disaster Preparedness and Response Planning

- Develop and implement comprehensive disaster management plans that include early warning systems and evacuation protocols.

b. Infrastructure Resilience

- Invest in building resilient infrastructure designed to withstand cyclonic impacts, including flood defenses and robust housing.

c. Community Awareness and Training

- Engage communities in disaster preparedness training, awareness campaigns, and drills to enhance local resilience.

d. Ecosystem Protection

- Promote the conservation and restoration of coastal ecosystems, such as mangroves and wetlands, to act as natural buffers against cyclones.